Lecture 06

Tradable Permits

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AEM 4510

Roadmap

1. How do tradable permit systems work in theory and in the real world?
2. What happens under a tradable permit system?

Tradable permits

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This is easy with one firm, but what if we have several, or hundreds?

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Optimal policy with multiple firms

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Regulator problem:

$$\min_{E_1,\ldots,E_N} \sum_{i=1}^{N} C_i(E_i) + D\left(\sum_{i=1}^{N} E_i\right)$$

Optimal policy with multiple firms

Take one FOC for each firm $i = 1, \ldots, N$:

$$\frac{\partial}{\partial E_i}\left[\sum_{j=1}^{N} C_j(E_j) + D\left(\sum_{j=1}^{N} E_j\right)\right] = 0$$

$$C_i'(E_i) + D'\left(\sum_{j=1}^{N} E_j\right) = 0$$

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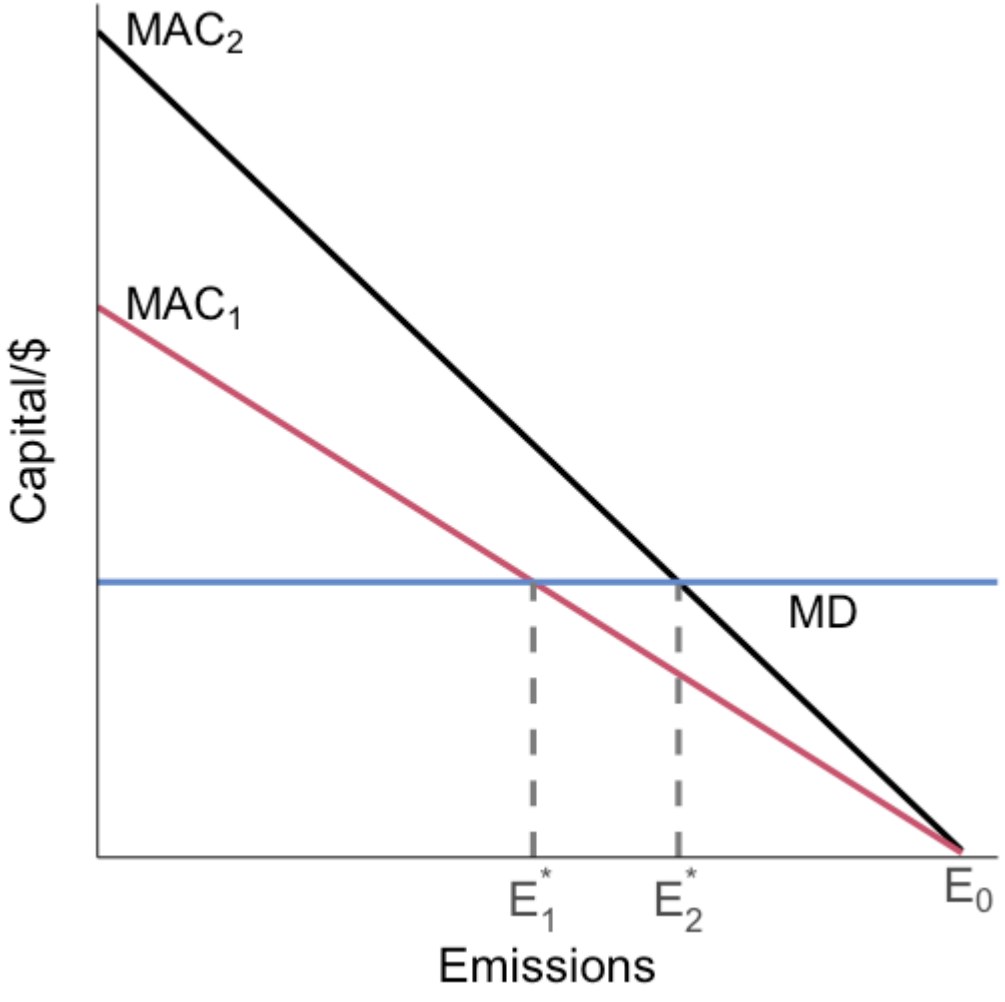
$$\underbrace{-C_i'(E_i)}_{MAC_i} = \underbrace{D'\left(\sum_{j=1}^{N} E_j\right)}_{MD}$$

At the optimum:

$$MAC_1 = MAC_2 = \cdots = MAC_N = MD$$

This gives the target condition that we try to achieve through a decentralized permit market

Optimal policy with multiple firms



Firm #2 is 'dirty': has higher MAC

Firm #1 is 'clean': has lower MAC

If we use a regular emission standard: it has to be firm-specific!

Mandate E_1^* for 1 and E_2^* for 2

This requires **a lot** of info and political capital on behalf of the regulator

Tradable permit systems

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Total emissions didn't change, but how much each firm gets to emit does through voluntary trade

Tradable permit systems: example

The US Acid Rain Program is the classic example

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Permit = license to create 1 ton of SO2

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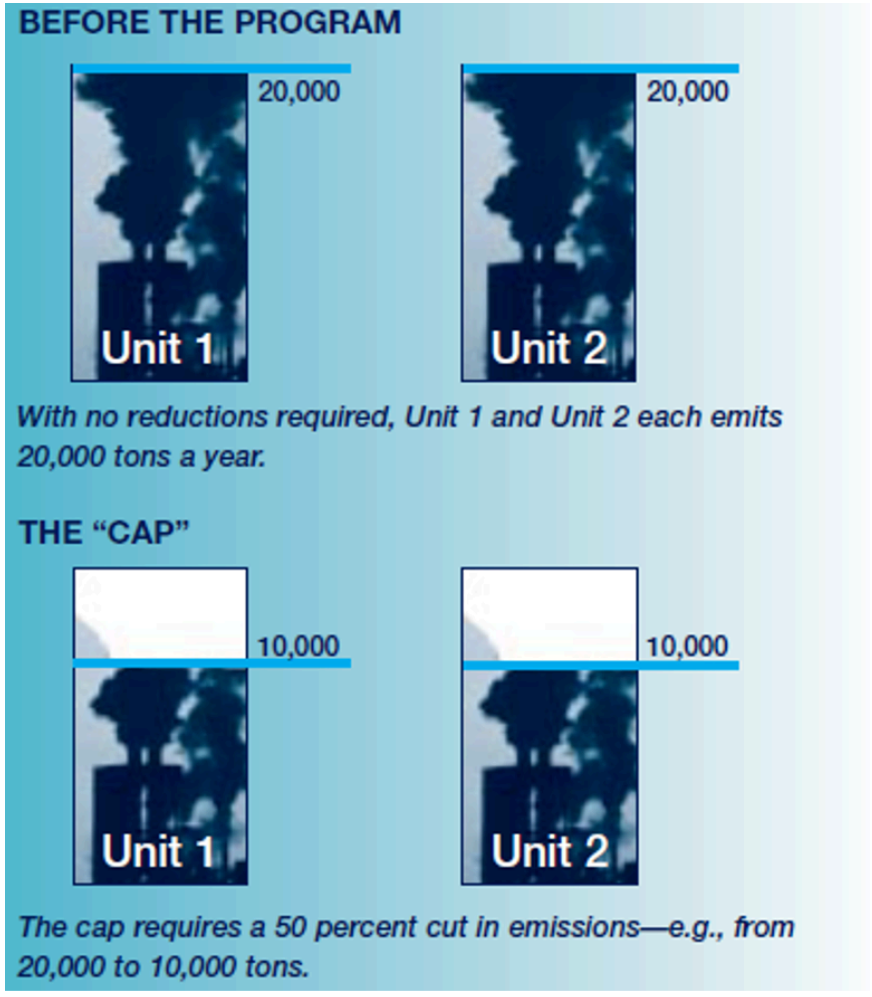
Phase I (1995-2000):

- 6.3 million permits issued per year
- affected 263 generating units at 110 dirtiest power plants

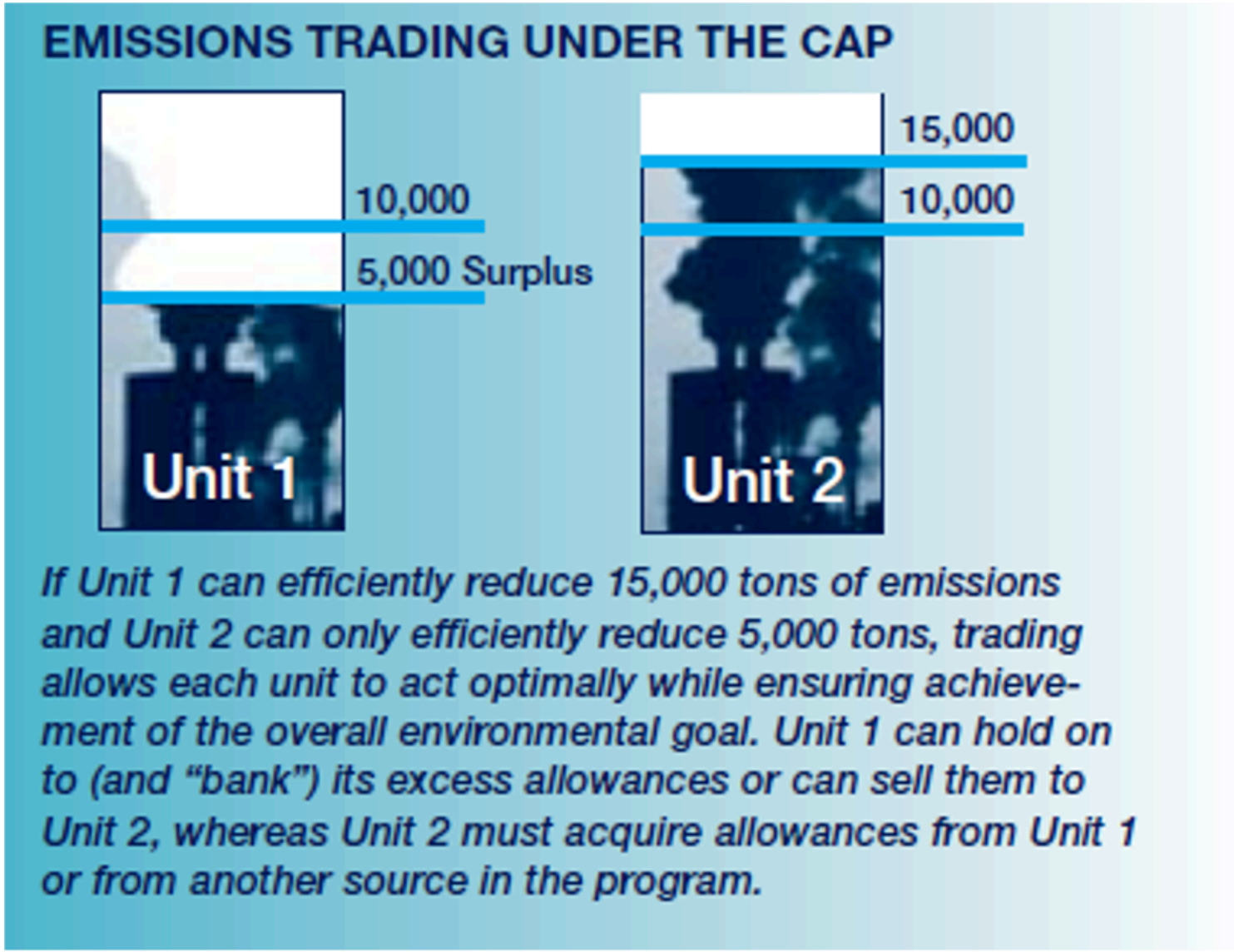
Phase II (2000+):

- 9 million permits issued per year
- affects all power plants over some minimum size

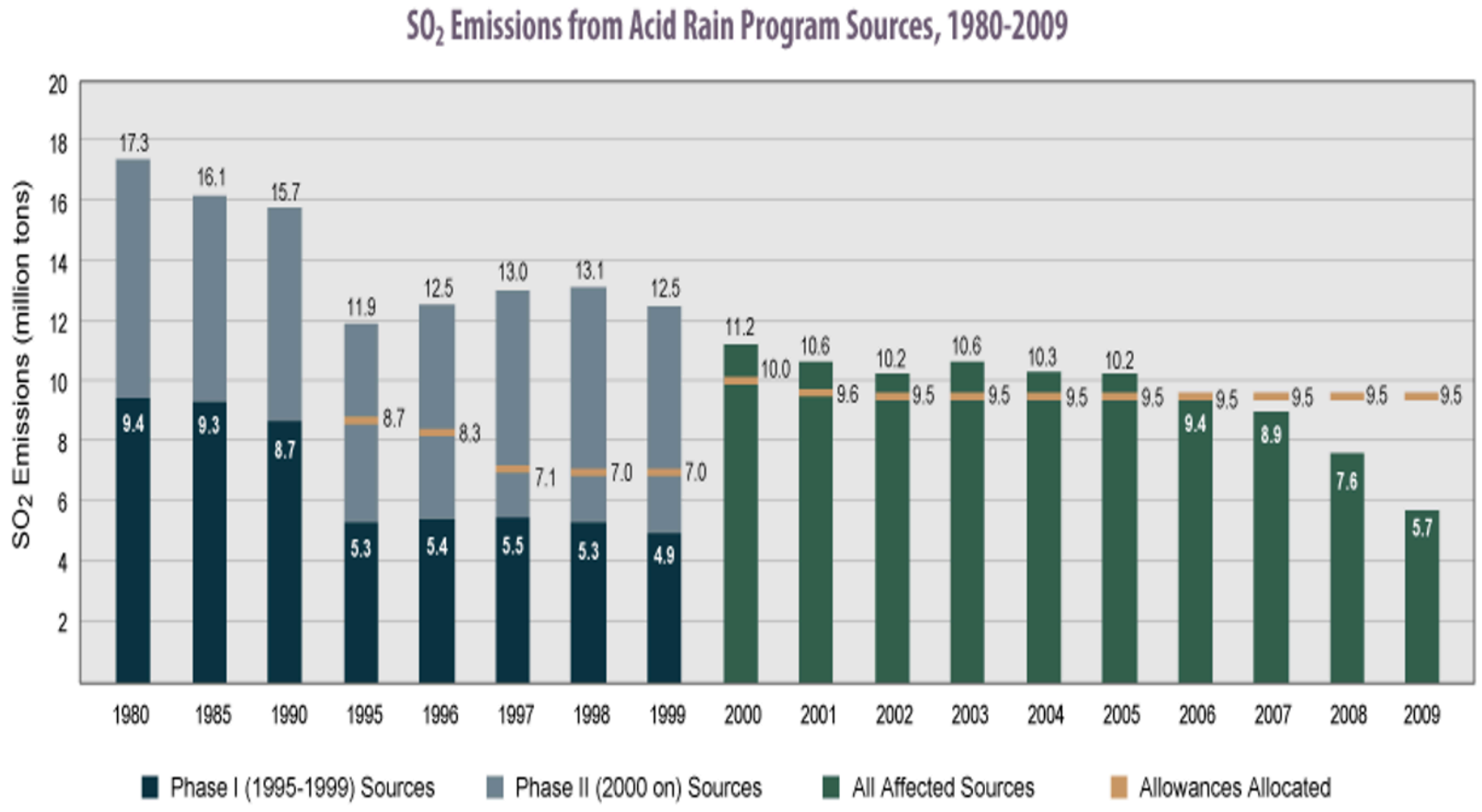
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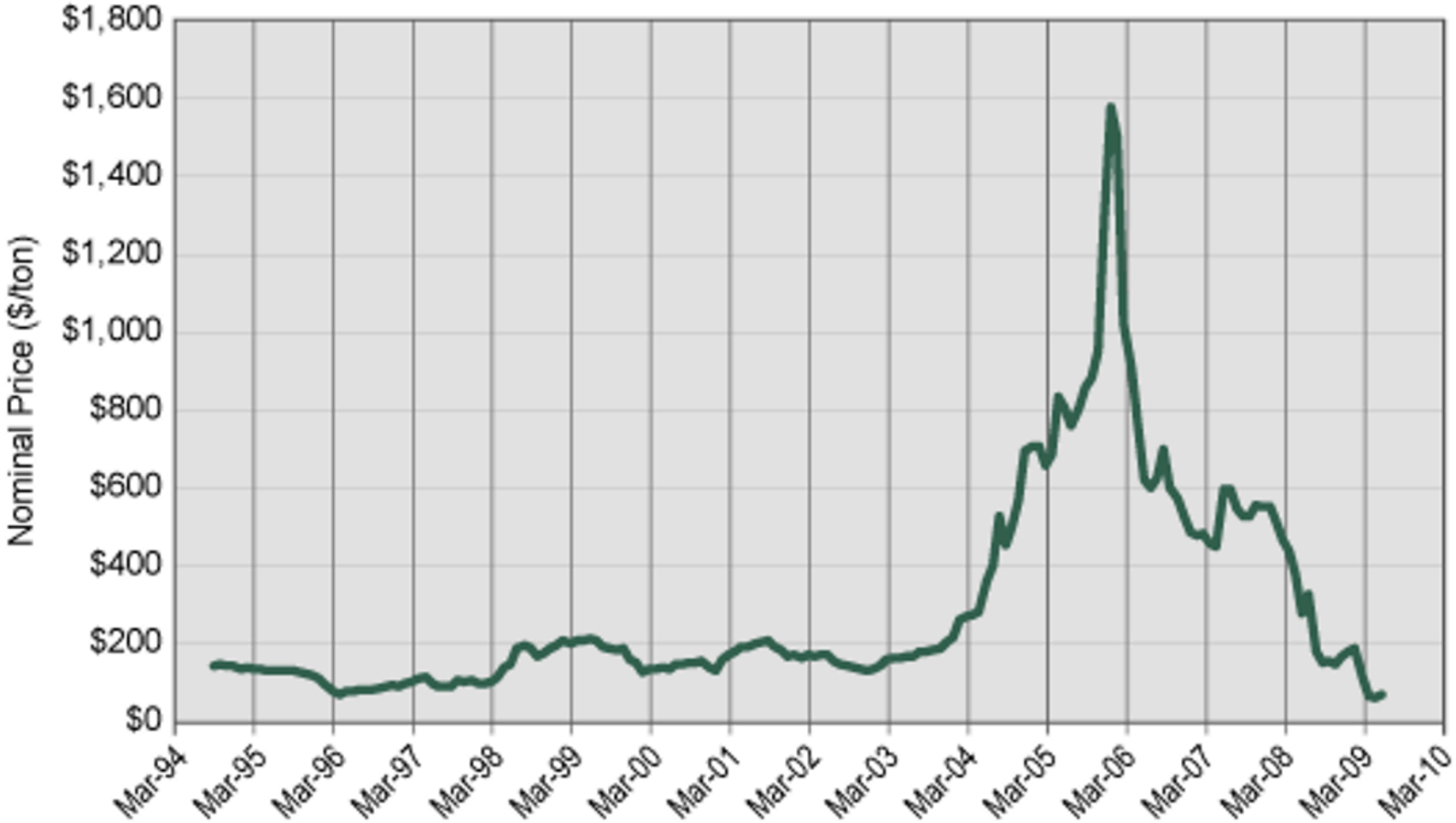


Tradable permit systems: example



Source: EPA, 2010

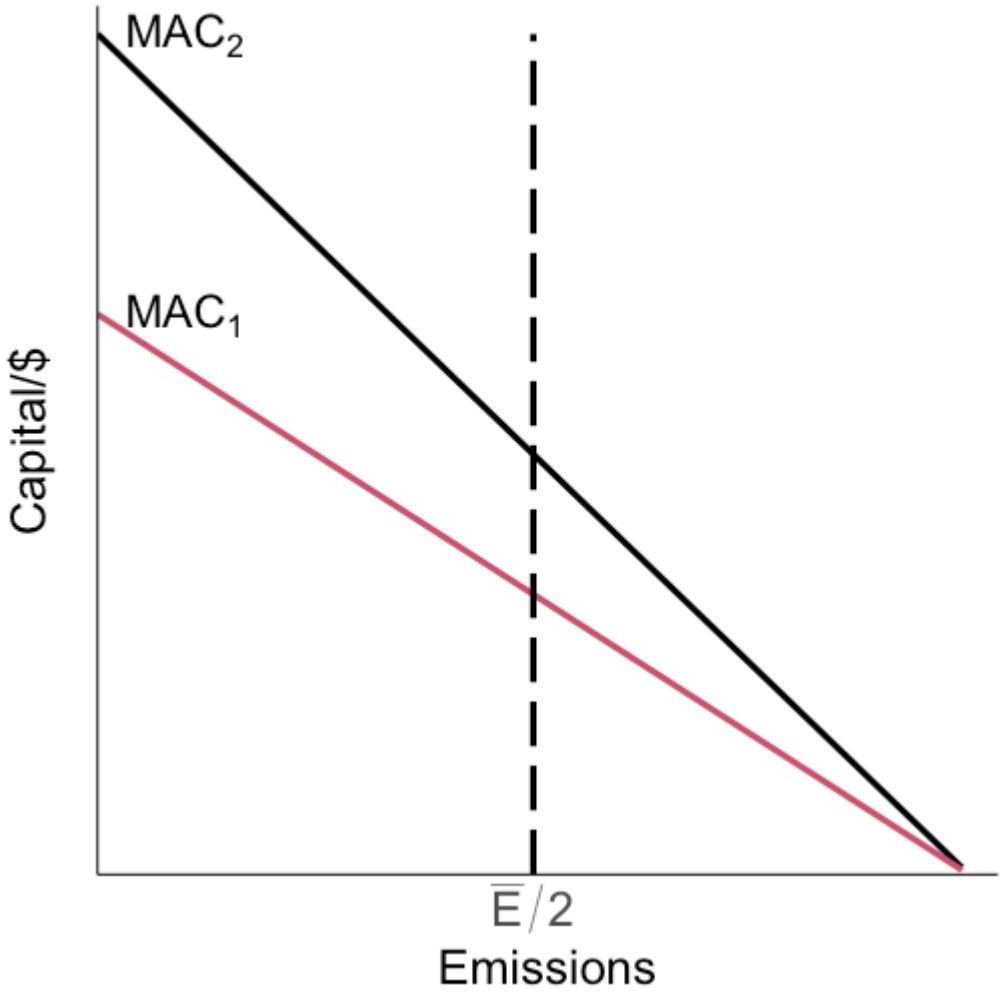
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Quantified benefits*:		
$PM_{2.5}$ mortality (U.S. and southern Canada)	$107,000	
$PM_{2.5}$ morbidity (U.S. and southern Canada)	$8,000	
Ozone mortality (eastern U.S.)	$4,000	
Ozone morbidity (eastern U.S.)	$300	
Visibility at parks (3 U.S. regions)	$2,000	
Recreational fishing in NY	$65	
Ecosystem improvements in Adirondacks (NY residents)	$500	
Total annual quantified benefits		$122,000
Quantified costs for U.S. power generation:		
SO_2 controls	$2,000	
NO_X controls	$1,000	
Total annual quantified costs		$3,000

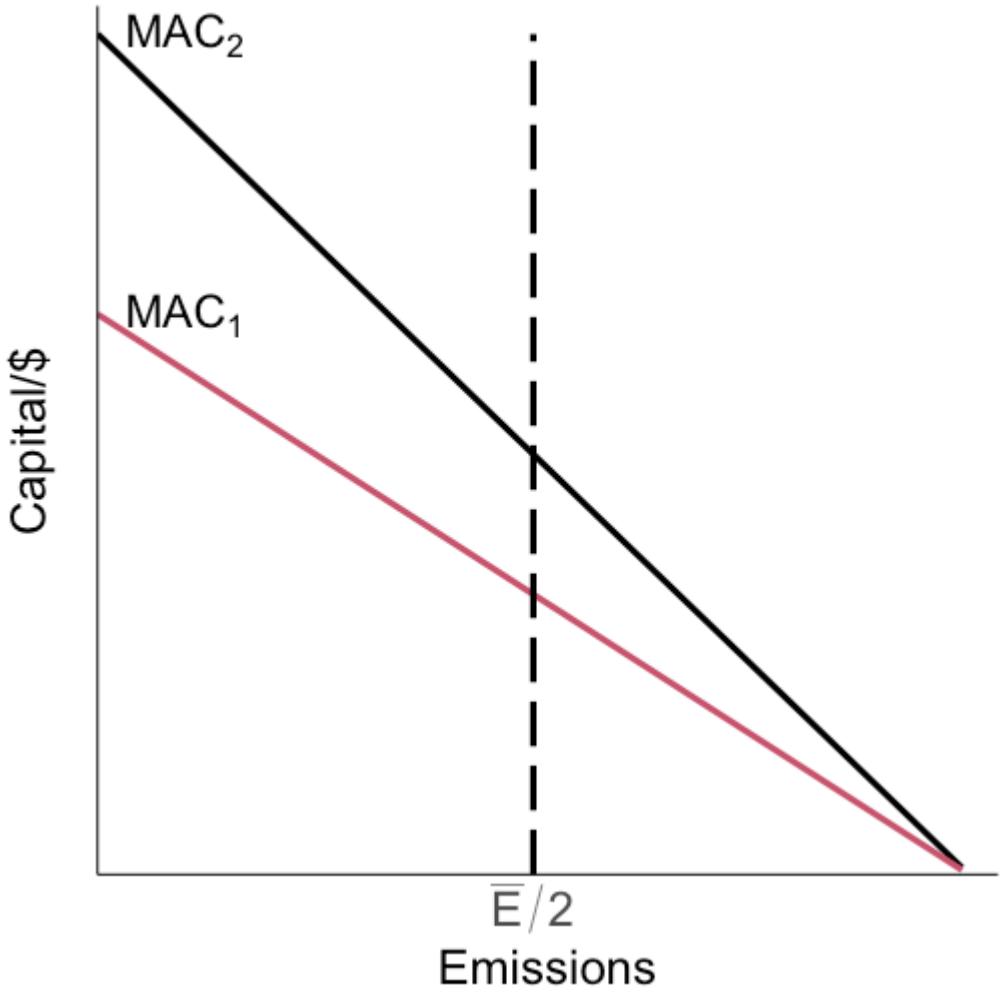
Tradable permits: graphical



Suppose we want to limit to $\bar{E}$ total emissions so each firm gets $\bar{E}/2$ permits, but cant trade them

Is it efficient?

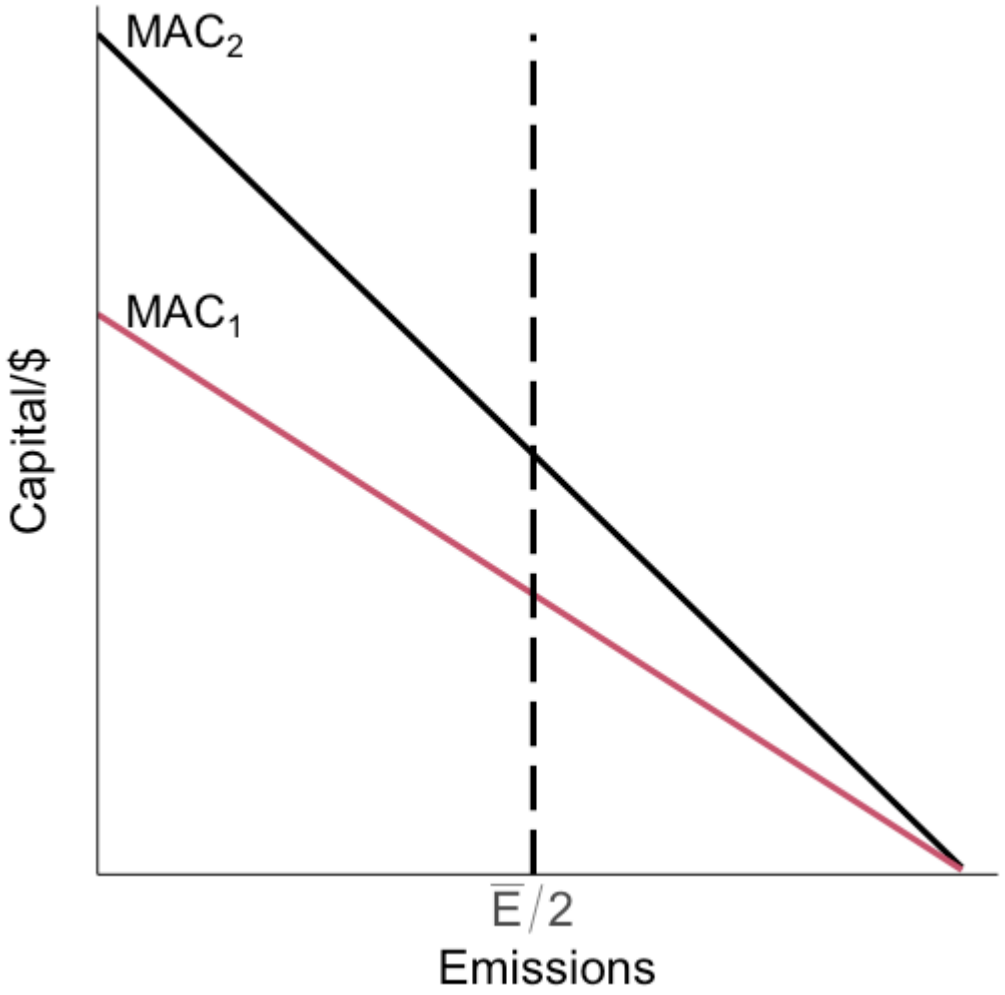
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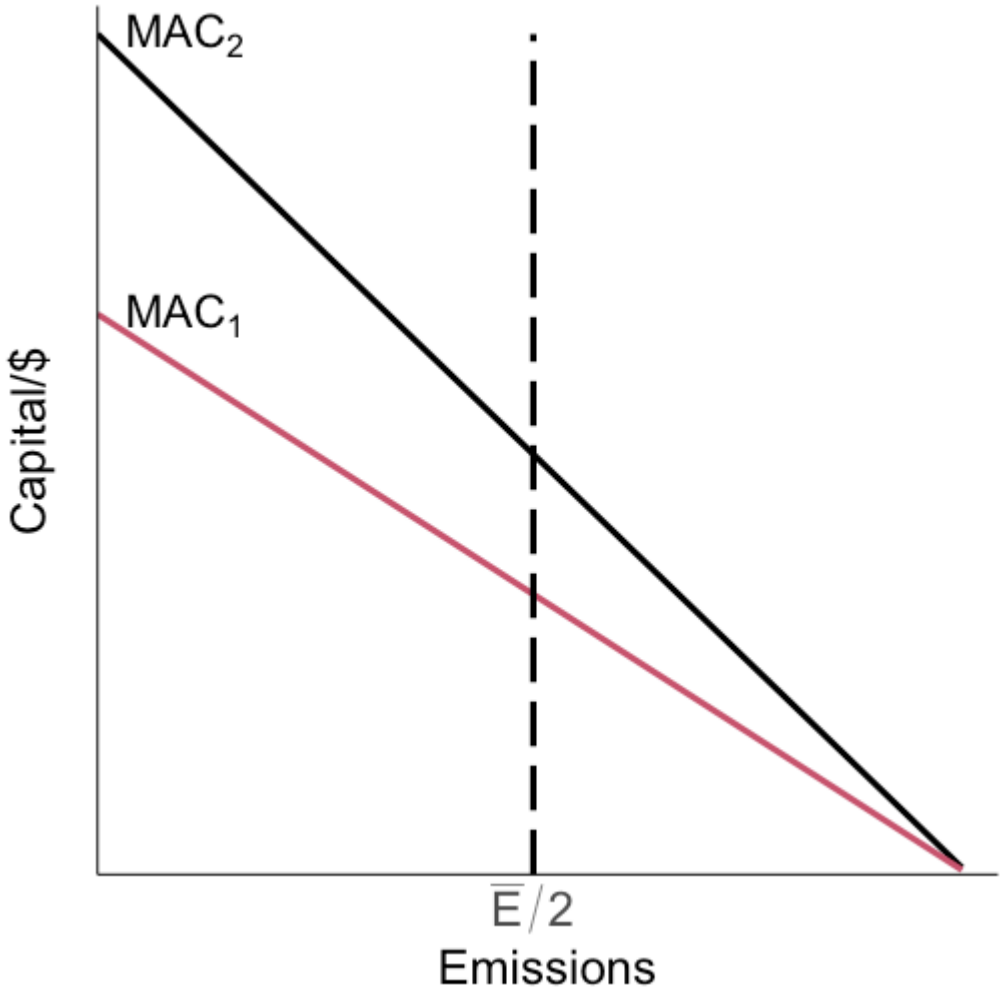


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Is it cost-effective?

Tradable permits: graphical



Suppose we want to limit to $\bar{E}$ total emissions so each firm gets $\bar{E}/2$ permits, but cant trade them

This can't be efficient (i.e. maximize social welfare given some MD curve)

It also can't be cost-effective: it doesn't minimize the cost of achieving $\bar{E}$ total emissions

Tradable permits: cost-effectiveness

For cost-effectiveness, we need total costs to be minimized for achieving a given level of emissions:

$$\min_{E_1,E_2} C_1(E_1) + C_2(E_2) \text{ subject to: } E_1 + E_2 = \bar{E}$$

This is the same problem as:

$$\min_{E_1,E_2} C_1(E_1) + C_2(\bar{E} - E_1)$$

which has a solution where:

$$-C_1'(E_1^*) = -C_2'(\bar{E} - E_1^*)$$

Tradable permits: cost-effectiveness

Cost-effectiveness requires:

$$-C_1'(E_1^*) = -C_2'(\bar{E} - E_1^*) \leftrightarrow MAC_1 = MAC_2$$

That marginal abatement costs are equal across all emitters

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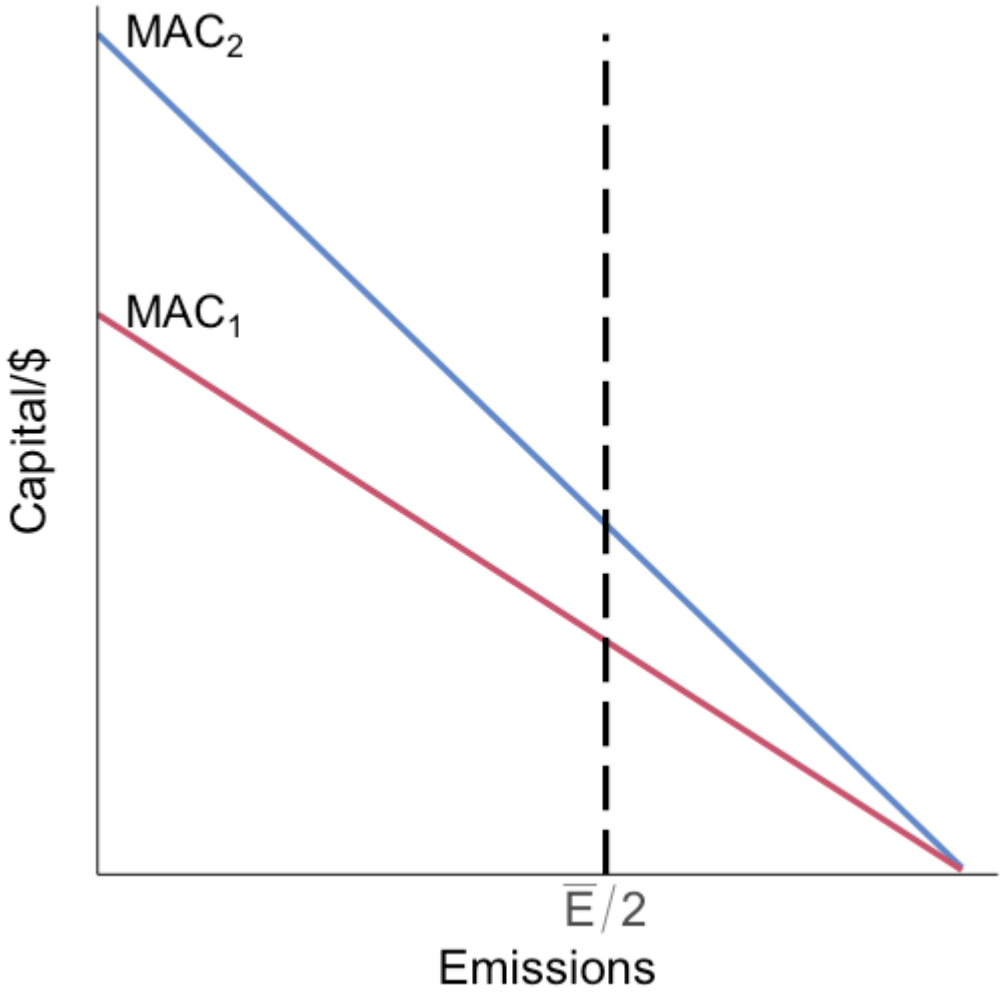
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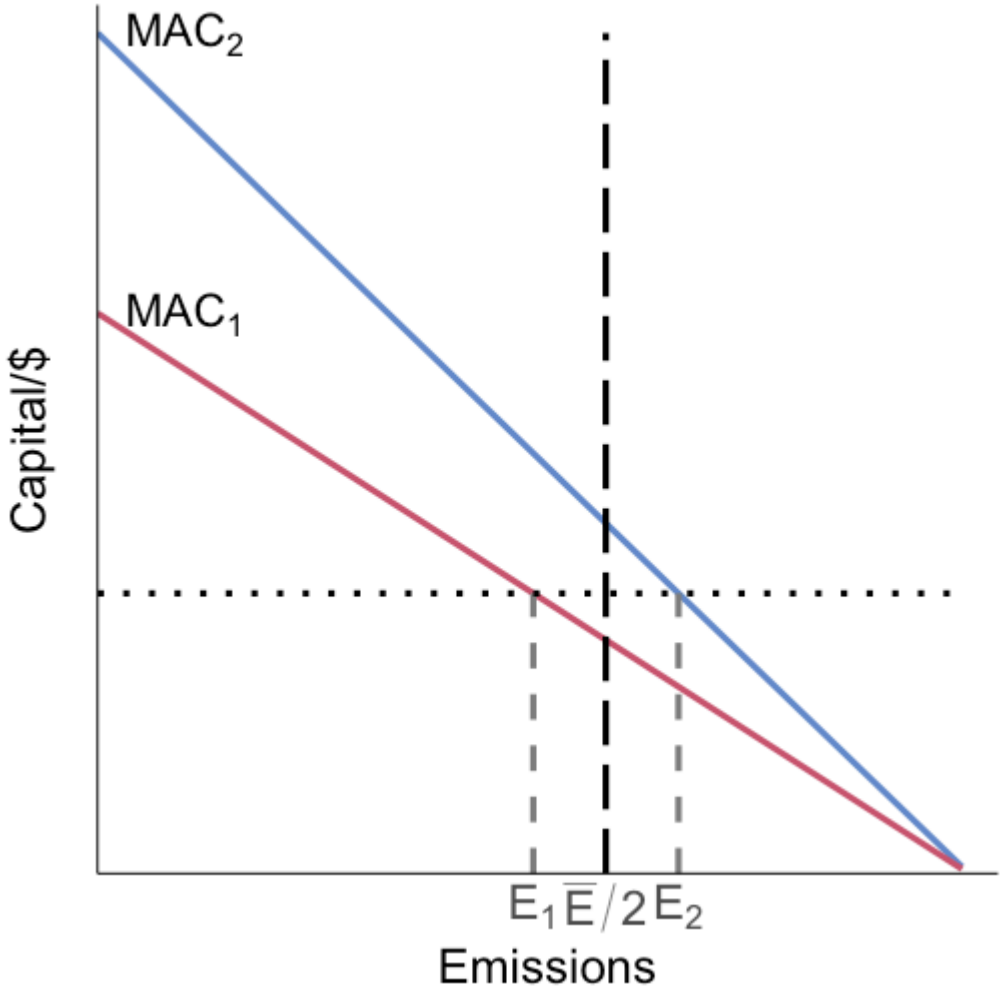
Let them trade the permits

Tradable permits: graphical



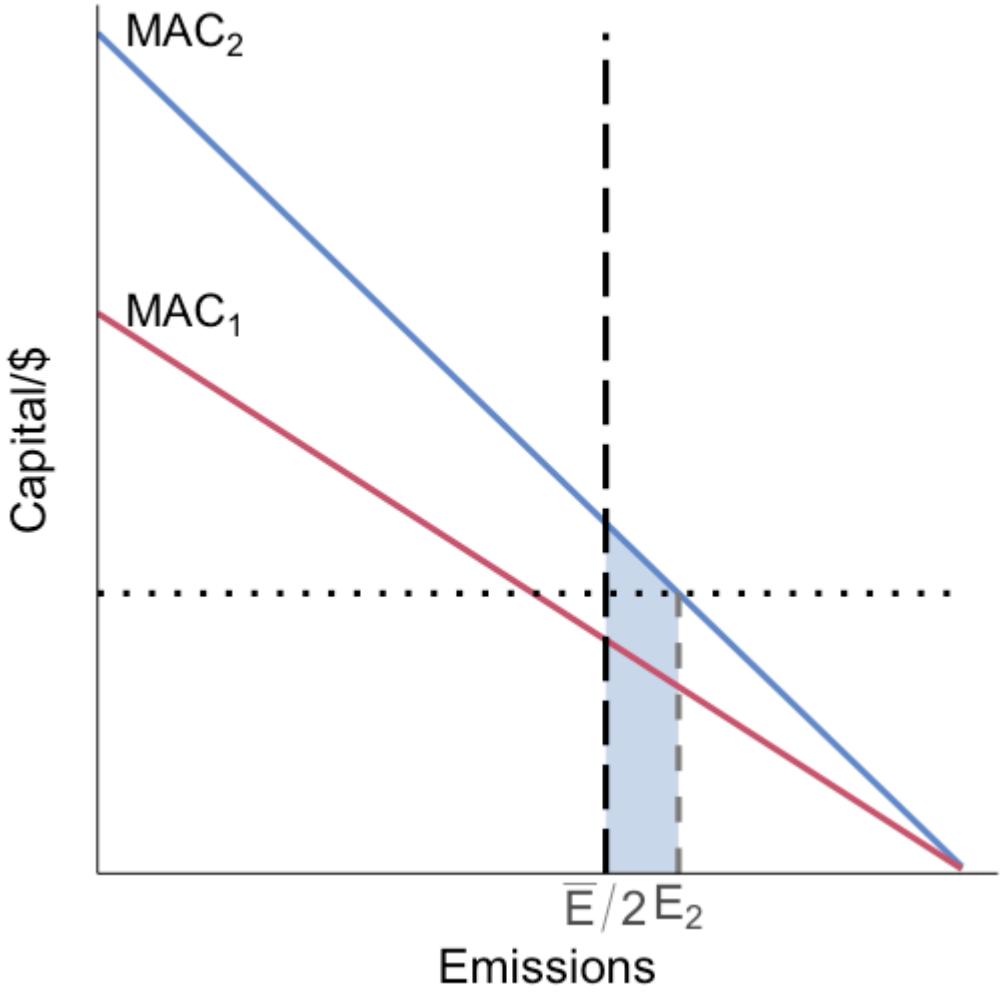
We can reduce costs by increasing abatement at which firm, and decreasing abatement at which firm?

Tradable permits: graphical



We can reduce costs by increasing emissions at high MAC firm 2 and decreasing emissions at low MAC firm 1 until they are equal

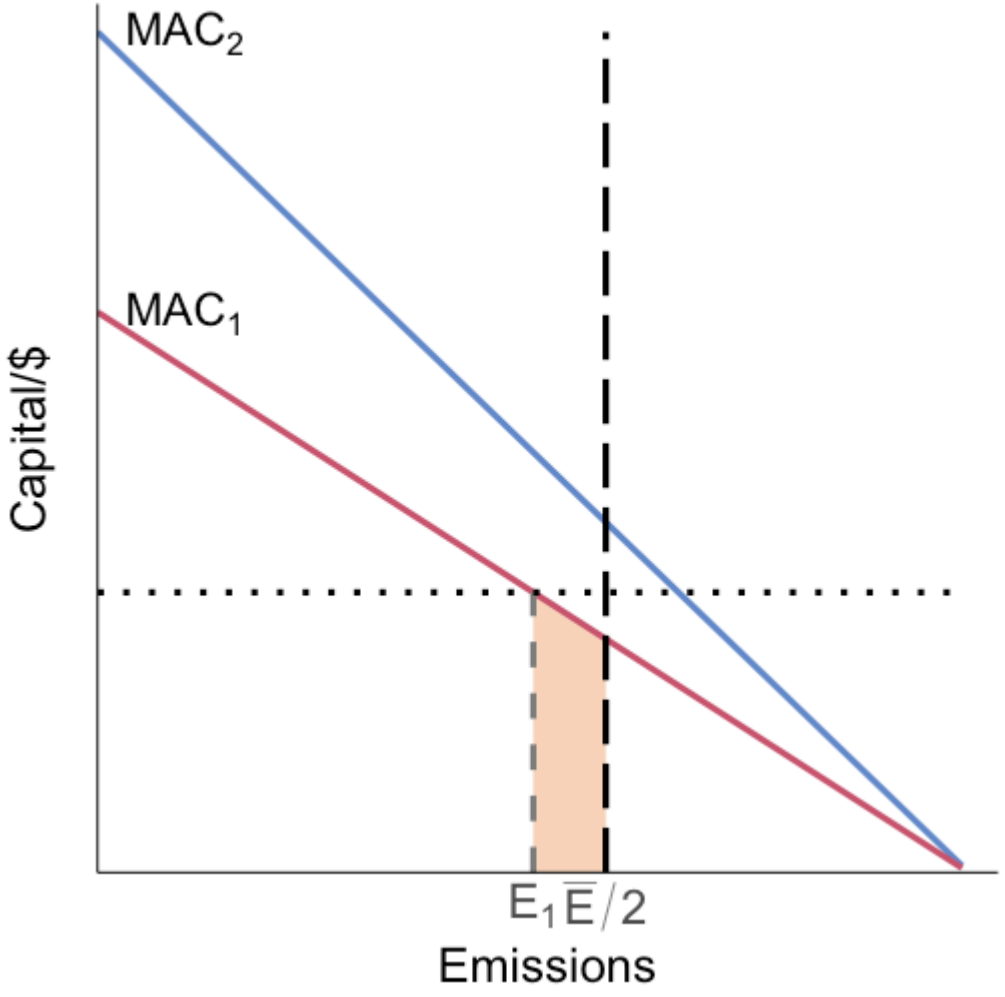
Tradable permits: graphical



Step 1: allow firm 2 (high MAC) to emit more, from $\bar{E}/2$ to E_2

The blue shaded area is firm 2's cost saving

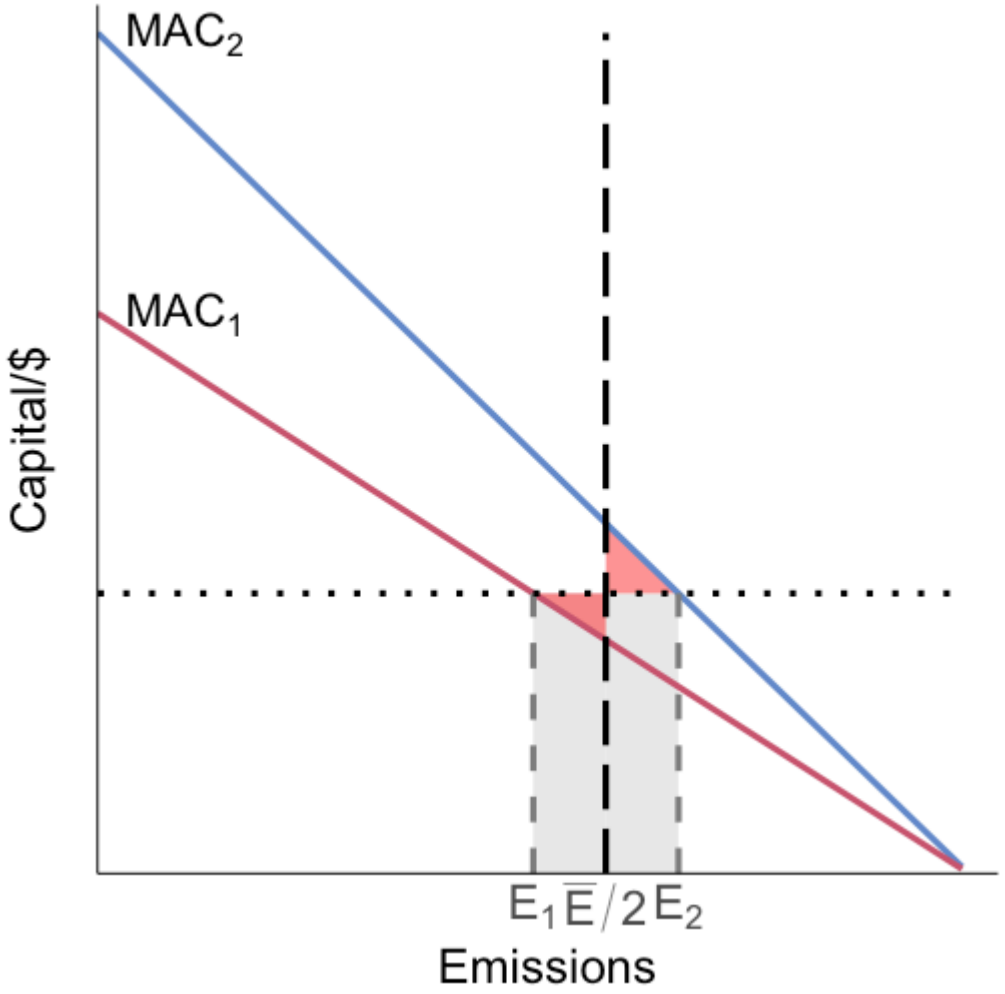
Tradable permits: graphical



Step 2: firm 1 (low MAC) emits less, from $\bar{E}/2$ to E_1

The orange shaded area is firm 1's cost increase

Tradable permits: graphical

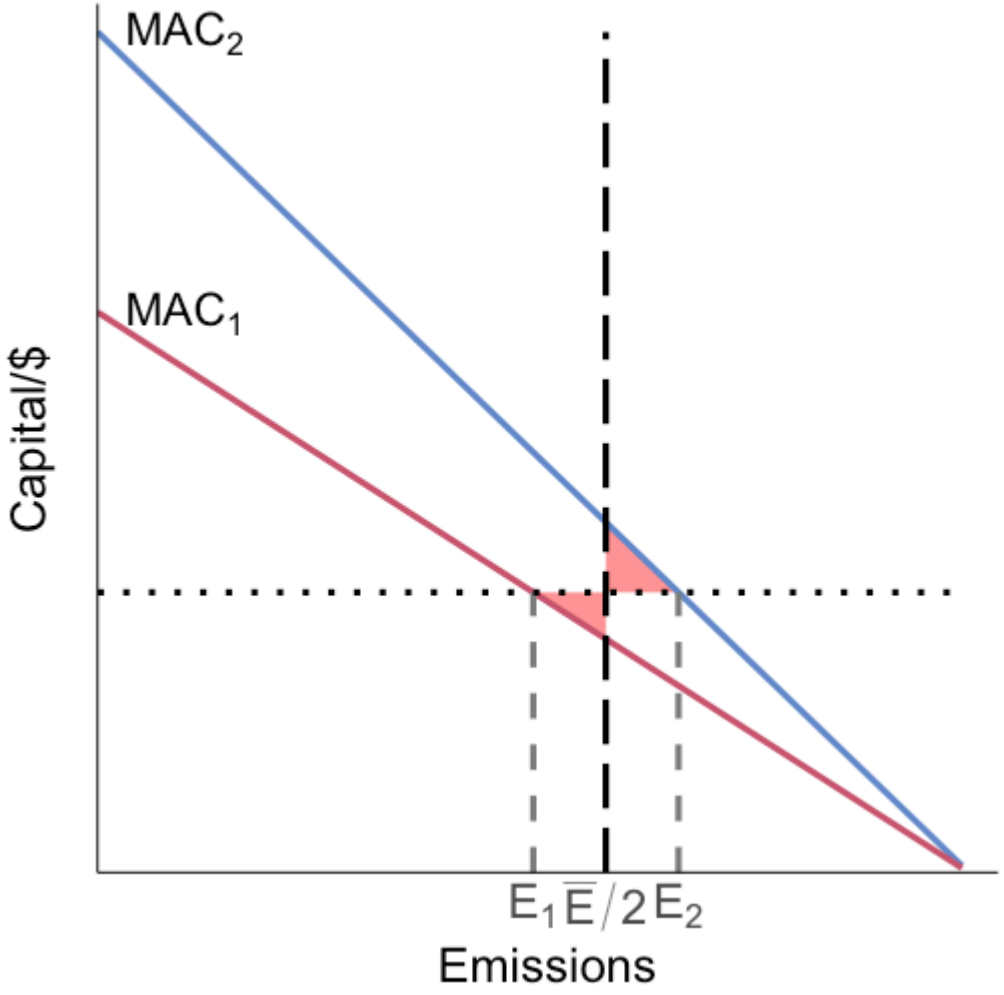


Step 3: net cost change from trading

The grey rectangles are equal (same height and same width ΔE), so they cancel out

What remains is the two red triangles: the net cost reduction from reallocation

Tradable permits: graphical

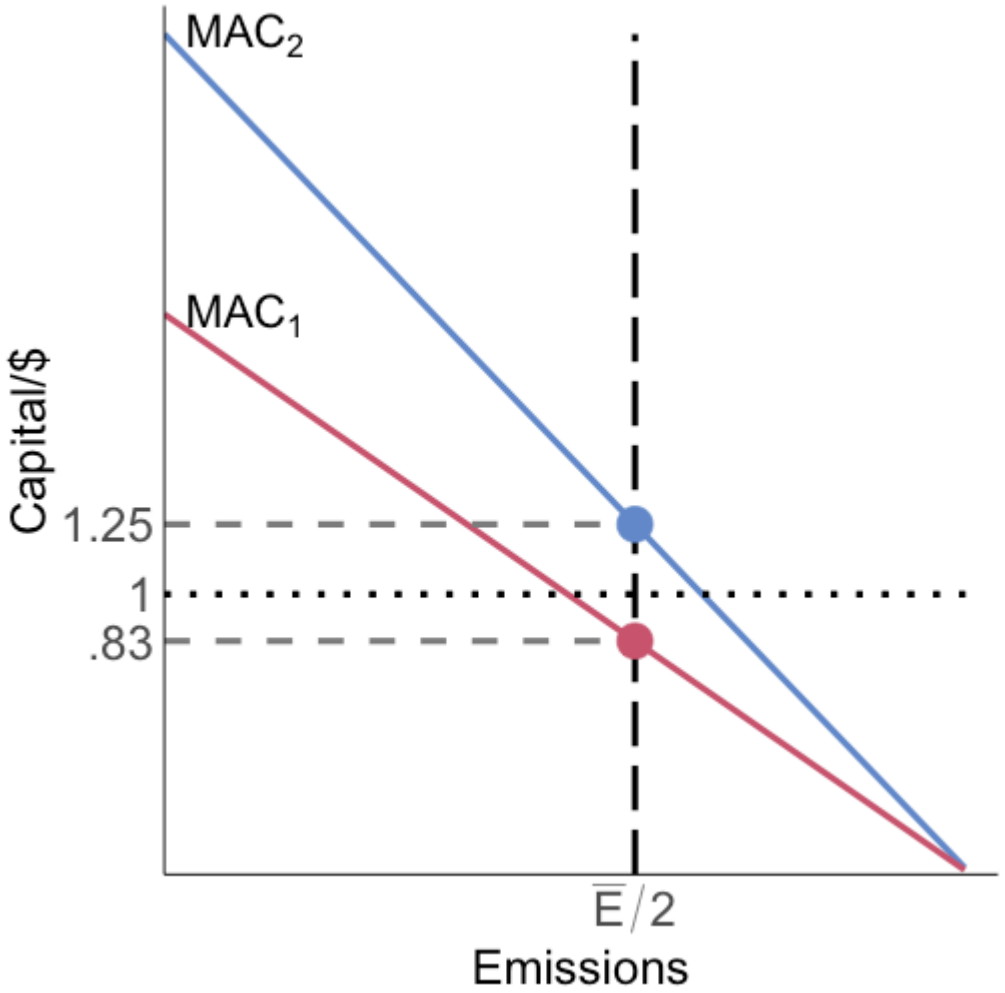


We can reduce costs by increasing emissions at high MAC firm 2 and decreasing emissions at low MAC firm 1 until they are equal

This allows us to recover DWL equal to the red area

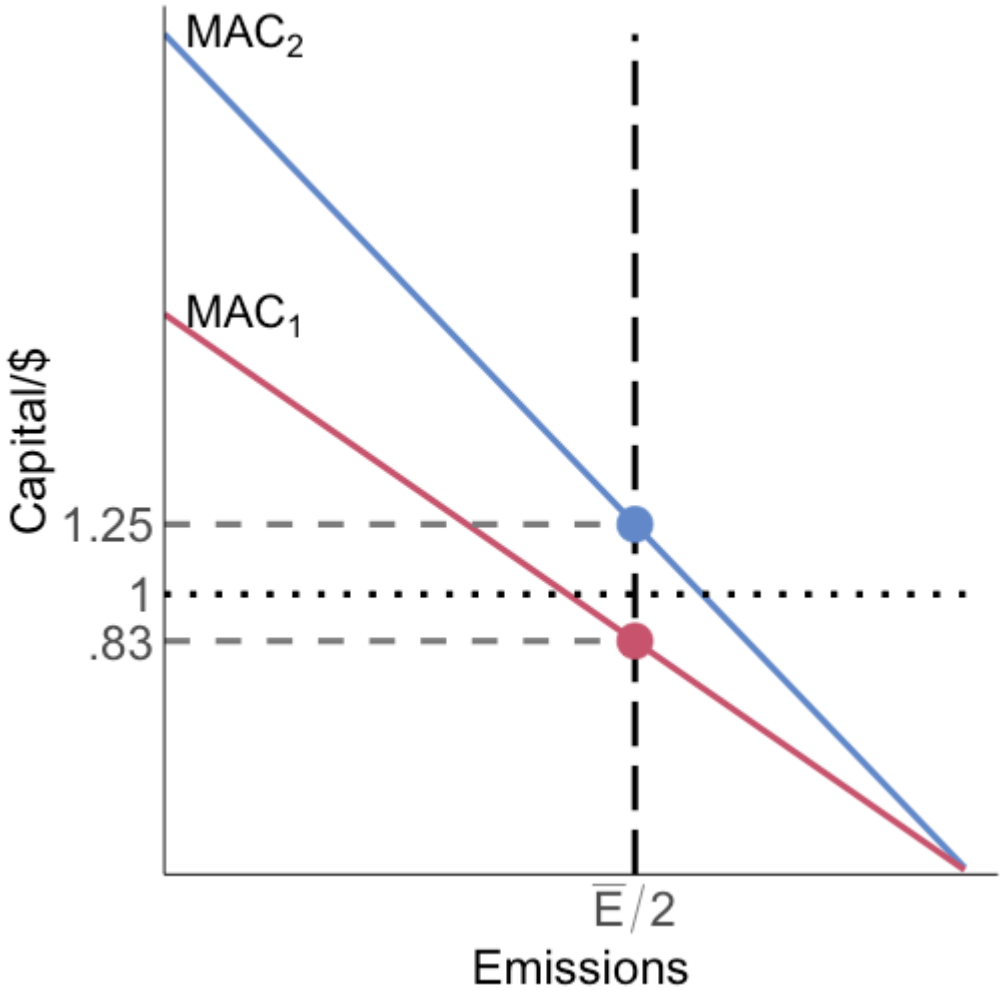
The red area is the difference in areas under MAC2 and MAC1 over the range of emissions changes

Tradable permits: graphical



We want to equalize MACs for cost-effectiveness, but does the permit market cause this to happen?

Tradable permits: graphical

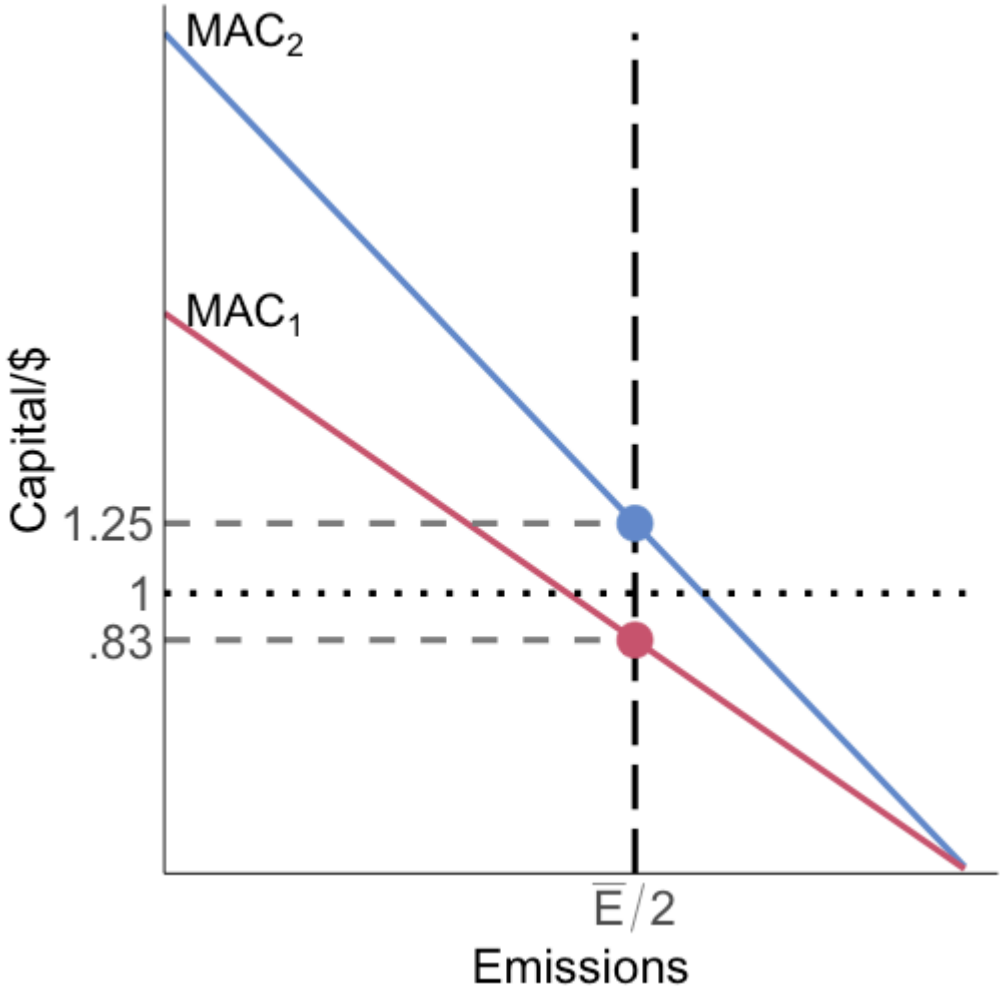


We want to equalize MACs for cost-effectiveness, but does the permit market cause this to happen?

Firm 2 is willing to pay a price up to the blue point (1.25) to be able to emit 1 more unit

Firm 1 can abate 1 more unit at cost equal to the red point (0.83)

Tradable permits: graphical

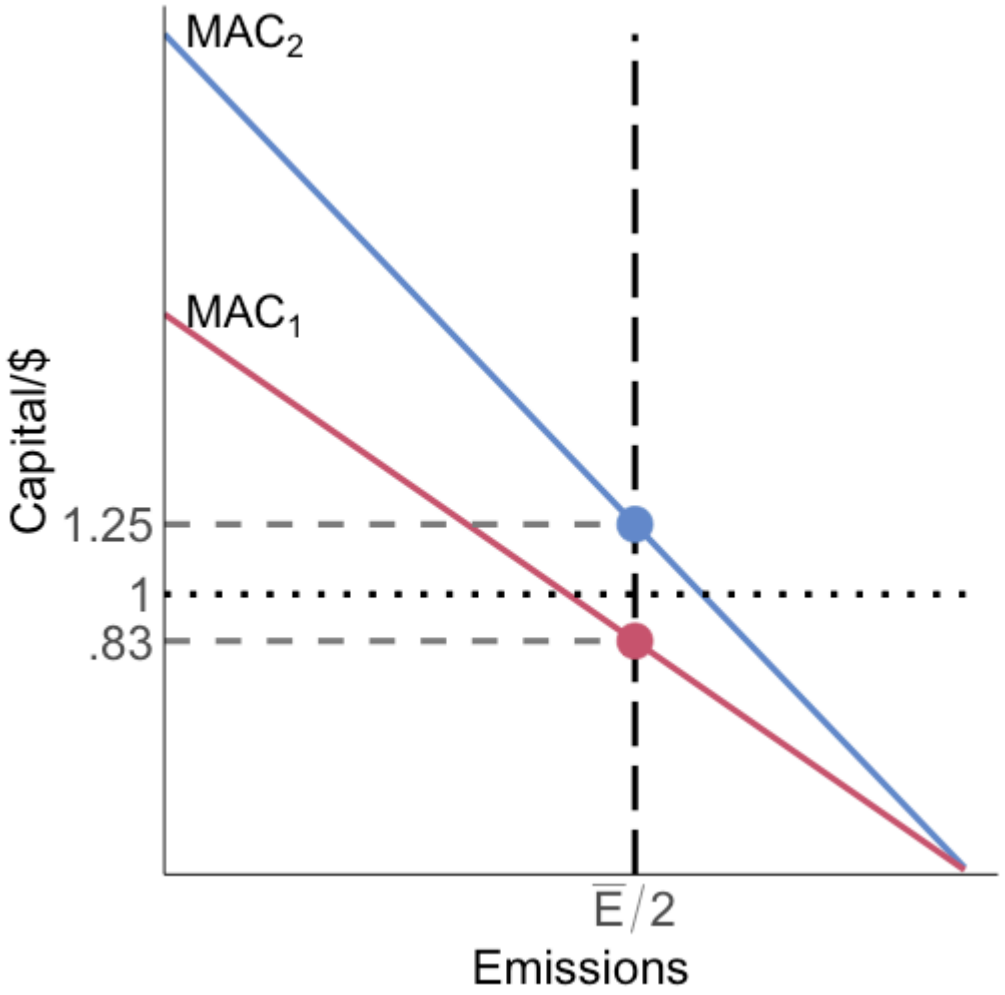


Firm 2 can buy the right to emit 1 unit of pollution from firm 1 for anywhere between 1.25 and 0.83 and **both will be better off** [very Coasean!]

These trades can be done until the MACs are equal at a value of 1

This would be the prevailing permit price in a tradable permit system

Tradable permits: graphical



An alternative way to think about it:

the prevailing permit price is the MC of freeing up one more permit, the MAC of the selling firm

or it is the MB of freeing up one more permit (avoided MAC), the MAC of the buying firm

Tradable permits: firm

We can also see this result mathematically

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Firms are price-takers in the permit market

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Suppose there is a permit price p in the competitive tradable permit market

Firms are price-takers in the permit market

Let's set up the firm problem: they want to minimize the cost of satisfying the policy

Tradable permits: firm

The firm's problem is then:

$$\min_{E} C(E) + pE$$

The firm's first-order condition to minimize costs is:

$$-C'(E^*) = p$$

Tradable permits: firm

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The firm's first-order condition to minimize costs is:

$$-C'(E^*) = p$$

The firm minimizes costs by choosing emissions E^* so that its MAC equals the permit price

Tradable permits: cost-effectiveness

This makes sense!

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The permit price is the MC of emitting, the MAC is the MB of emitting (reduced abatement cost)

Costs are minimized when these two things are equal

Tradable permits: cost-effectiveness

What else does firm behavior tell us about permits?

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If firms all set their MACs equal to p then all their MACs are equal to one another, **we have cost-effectiveness:**

$$-C_1'(E_1^*) = -C_2'(E_2^*) = \cdots = -C_N'(E_N^*) = p$$

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Taxes and permits both achieve cost-effectiveness

Why?

Because firms treat permit prices and a tax identically in decisionmaking

Tradable permits: cost-effectiveness

Tradable permit systems are **always** cost-effective: whatever emissions limit you set, it will be achieved at least-cost

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This does not mean that it is necessarily efficient!

Tradable permits: efficiency

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To keep things simple suppose MD is constant and the same across all firms: $MD = d$

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To keep things simple suppose MD is constant and the same across all firms: $MD = d$

If we can set $\bar{E}$ such that the equilibrium permit price $p = d$, then we also have efficiency

Tradable permits in practice

Knowing MD is often difficult in practice

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Tradable permit systems are nice because we can just let politicians choose a $\bar{E}$ that is politically feasible, and then the permit market ensures that we get the associated emissions reductions at least-cost

What often happens in practice is $\bar{E}$ starts high, giving us a low p, and then $\bar{E}$ gets ratcheted down over time

Tradable permits: numerical example

Suppose two firms trade permits with total cap:

$$E_1 + E_2 = \bar{E} = 90$$

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Abatement cost functions:

$$C_1(E_1) = \frac{1}{2}(90 - E_1)^2, \qquad C_2(E_2) = \frac{1}{4}(150 - 2E_2)^2$$

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Abatement cost functions:

$$C_1(E_1) = \frac{1}{2}(90 - E_1)^2, \qquad C_2(E_2) = \frac{1}{4}(150 - 2E_2)^2$$

So marginal abatement costs are:

$$MAC_1(E_1) = -C_1'(E_1) = 90 - E_1$$

$$MAC_2(E_2) = -C_2'(E_2) = 150 - 2E_2$$

Tradable permits: numerical example

In a competitive permit market:

$$MAC_1(E_1^*) = p, \qquad MAC_2(E_2^*) = p, \qquad E_1^* + E_2^* = \bar{E} = 90$$

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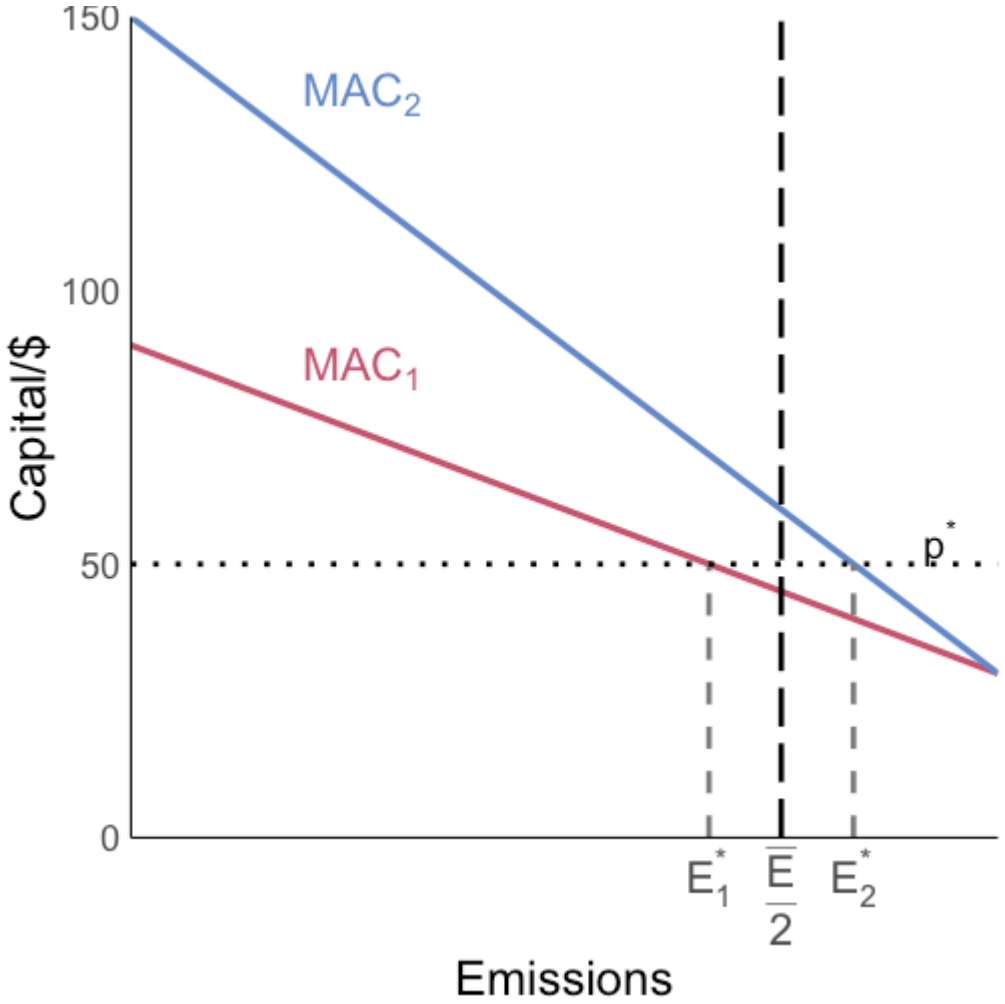
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$$E_1^* = 40, \qquad E_2^* = 50, \qquad p^* = 50$$

Firm 2 has higher MAC at an equal split, so it buys permits and emits more in equilibrium.

Tradable permits: numerical example



Numerical solution:

$$E_1^* = 40, \quad E_2^* = 50, \quad p^* = 50$$

Starting from equal allocation ($\frac{\bar{E}}{2} = 45$ each):

- firm 1 sells permits and emits less
- firm 2 buys permits and emits more

Political economy of permits

Firms are often more willing to accept a higher p than τ because they are often endowed with (some) permits for free

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Firms are often more willing to accept a higher p than τ because they are often endowed with (some) permits for free

This means that *in practice*, we might expect to get greater emissions reductions under a permit system than a tax because of these political economy reasons

This is one of the key reasons the 1990 CAA amendments were able to be passed

Permit market challenges

How we do initially allocate permits?

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Lottery: Randomly assign permits

Grandfathering: give permits to existing firms based on historical emissions

Permit auctions

Recall the firm's problem when a permit costs p:

$$\min_E \; C(E) + pE \quad \Rightarrow \quad MAC(E) = p$$

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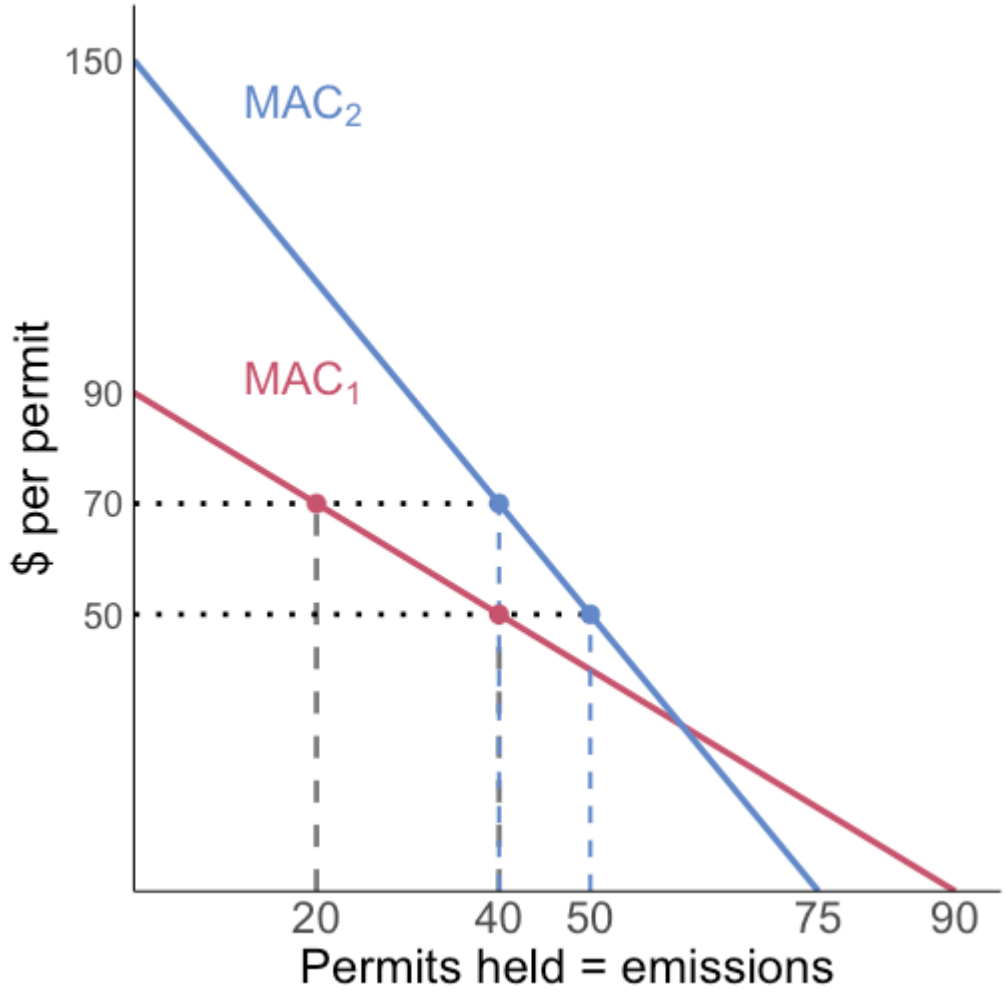
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The MAC curve is the firm's demand curve for permits

Each permit lets the firm emit one more unit, so it is worth the abatement cost it avoids

Permit auctions: MAC is permit demand



Firm 1: $MAC_1 = 90 - E_1$

At $p = 70$: wants 20 permits

At $p = 50$: wants 40 permits

Firm 2: $MAC_2 = 150 - 2E_2$

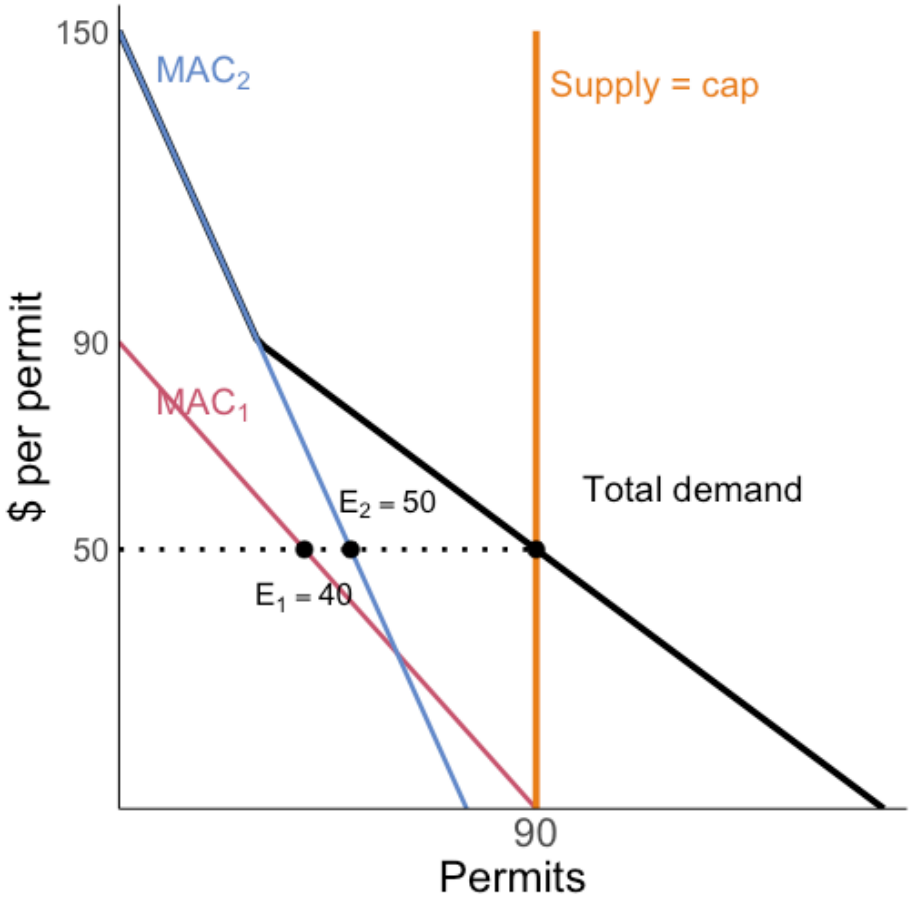
Setting $MAC_2 = p$ gives

$E_2 = 75 - p/2$

At $p = 70$: wants 40 permits

At $p = 50$: wants 50 permits

Permit auctions: supply meets demand



Supply: the cap, $\bar{E} = 90$

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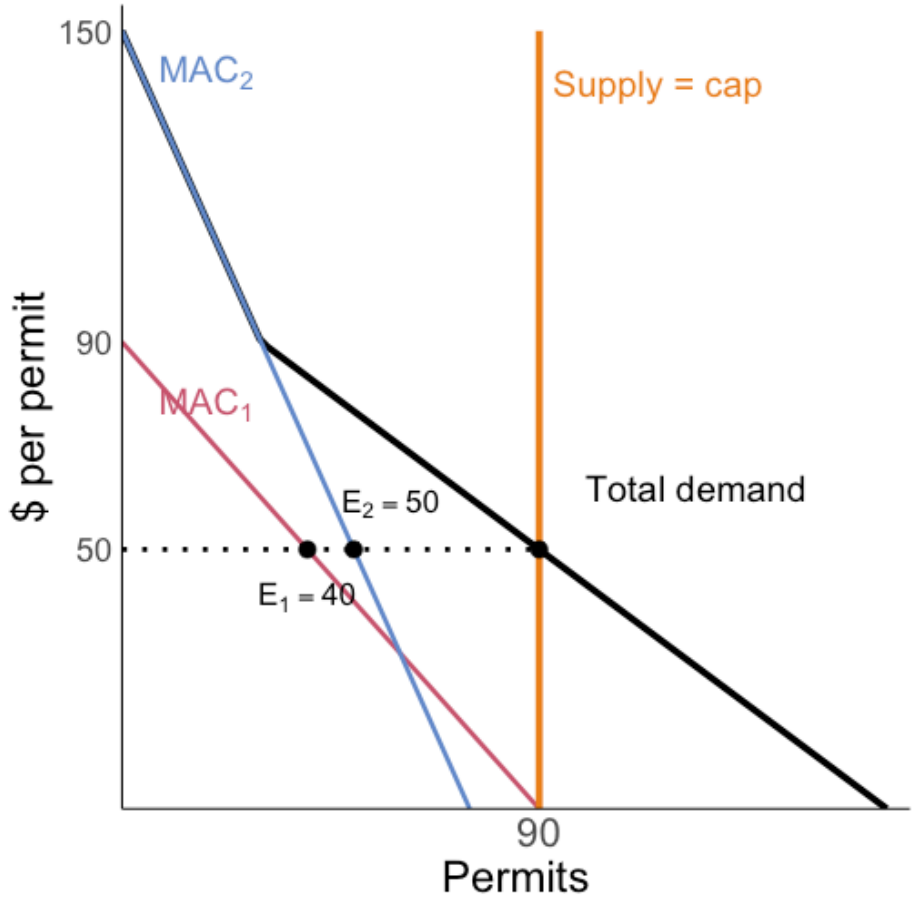
At price p, it wants $75 - p/2$ permits

Demand: add up the firms' demands[1]

$$(90 - p) + (75 - p/2)$$

[1] For $p \leq 90$; above that, firm 1 wants no permits.

Permit auctions: supply meets demand



Supply: the cap, $\bar{E} = 90$

Demand for $p \leq 90$:

$$(90 - p) + (75 - p/2)$$

Set equal: $165 - 1.5p = 90$, so $p = 50$

$E_1 = 40$, $E_2 = 50$: **the trading equilibrium**

Permit auctions: the rules

Major permit auctions (RGGI, California-Quebec, EU ETS) are **uniform-price sealed-bid** auctions. These are our Round 1 rules:[1]

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1. Each firm privately bids the most it would pay for each permit
2. The regulator ranks all bids from highest to lowest
3. The top $\bar{E}$ bids win
4. **Every** winner pays the same price: the lowest winning bid

[1] The EU ETS and California-Quebec also charge the lowest winning bid; RGGI charges the highest losing bid; EPA's Acid Rain Program auctions charged each winner its own bid.

Permit auctions: example

Three firms bid for **5 permits**:

	Permit 1	Permit 2	Permit 3	Permit 4
Firm A	12	9	6	3
Firm B	10	7	4	1
Firm C	8	5	2	

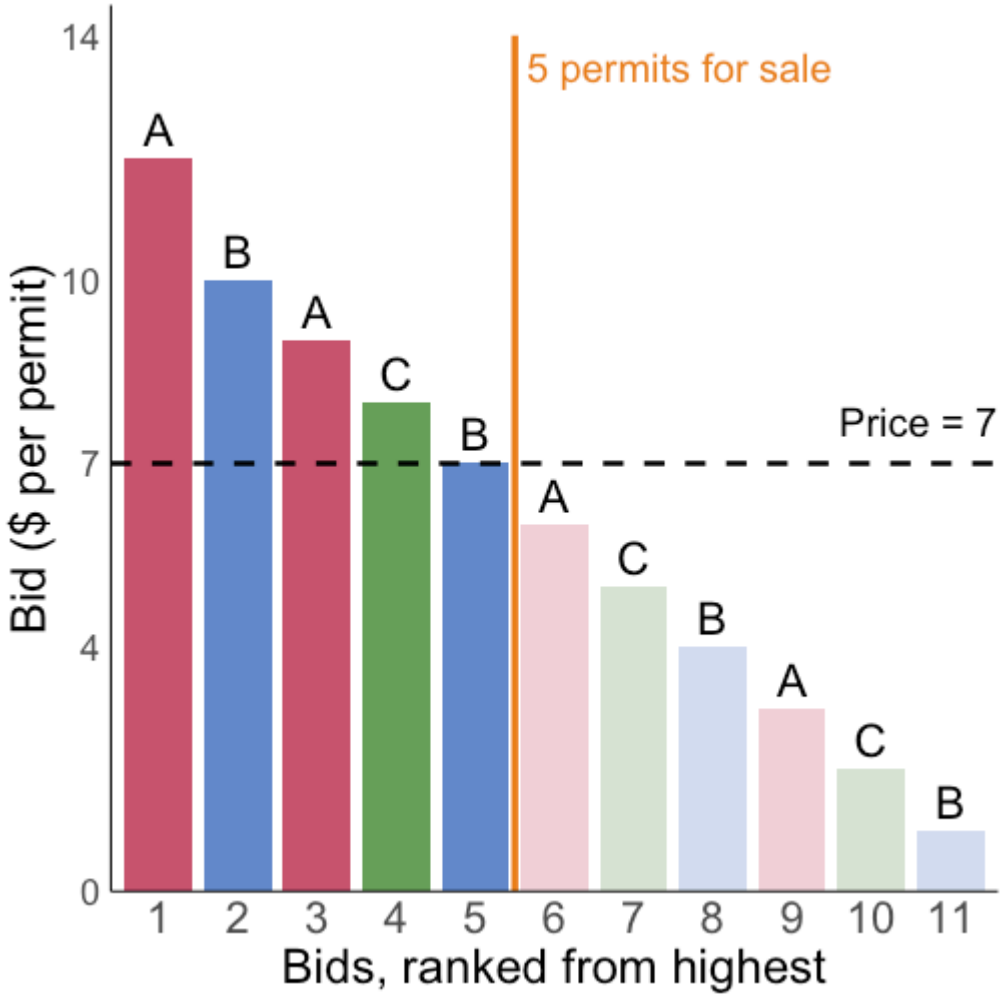
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Which bids win? What is the price? What does each firm pay?

Permit auctions: example



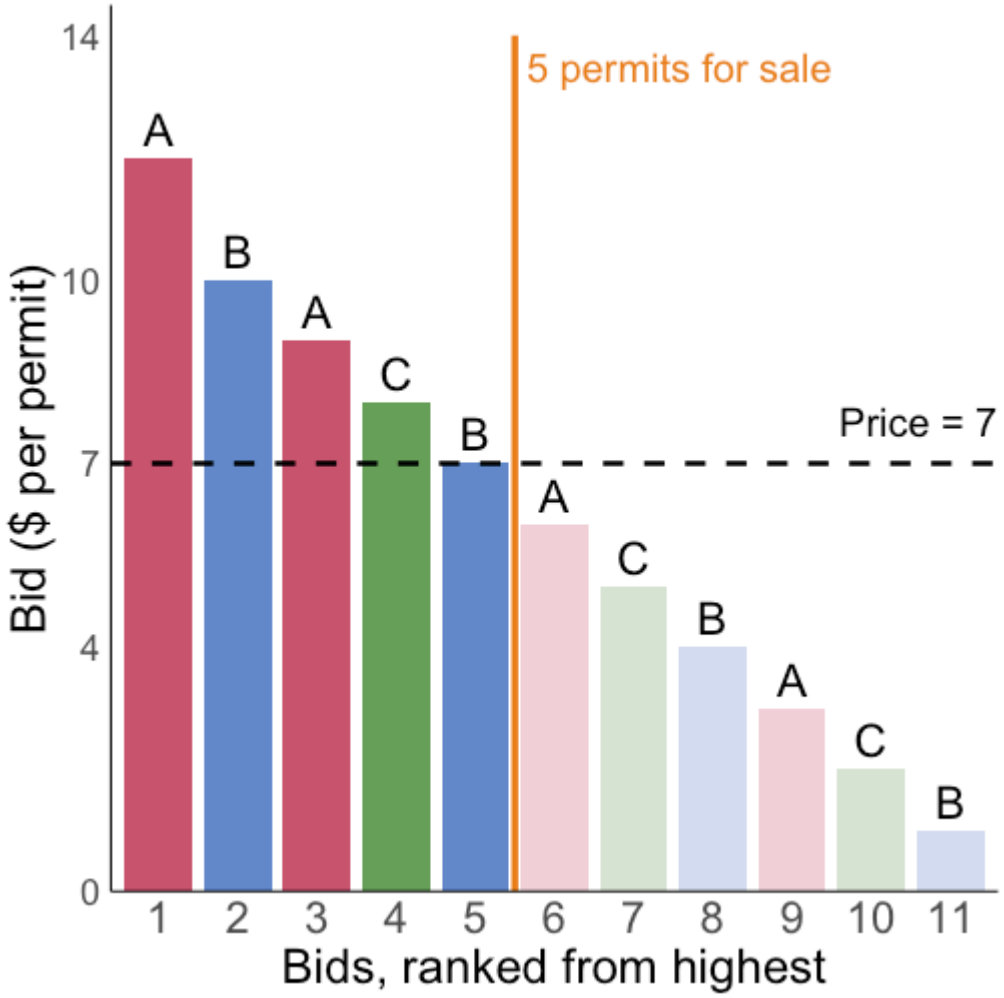
Top 5 bids win: 12, 10, 9, 8, 7

Price = lowest winning bid = **7**

All winners pay only **7 per permit**, even when they bid higher

Your bid mostly decides **whether** you win a permit; the price is the lowest winning bid, usually someone else's

Permit auctions: the gain to the firm



A whole permit avoids one unit of abatement. Its value is the **area under MAC** over that unit:

$$\text{Gain} = \text{avoided abatement cost} - p$$

If a permit saves 12 in abatement cost and costs 7, the firm gains **5**

The bars show submitted bids

Let's trade: the permit market game

The permit market game: auctions and trading

Round 1: uniform-price auction. Every winning permit costs the lowest winning bid

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Round 3: transaction costs. Restart Round 2 trading: same starting permits and post-shock MACs, plus a **$3 buyer fee per permit**

The permit market game: what should happen?

- Different bidding incentives under uniform-price and pay-as-bid auctions
- Trades from low-MAC to high-MAC firms after costs change
- Lower total abatement costs as firms trade toward the cost-effective allocation
- In Rounds 1 and 2, equal MACs when the class reaches that allocation
- In Round 3, the $3 fee blocks trades whose abatement-cost savings are less than $3

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After trading closes, compare the MAC histogram with the cost-effective price line

The permit market game: scoring

$$\text{Score} = \text{avoided abatement cost} + \text{sales} - \text{purchases} - \text{fees}$$

Avoided cost is relative to abating all baseline emissions, using your **post-shock MAC**. Purchases include auction payments

Leaderboard points = your score minus your firm's benchmark score, summed over rounds

The benchmark values your cost-effective permits at the no-fee price. Round 3 reuses Round 2's auction payments and deducts $3 per permit bought

The permit market game: class cost reductions

$$100 \times \frac{C_{\text{start}} - C_{\text{end}}}{C_{\text{start}} - C^*}$$

- C_{start}: total abatement cost at the auction allocation
- C_{end}: total abatement cost after trading
- C^*: cost-effective total abatement cost

All three use the same **post-shock MACs** and exclude permit payments

100% means the no-fee gap closed. A zero initial gap has no percentage

Round 3 also reports **abatement cost + transaction costs**

The permit market game: bids and orders

Auction: enter one price per permit, in any order. The game ranks bids highest to lowest

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Open market: submit a buy or sell order with a price and a **positive whole-number quantity**

Join the permit market game

Scan to open the student login, then join with your team name

Or open **aem4510.ivanrudik.com/permits**

After the permit game

Debrief: monopsony power in the auction

Monopsony power: a large buyer can influence the price it pays by reducing its demand

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This **demand reduction** uses monopsony power. It can leave high-value buyers with too few permits

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In our game, MACs shift after each auction, creating new gains from trade

Firms whose MAC rose can buy from firms whose MAC fell. Trading can also correct an auction's misallocation

Debrief: pay as you bid

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Bids can therefore differ from avoided abatement costs, leaving gains from later trading

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Pay-as-bid gives even small buyers a reason to shade bids down, because each winning bid sets its own payment

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We can decentralize trading market to cut down on transaction costs

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Do trades need to be validated by central authority to ensure permit validity?

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Lots of these costs are fixed, prohibit small trades

Permit systems and heterogeneous MDs

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How well do permit systems perform with heterogeneous MD?

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Permit systems and heterogeneous MDs

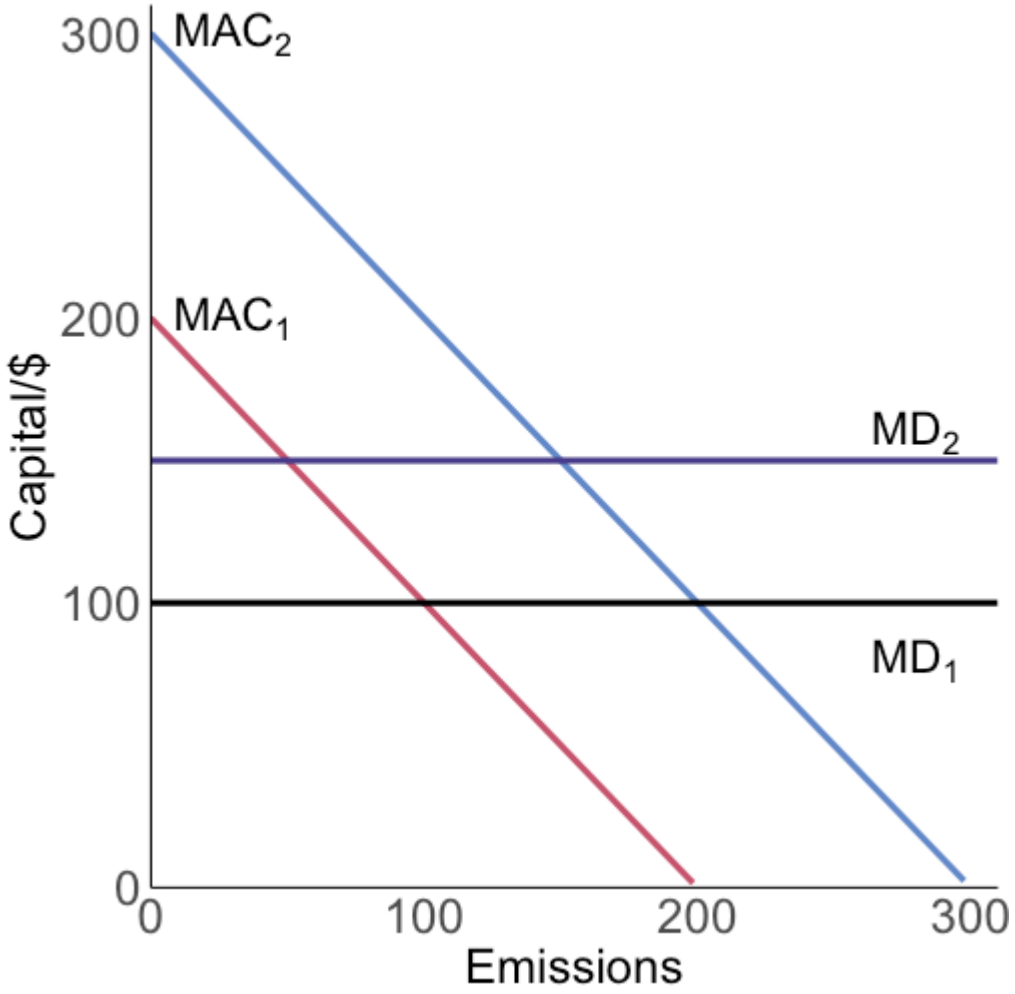
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If $MD_1 \neq MD_2$ then the permit system does **not** deliver efficiency!

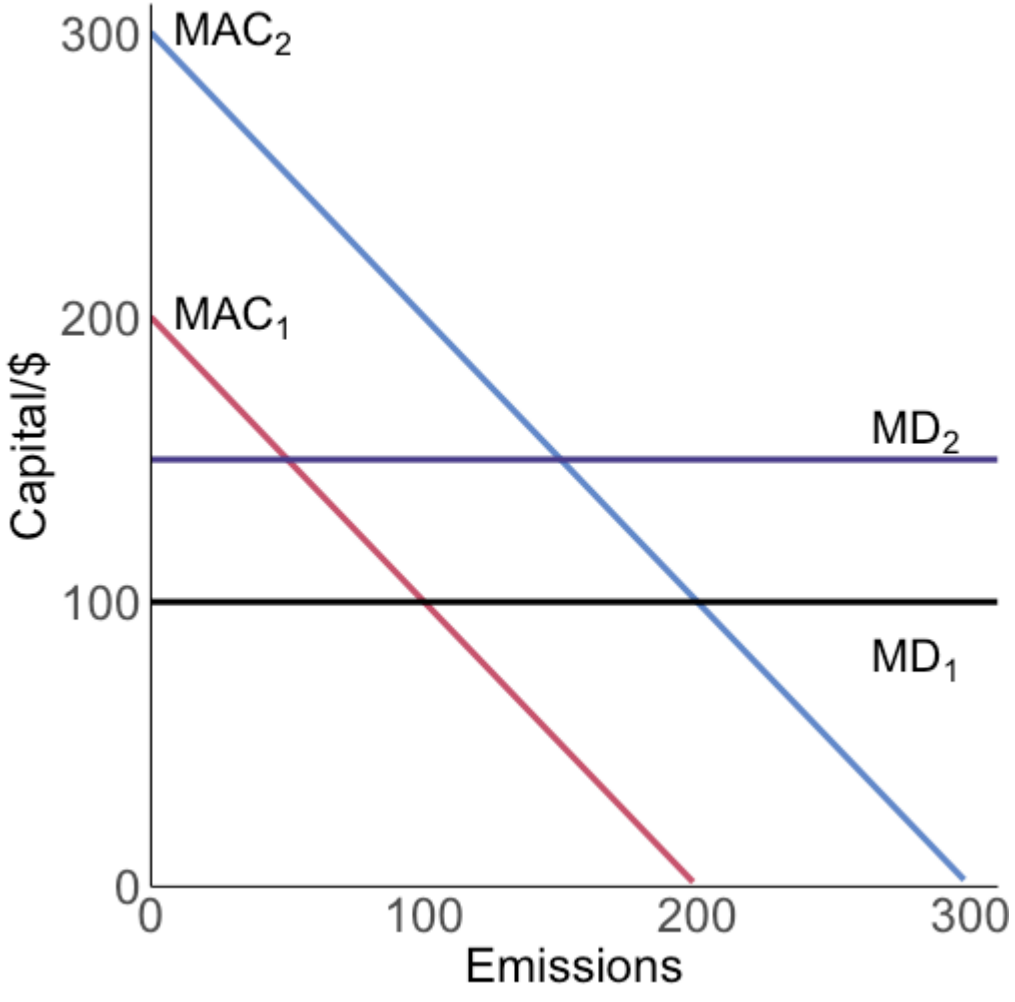
Permit systems and heterogeneous MDs: graphical



Suppose we have the two firms with different MACs and MDs:

- $MAC_1 = 200 - E_1$
- $MAC_2 = 300 - E_2$
- $MD_1 = 100$
- $MD_2 = 150$

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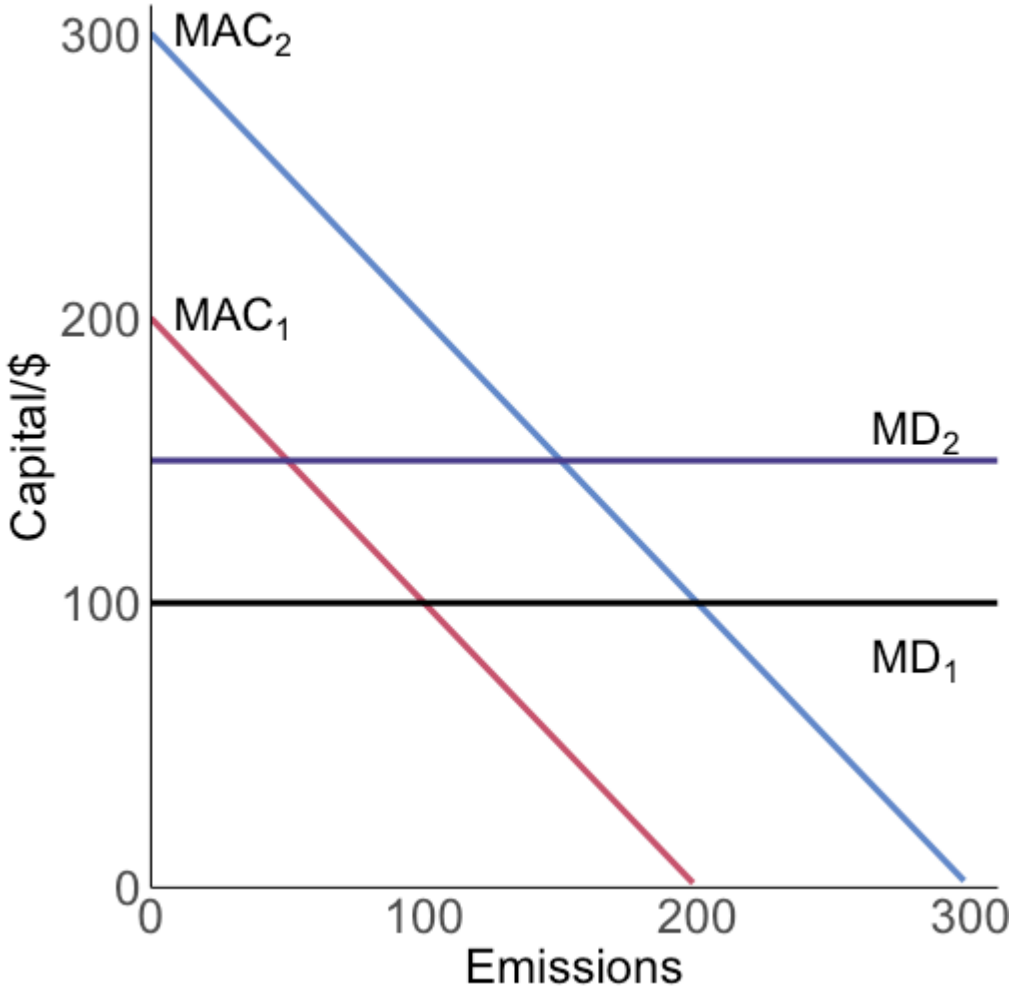
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The efficient emissions allocation is: $E^* = 250 : E_1^* = 100, E_2^* = 150$

The regulator sets $\bar{E} = 250$

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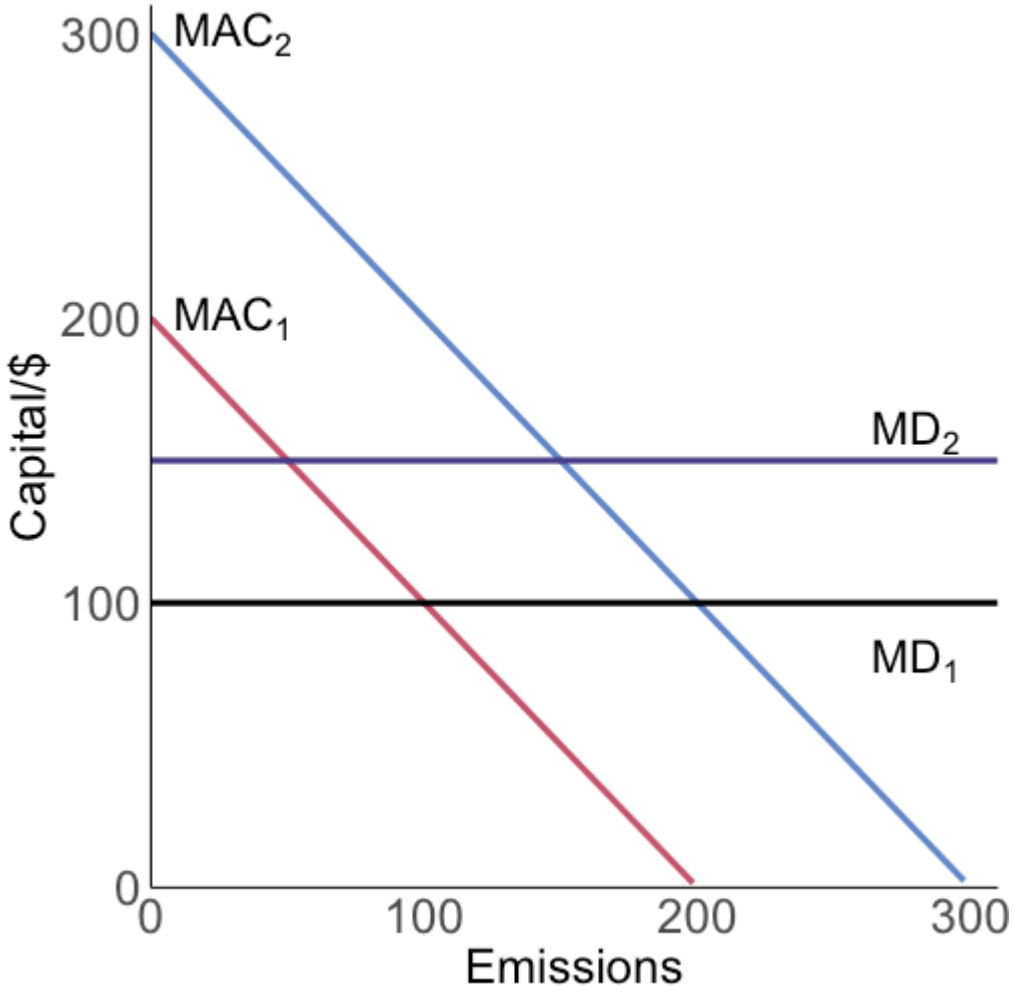
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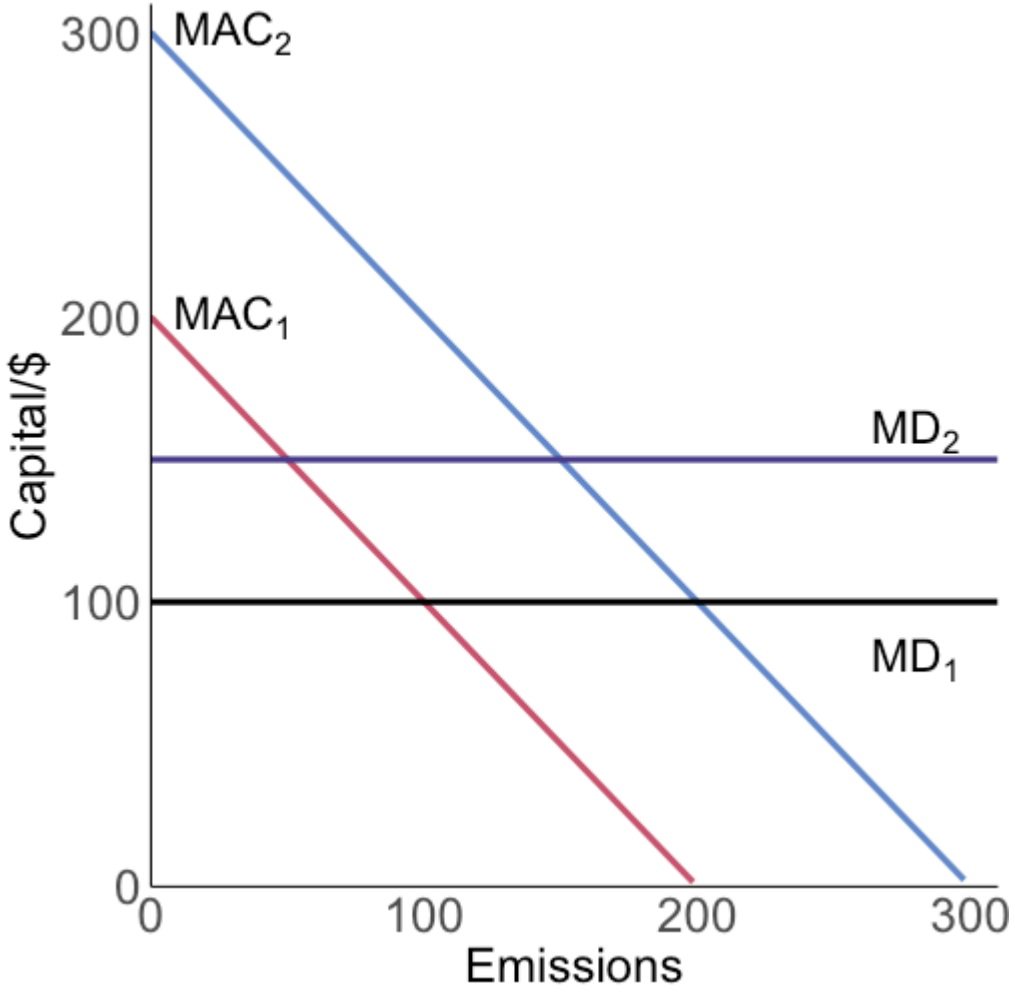
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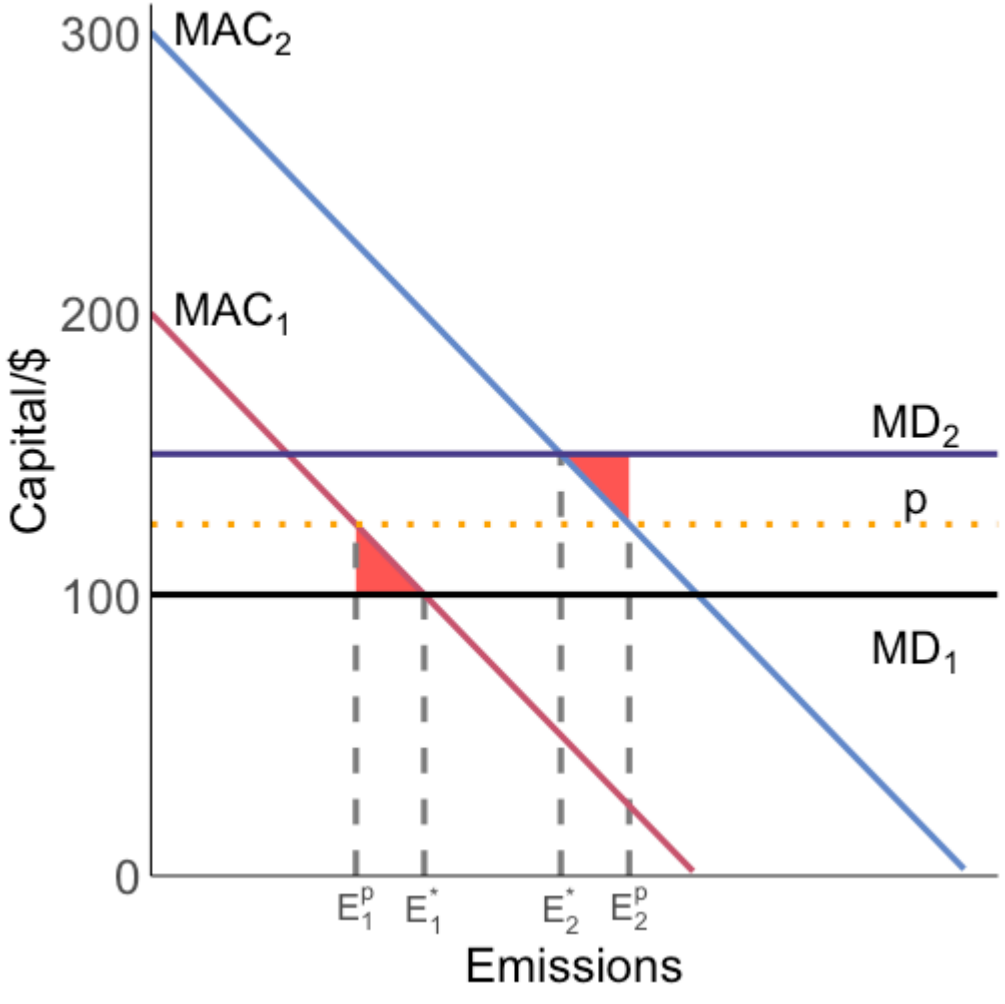
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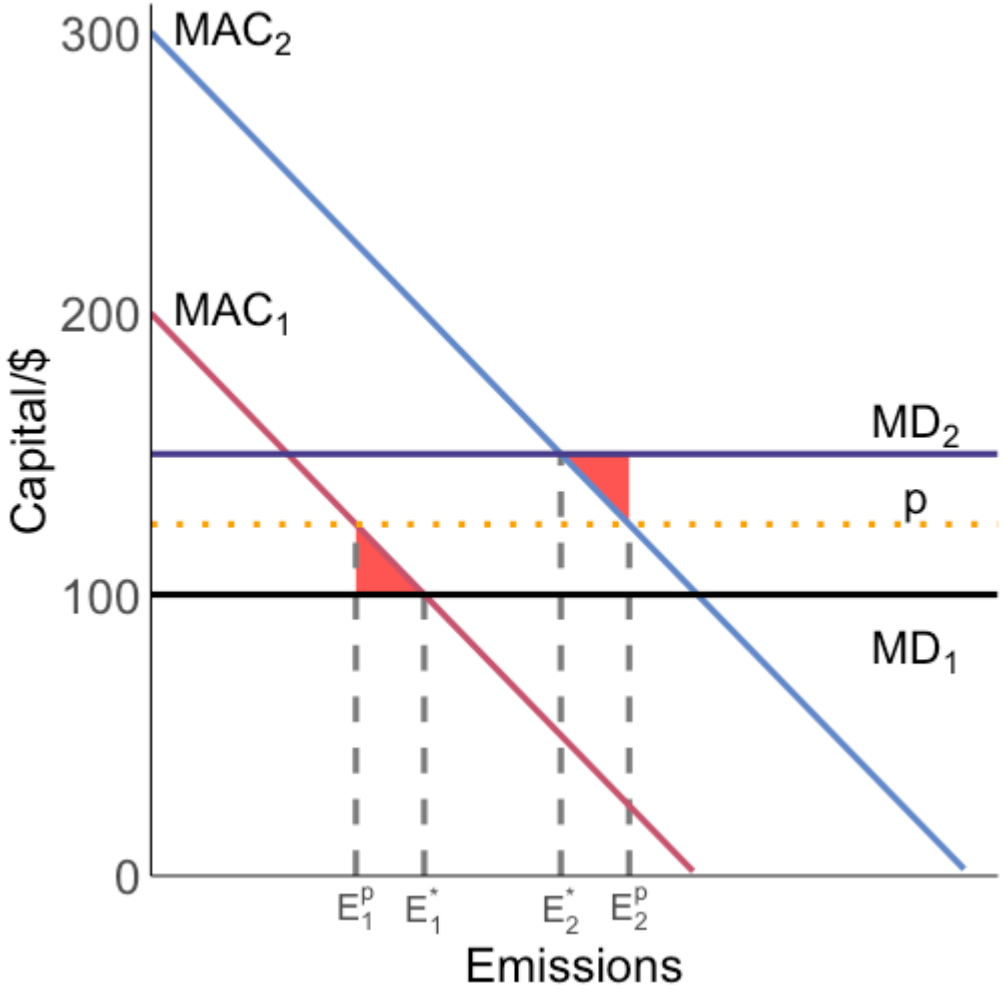
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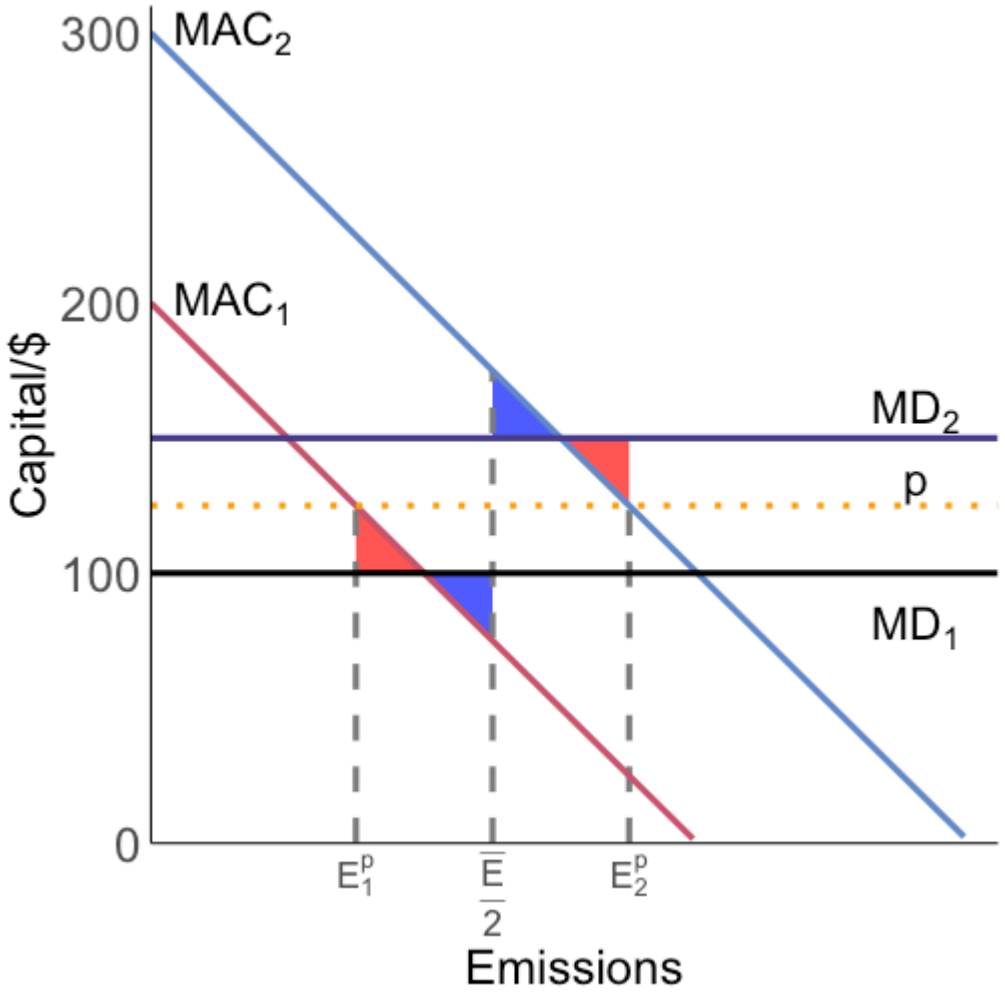


Relative to the optimal allocation, the permit system has DWL equal to the **red area**

The permit allocation is not an **efficient** allocation, but is it a **Pareto improvement** over:

1. No policy?
2. A uniform standard of $\bar{E}/2$?

Permit systems and heterogeneous MDs: graphical

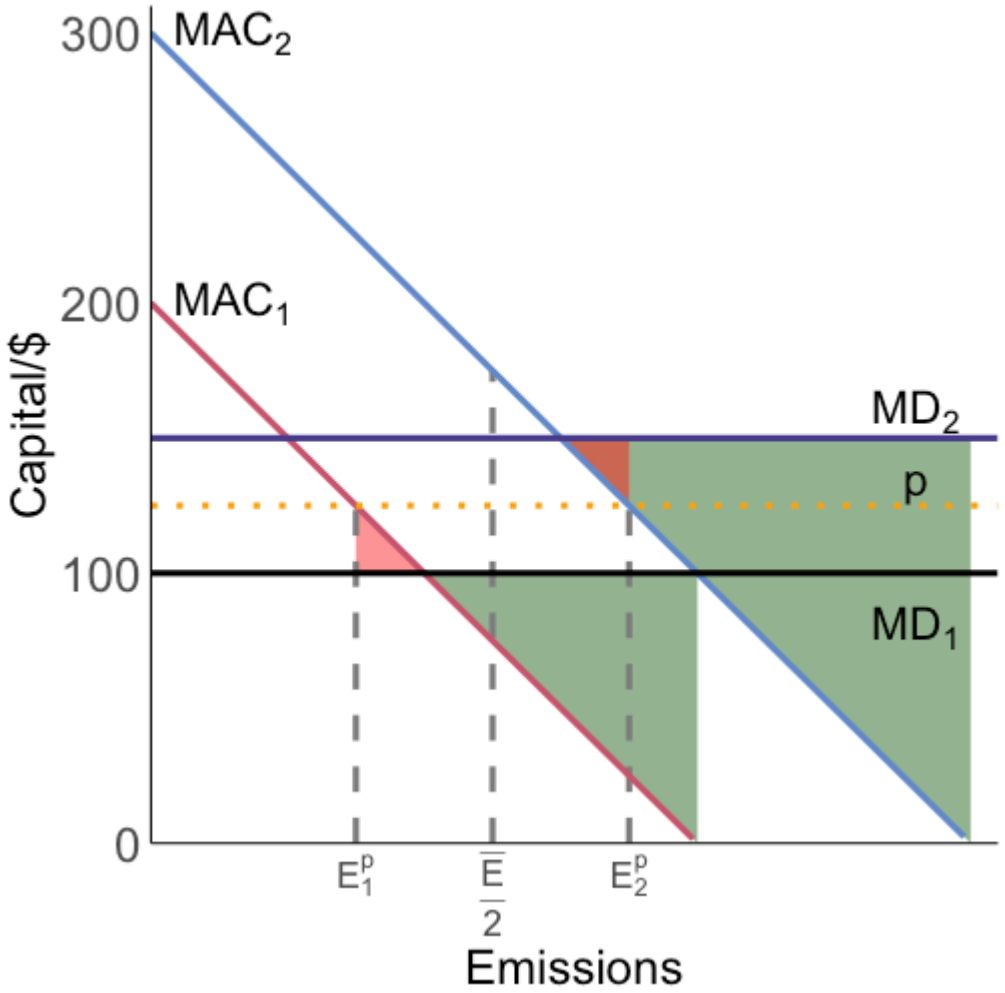


The blue area is the DWL under the uniform standard

In this specific case, a uniform standard and the permit system have the same efficiency since the red and blue areas are equal

The only difference is what kind of welfare loss is occurring where

Permit systems and heterogeneous MDs: graphical

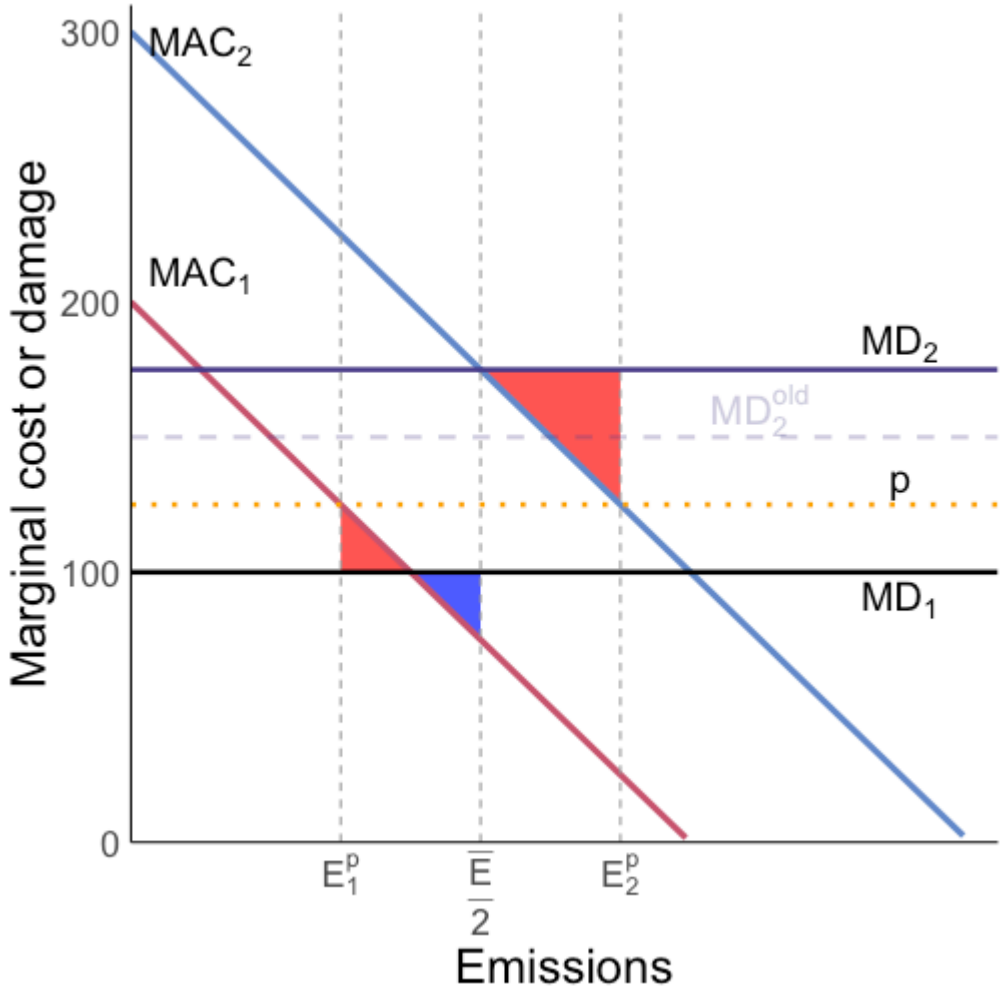


The DWL without any policy is the two large green triangles

These are clearly larger than the DWL under the permit system

The permit system can deliver a welfare improvement

When the uniform standard has higher welfare



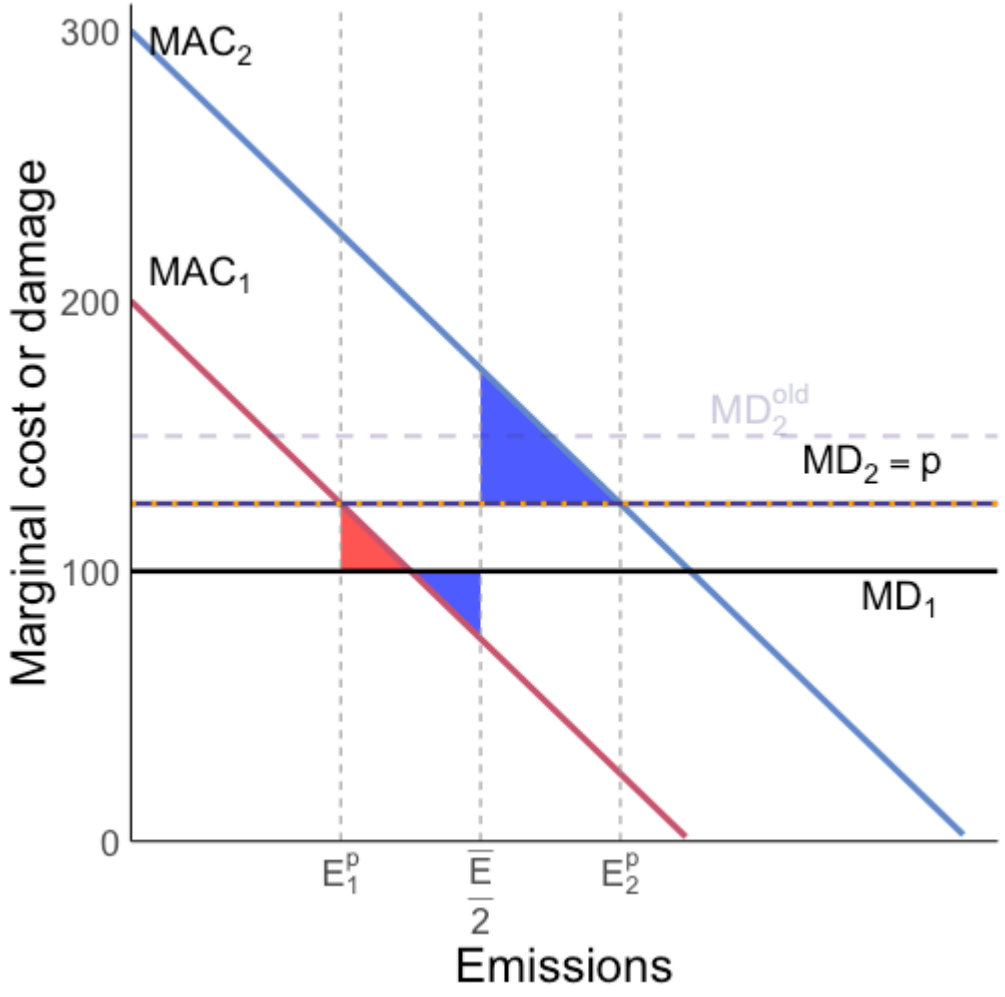
The welfare ranking depends on the MACs and MDs

Raise MD_2 (solid line); the dashed line shows its previous level

Red permit losses exceed **blue standard losses**

The uniform standard has higher welfare

When permit trading has higher welfare



Lower MD_2 (solid line); the dashed line shows its previous level

Blue standard losses exceed **red permit losses**

Permit trading has higher welfare

Changing the MACs and MDs can make either policy better, or leave them tied

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Another way is **zonal trading**: firms can only trade in similar MD areas

Trading ratios: Acid Rain Program

Below are estimates of efficient trading ratios for the Acid Rain Program

TABLE 3—TRADING RATIOS BETWEEN SOURCES AT EACH QUANTILE FOR SO_2

Quantile	Source location (county, state)	1st	25th	50th	75th	99th	99.9th
1st	Josephine, OR	1.0	0.4	0.2	0.2	0.1	0.02
25th	Polk, TX	2.5	1.0	0.6	0.4	0.1	0.1
50th	Grant, AR	4.5	1.8	1.0	0.7	0.2	0.1
75th	Marion, SC	6.0	2.4	1.3	1.0	0.3	0.1
99th	Allegheny, PA	19.2	7.7	4.3	3.2	1.0	0.4
99.9th	Hudson, NJ	49.4	19.6	11.0	8.2	2.5	1.0

Note: The trading ratio represents the number of tons from the column source required to offset one ton from the row source.

Muller and Mendelsohn (2009)

PM2.5 damages

Trading ratios are required because damages are heterogeneous across space

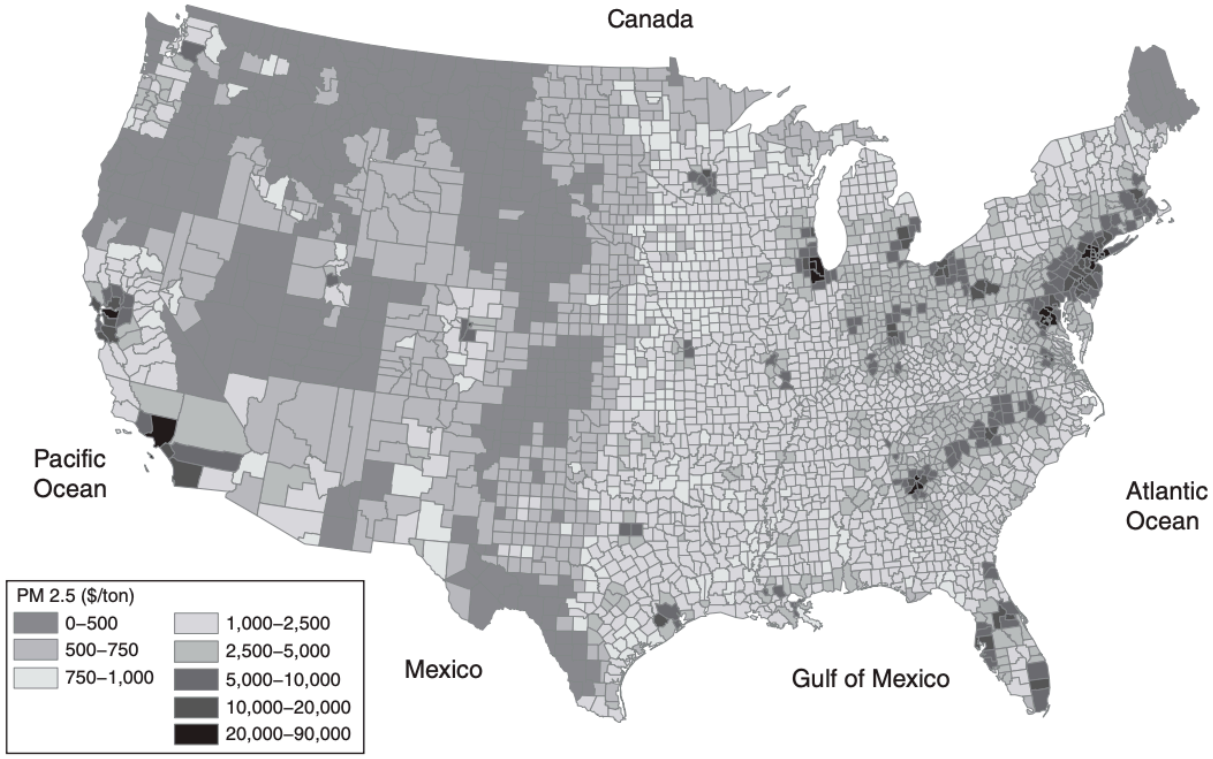
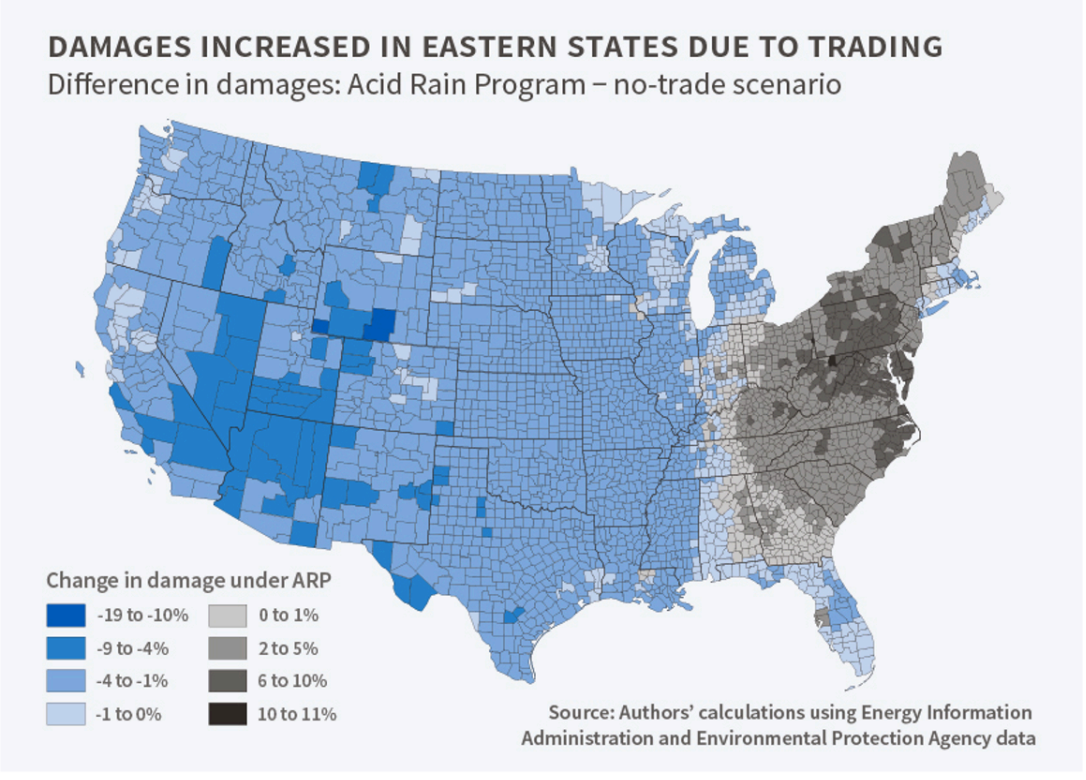


FIGURE 1. MARGINAL DAMAGE OF EMISSIONS FOR GROUND LEVEL SOURCES OF $PM_{2.5}$ ($/TON/YEAR)

Damages caused by ARP

The Acid Rain Program **increased** damages in the eastern US



Chan et al. (2018)

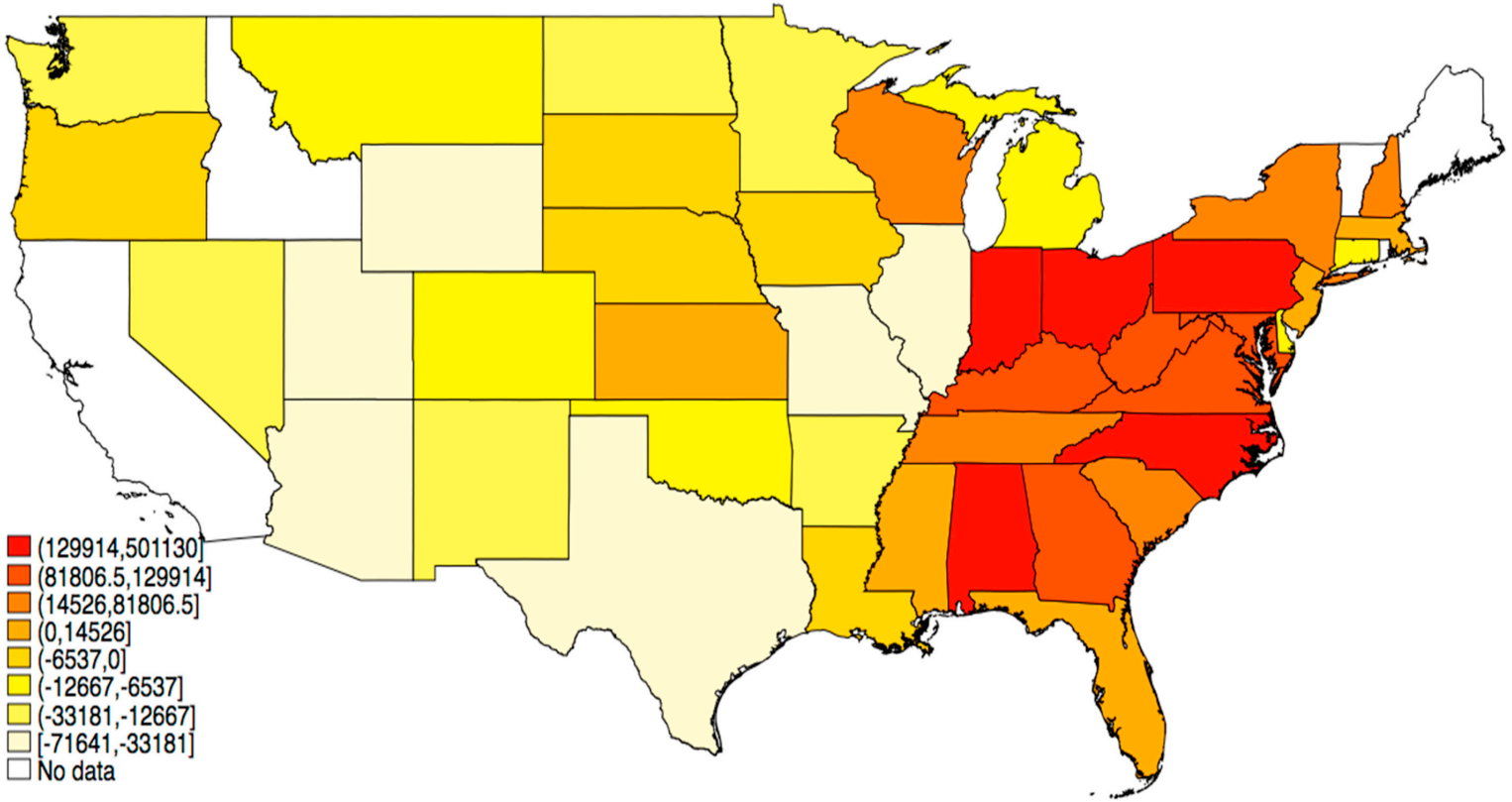
Damages caused by ARP

Chan et al. (2018) JEEM:

> We also compare health damages associated with observed SO2 emissions from all ARP units in 2002 with damages from a no-trade counterfactual. Damages under the ARP are $2.1 billion (1995)$ higher than under the no-trade scenario, reflecting allowance transfers from units in the western US to units in the eastern US with larger exposed populations.

Damages caused by ARP

Redder: trading lead to greater emissions vs no trading

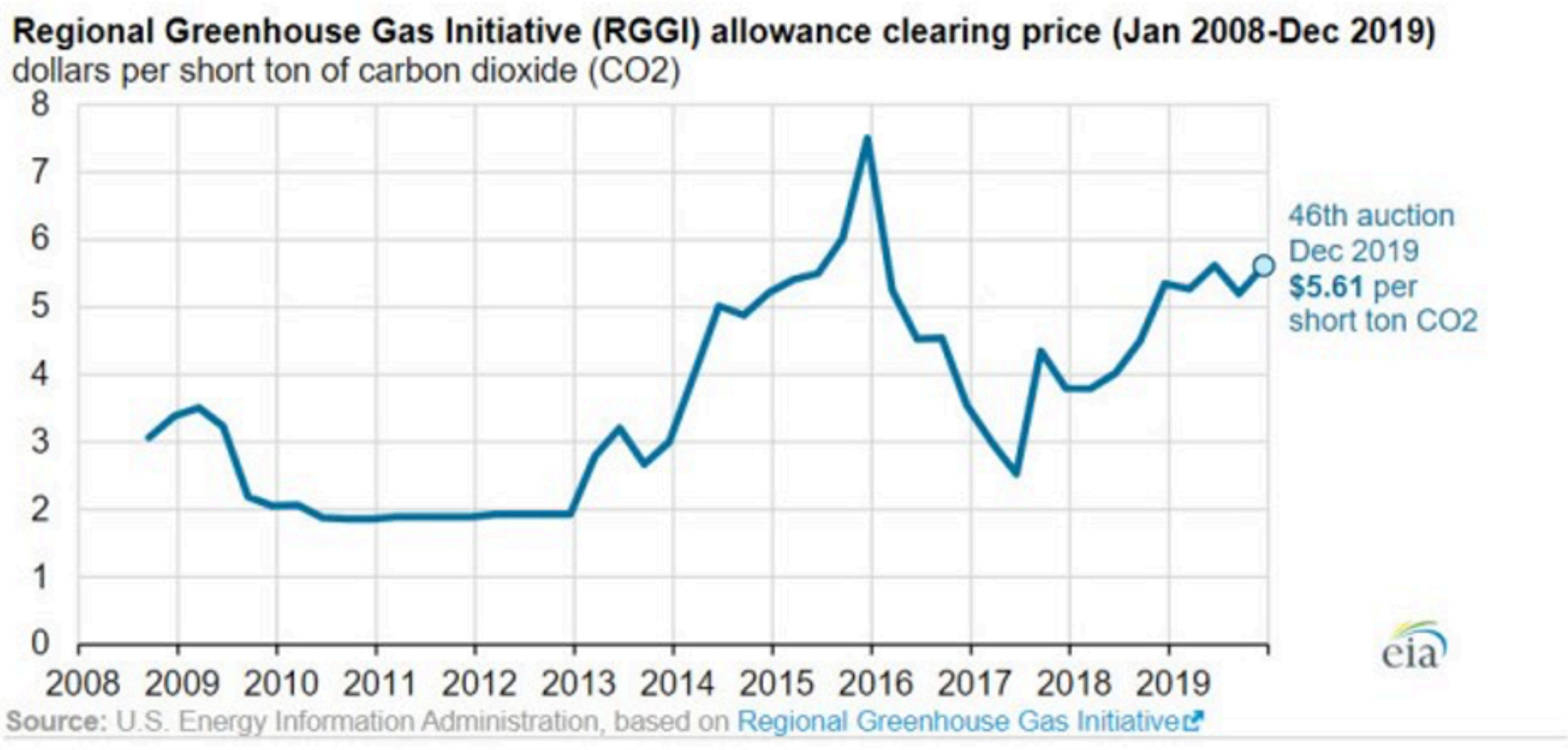


Other permit market examples

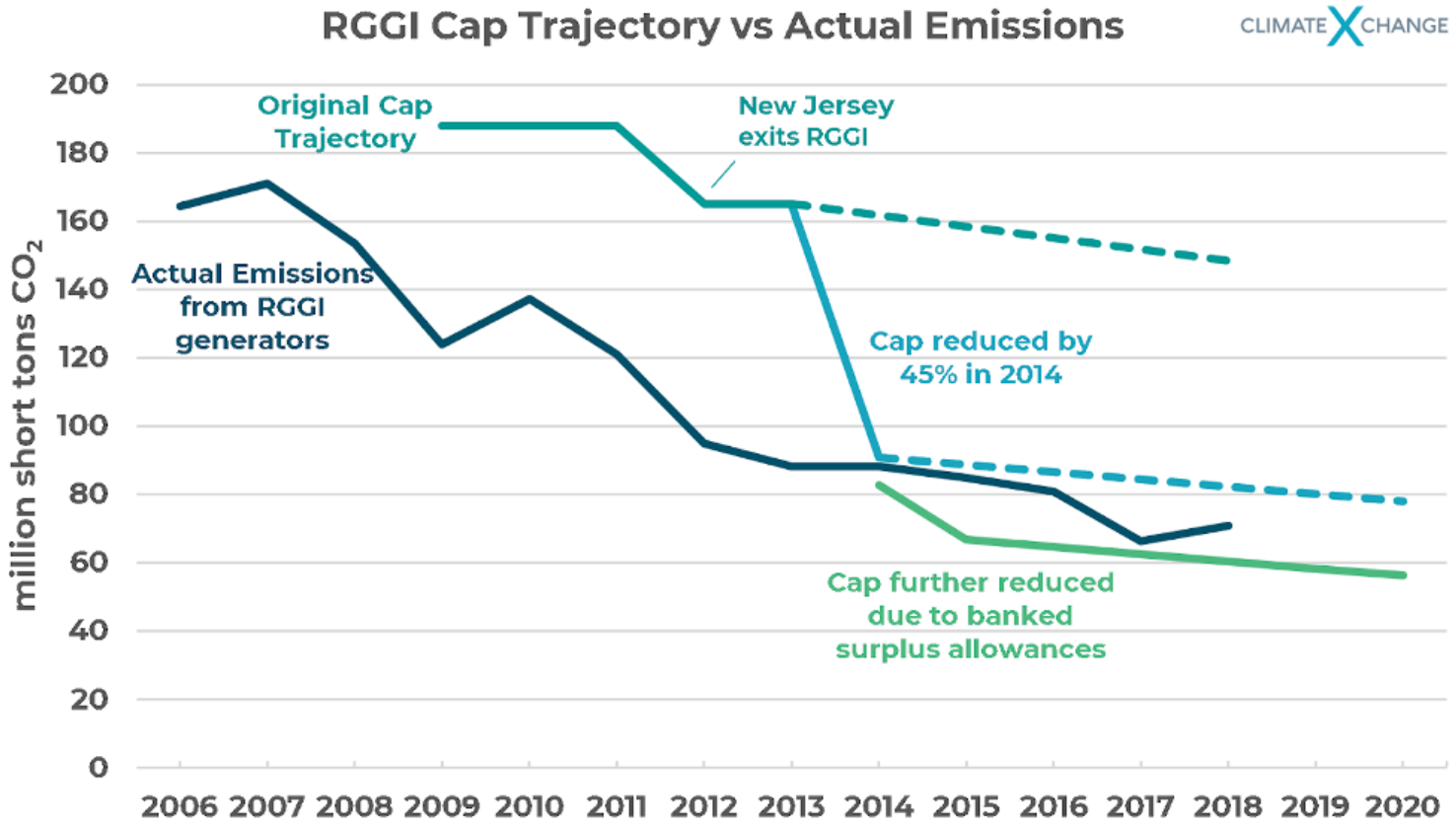
Tradble permit systems are increasingly common:

1. Acid Rain Program
2. NOx Budget Program
3. Regional Greenhouse Gas initative
4. California AB32
5. EU Emission Trading System
6. China's National Carbon Cap and Trade

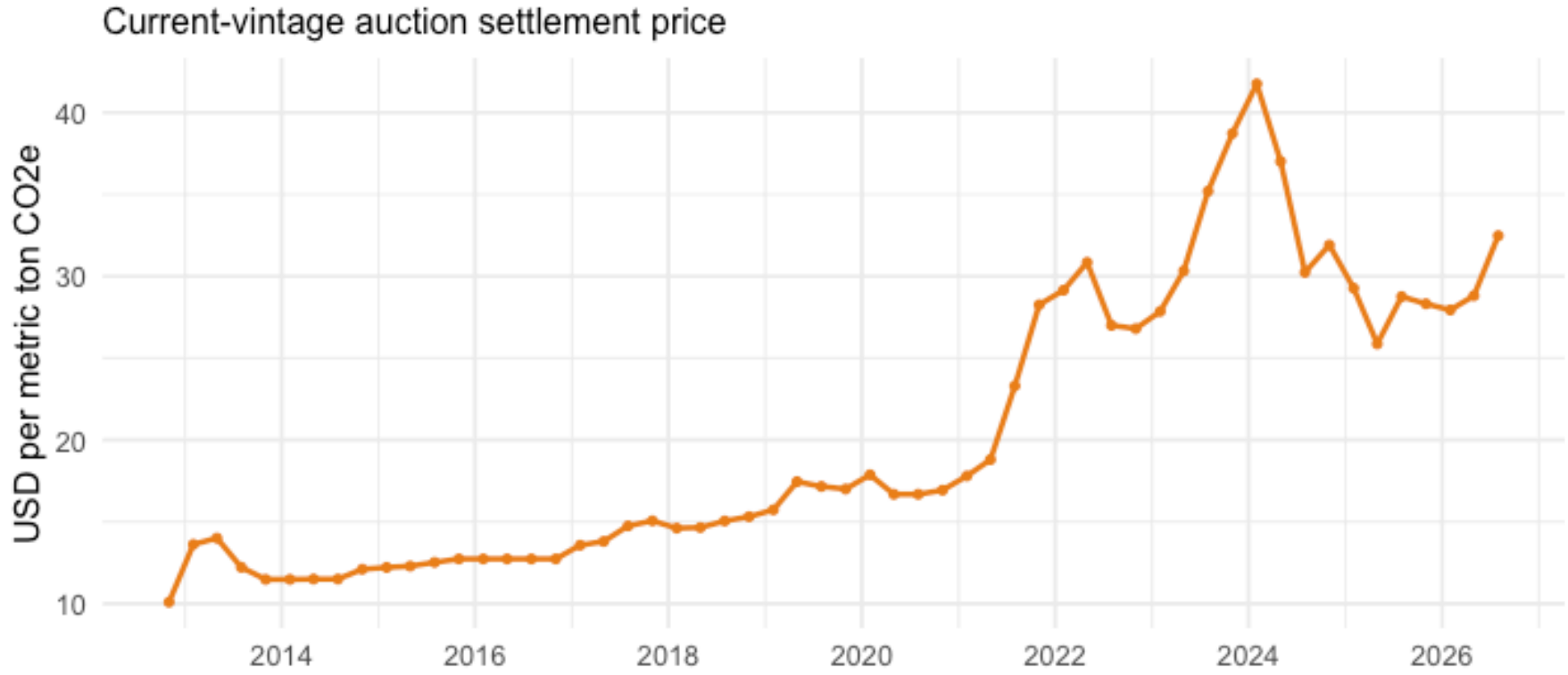
RGGI


Regional Greenhouse Gas Initiative (RGGI) allowance clearing price (Jan 2008-Dec 2019)
dollars per short ton of carbon dioxide (CO2)
8
7
6
5
4
3
2
1
0
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
46th auction
Dec 2019
$5.61 per
short ton CO2
eia
Source: U.S. Energy Information Administration, based on Regional Greenhouse Gas Initiative

RGGI


RGGI Cap Trajectory vs Actual Emissions
CLIMATE CHANGE
million short tons CO_2
200
180
160
140
120
100
80
60
40
20
0
2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020
Original Cap Trajectory
New Jersey exits RGGI
Actual Emissions from RGGI generators
Cap reduced by 45% in 2014
Cap further reduced due to banked surplus allowances

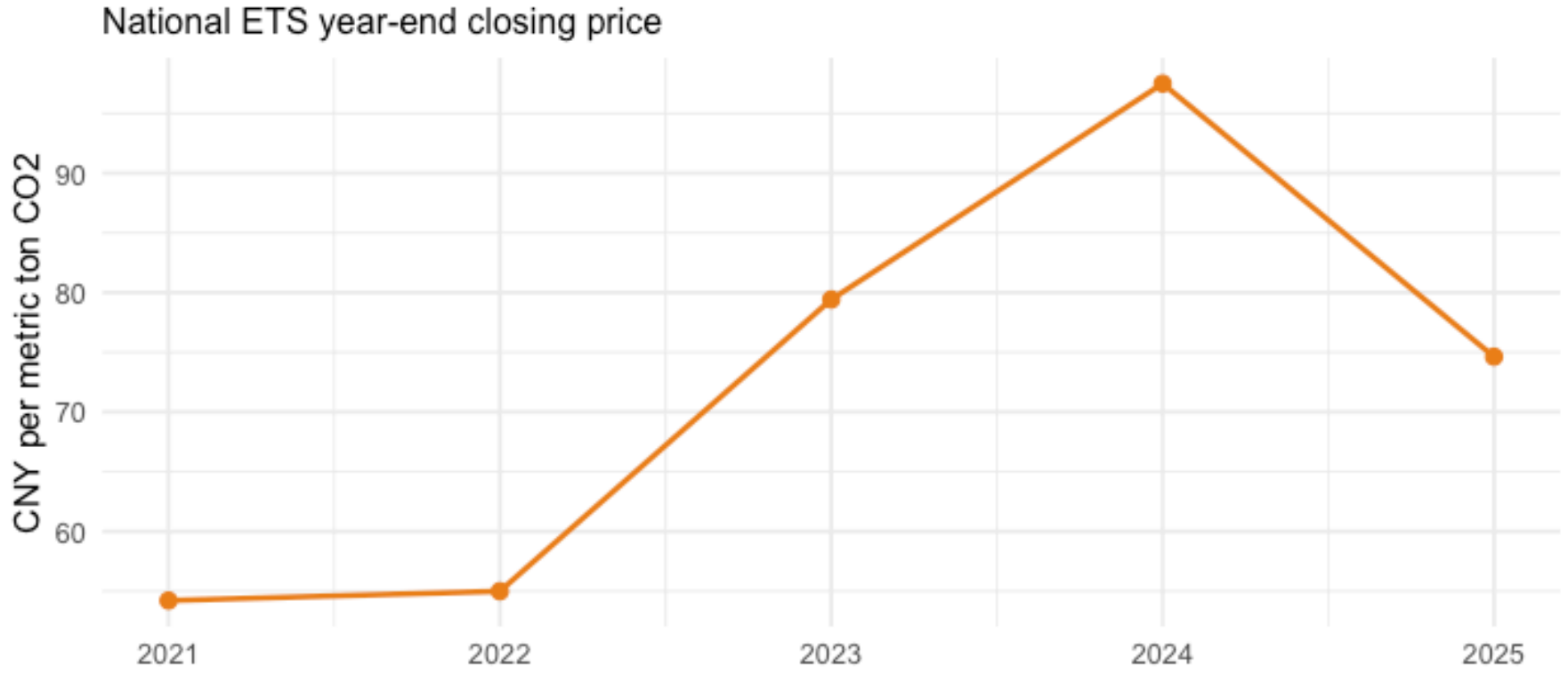
AB32


Current-vintage auction settlement price
USD per metric ton CO2e
40
30
20
10
2014
2016
2018
2020
2022
2024
2026

EU-ETS


Monthly mean of primary auction settlement prices
EUR per metric ton CO2
0
25
50
75
2014
2016
2018
2020
2022
2024
2026

China ETS


National ETS year-end closing price
CNY per metric ton CO2
90
80
70
60
2021
2022
2023
2024
2025

Banking and borrowing in permit markets

Banking and borrowing in permits

In cap-and-trade, permits may be shifted across time if we allow for banking and borrowing:

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Banking: save unused permits for future compliance

Borrowing: use future permits early, then make up the shortfall later

Without borrowing, the firm can only use permits it already has

The firm's permit balance

$$B_t = B_{t-1} + \bar{E}_t - E_t$$

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- $\bar{E}_t$: new permits you receive in year t
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B_t is what remains after adding new permits and subtracting those used

- $B_t > 0$: extra permits you are saving for later
- $B_t < 0$: borrowed permits you must give back later

Banking and borrowing in permits

Consider how one firm allocates its abatement effort over two periods:

$$\min_{E_1, E_2} C_1(E_1) + C_2(E_2)$$

$$\text{subject to: } E_1 + E_2 = \bar{E}_1 + \bar{E}_2$$

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A firm equalizes its MAC over time!

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Intuition:

Standard cap-and-trade allows **within-period trade across firms**

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Standard cap-and-trade allows **within-period trade across firms**

Banking and borrowing add **across-period trade within a firm**

So a firm can effectively trade with itself over time

This shifts abatement away from high-MAC periods and toward low-MAC periods

Banking and borrowing in permits

Why might a firm's MAC change over time?

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5. Learning-by-doing improves operational efficiency

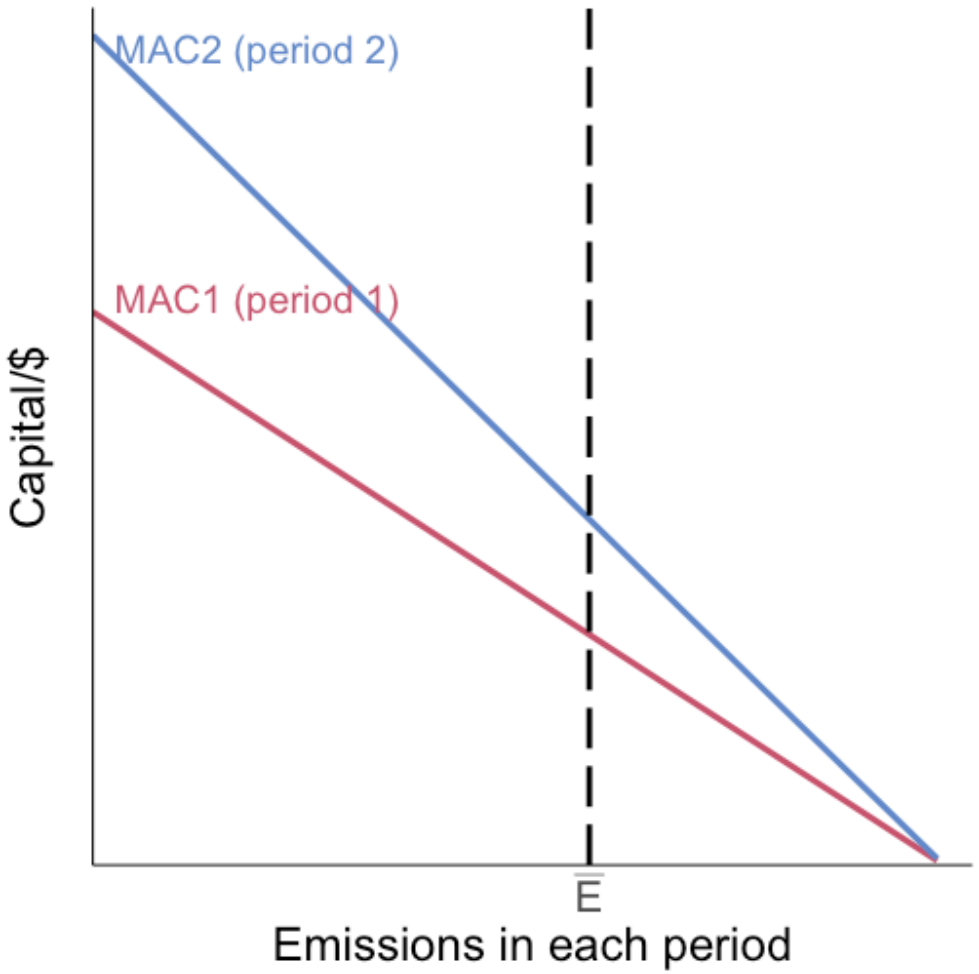
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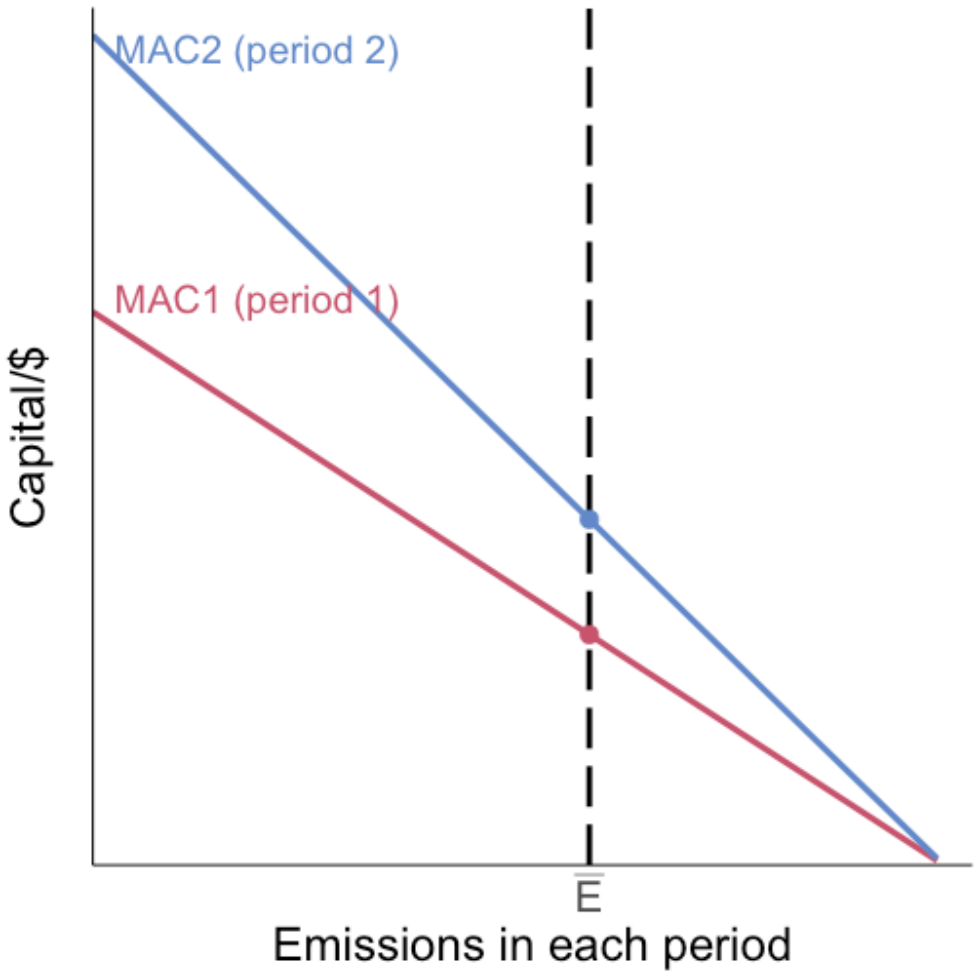
With these shocks, banking/borrowing lowers compliance cost relative to a rigid period-by-period cap

Banking and borrowing in permits



Compare the firm's MAC at each period's original emissions limit

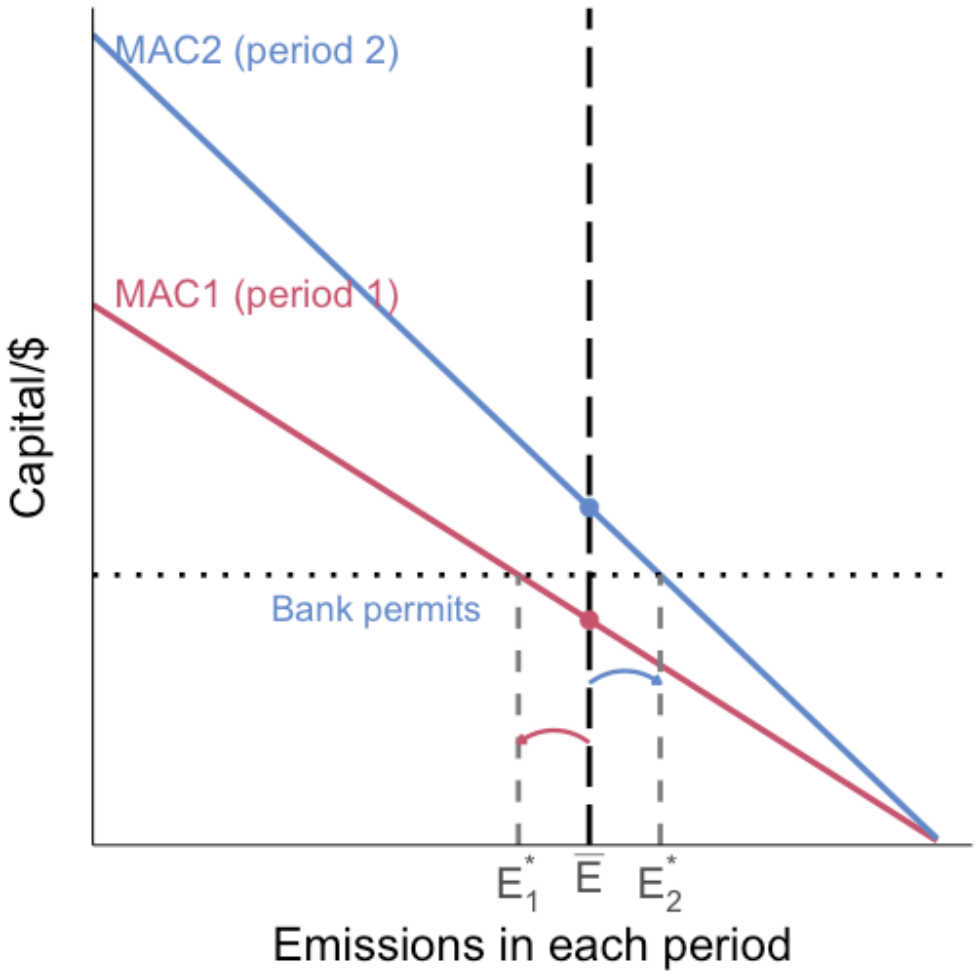
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If $MAC_2(\bar{E}_2) > MAC_1(\bar{E}_1)$, period 2 abatement is more expensive

Banking and borrowing in permits

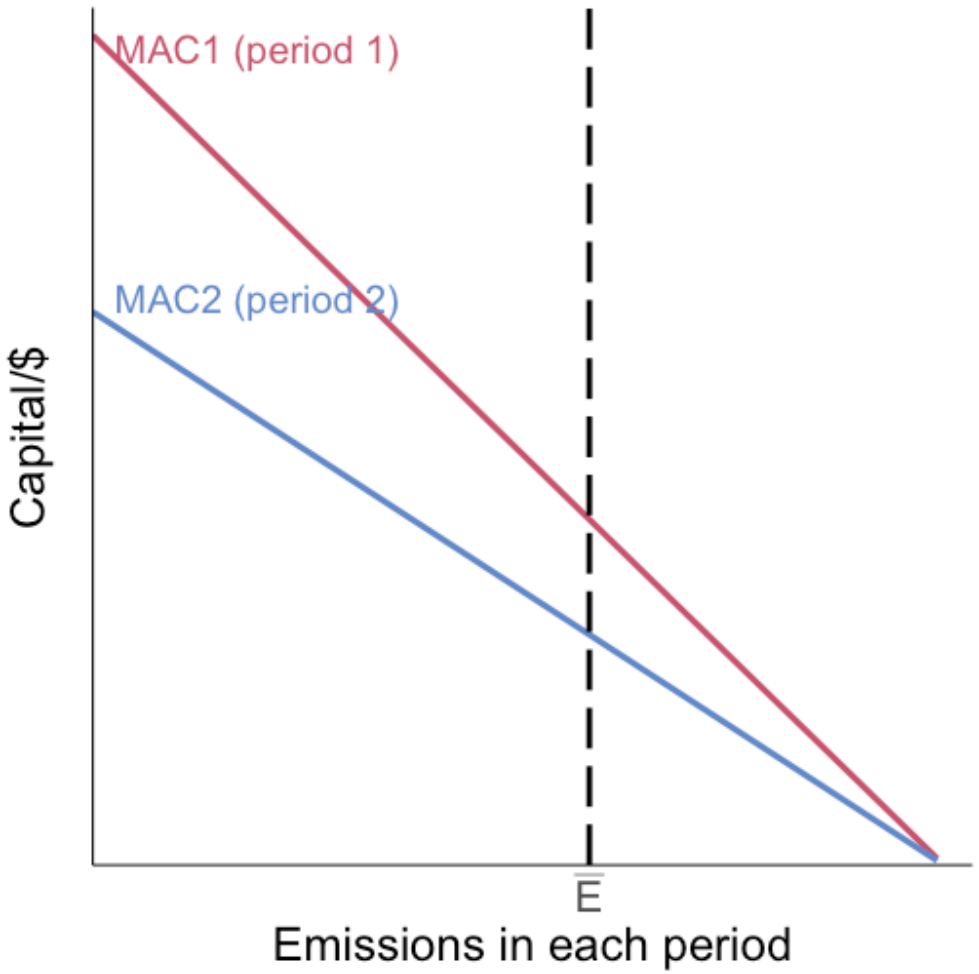


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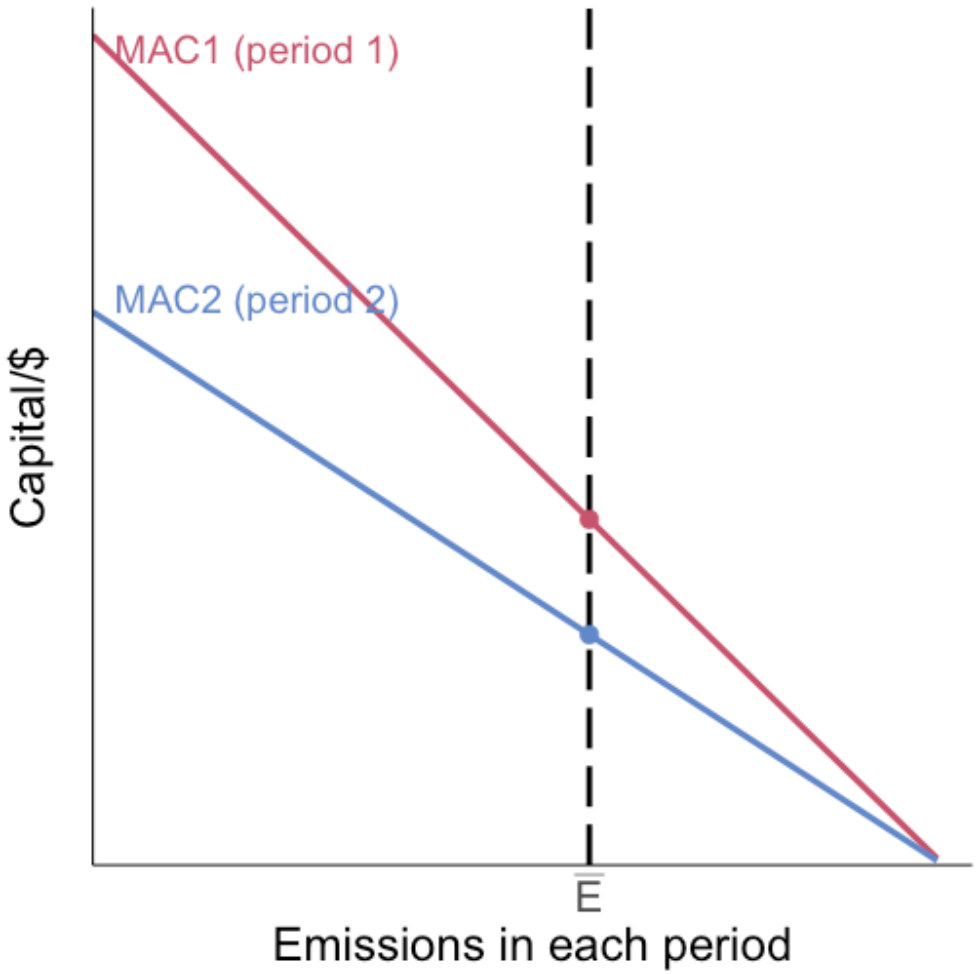
Banking is optimal: $E_1^* < \bar{E}_1$ and $E_2^* > \bar{E}_2$

Banking and borrowing in permits



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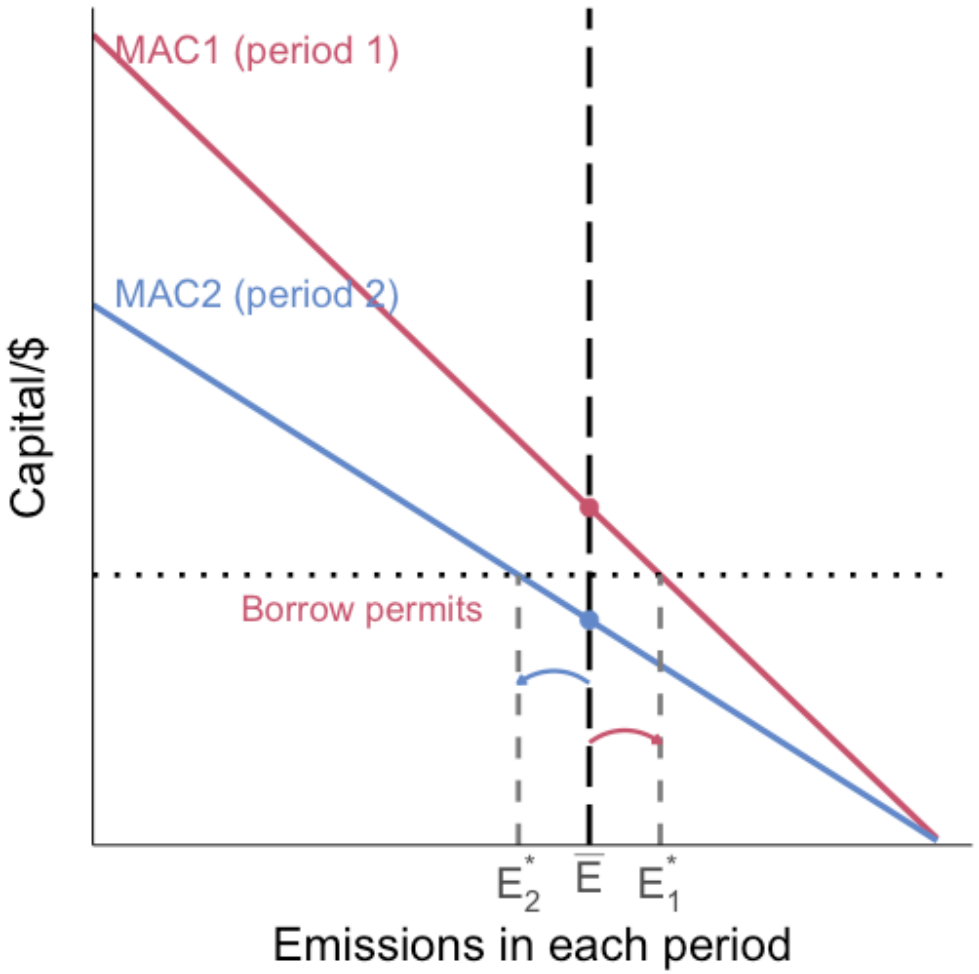
Banking and borrowing in permits



Compare the firm's MAC at each period's original emissions limit

If $MAC_1(\bar{E}_1) > MAC_2(\bar{E}_2)$, period 1 abatement is more expensive

Banking and borrowing in permits

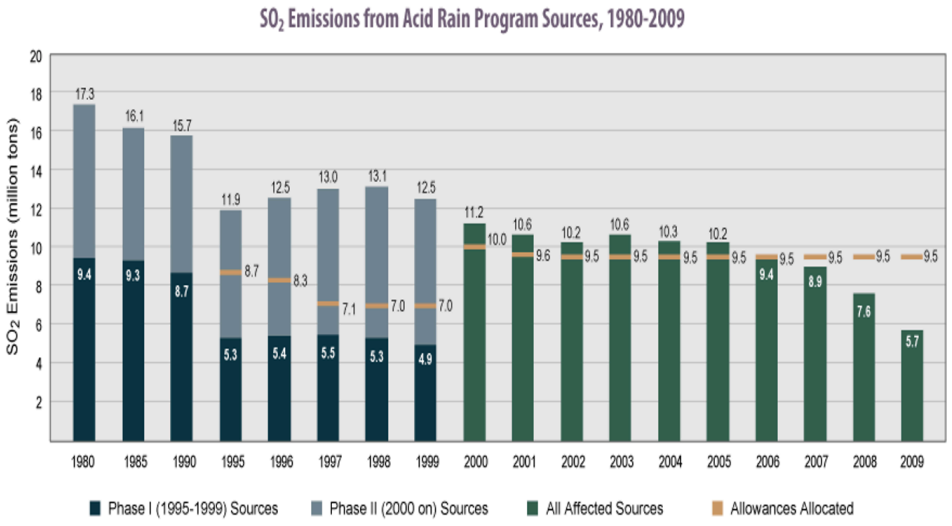


Compare the firm's MAC at each period's original emissions limit

If $MAC_1(\bar{E}_1) > MAC_2(\bar{E}_2)$, period 1 abatement is more expensive

Borrowing is optimal: $E_1^* > \bar{E}_1$ and $E_2^* < \bar{E}_2$

Banking and borrowing in practice: Acid Rain Program

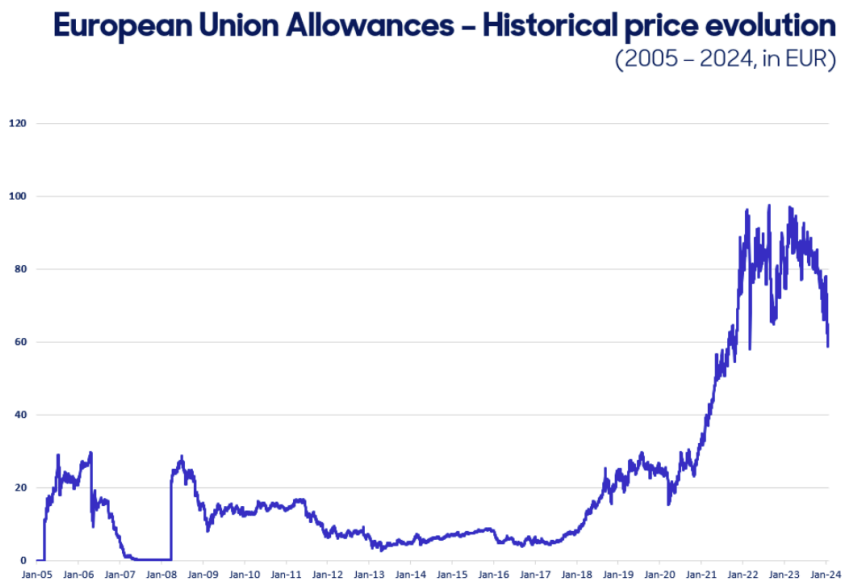


Source: EPA, 2010

U.S. SO_2 trading (Acid Rain Program) allowed banking across years

Firms over-complied in the 1990s and built an allowance bank before tighter caps

Banking and borrowing in practice: EU-ETS



Phase I (2005-2007): limited inter-phase banking; surplus allowances led to a price collapse near phase end

From Phase II onward, banking became a key margin for compliance planning

Large cumulative surplus carried forward across phases and affected prices

Banking and borrowing: design lessons

1. Banking is widely used because it lowers costs while preserving a cumulative cap
2. Borrowing is often constrained because of default, enforcement, and credibility concerns
3. Rules on banking/borrowing strongly affect allowance price dynamics
4. Without banking across phase boundaries, end-of-phase price volatility can be severe

The permit game: banking and borrowing

The permit game: decisions across rounds

We now allow permits to move between the two rounds

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Bank: emit less than your permits cover in Round 1 and save the unused permits for Round 2

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The permit game: decisions across rounds

We now allow permits to move between the two rounds

Bank: emit less than your permits cover in Round 1 and save the unused permits for Round 2

Borrow: emit more than your permits cover in Round 1 and repay the difference with Round 2 permits

Use the emissions choice in Round 1. Unrepaid borrowing pays a penalty at the end of Round 2

The permit game: predictions across rounds

How should expected Round 2 conditions change your Round 1 decisions?

The permit game: predictions across rounds

How should expected Round 2 conditions change your Round 1 decisions?

- Bank when permits are more valuable in Round 2
- Borrow when permits are more valuable in Round 1, accounting for repayment
- Trading across rounds brings marginal abatement costs closer together

The permit game: predictions across rounds

How should expected Round 2 conditions change your Round 1 decisions?

- Bank when permits are more valuable in Round 2
- Borrow when permits are more valuable in Round 1, accounting for repayment
- Trading across rounds brings marginal abatement costs closer together

Compare these results with the game without banking and borrowing

The trading cost-reduction percentage is omitted here. Separate-round benchmarks do not measure cost-effectiveness across time

Rejoin the permit game: banking and borrowing

Scan to open the student login, then join with your team name

Or open **aem4510.ivanrudik.com/permits**

Debrief: banking and borrowing

Did teams bank before the tighter Round 2 cap? Did any teams borrow instead?

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Compare your MAC across rounds: could shifting one more permit have reduced your total abatement cost?

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Banking and borrowing extend the gains from trade across time

Comparison of standards, taxes, permits

What do we know so far

So far we have seen that:

1. Standards, taxes, and tradable permits can all achieve the efficient allocation
2. Taxes and tradable permits are cost-effective **no matter what**
   - (all firms set MAC = τ and MAC = p)

What do we know so far

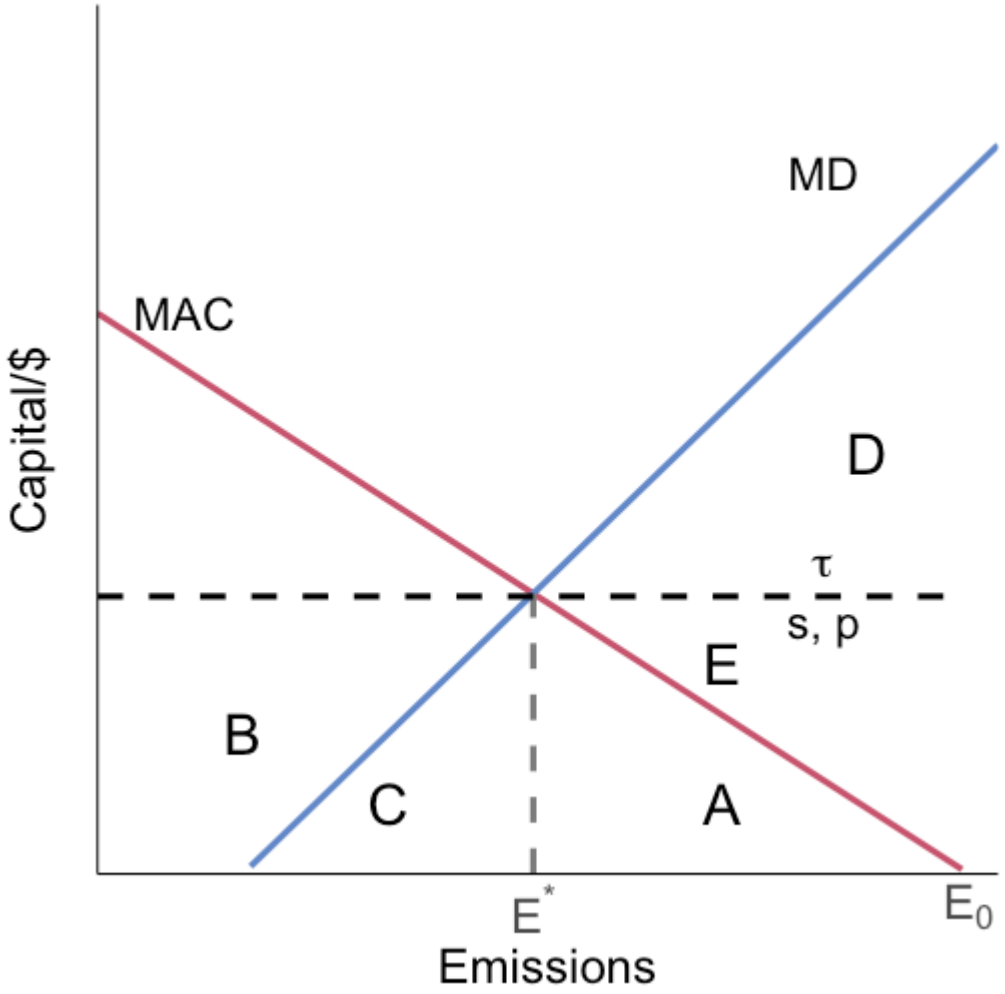
So far we have seen that:

1. Standards, taxes, and tradable permits can all achieve the efficient allocation
2. Taxes and tradable permits are cost-effective **no matter what**
   - (all firms set MAC = τ and MAC = p)

This still leaves a few questions to answer:

1. What are the equity effects?
2. What are the output effects?
3. What are the administrative burdens?
4. What are the **dynamic** incentives under these policies?

The equity set up

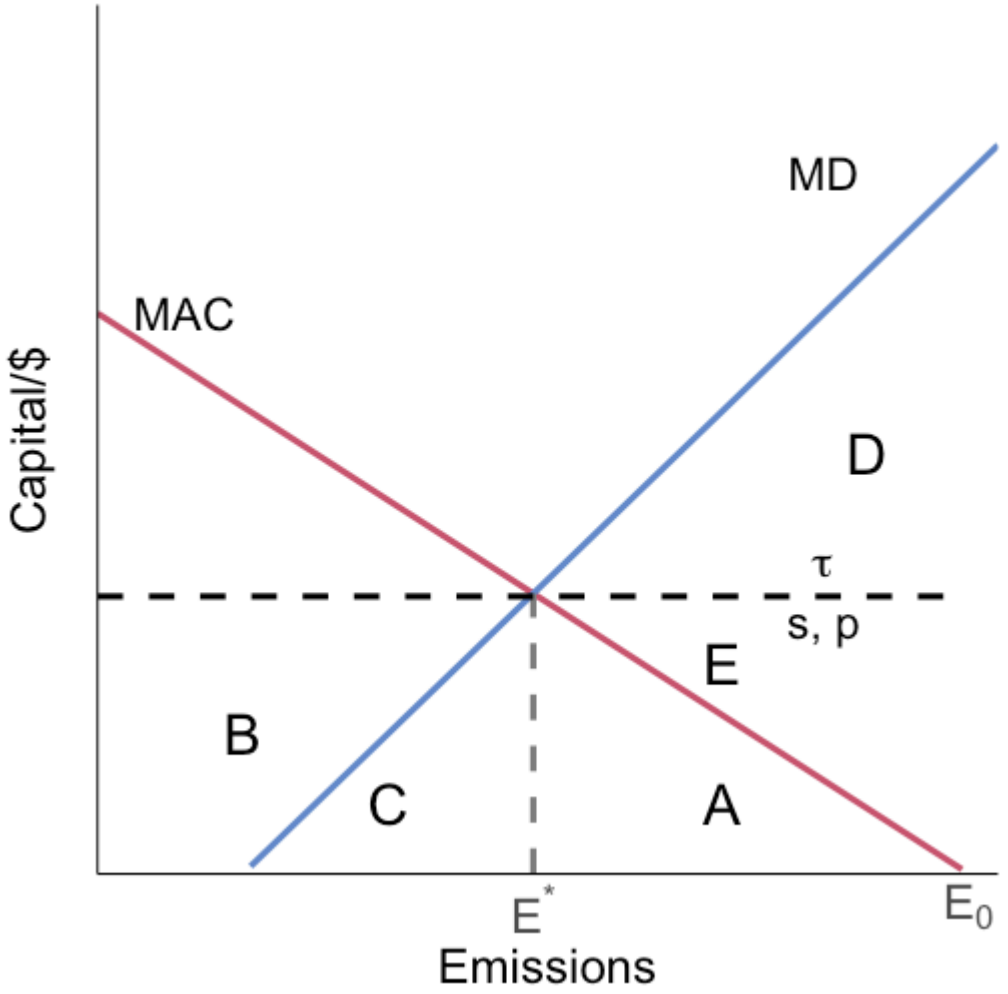


Lets consider this our base set up for 1 firm

The regulator can achieve E^* through:

- an emission standard of E^*
- a tax of τ
- an abatement subsidy of s
- "tradable permit" cap of E^* with auctioned permits at price p

The equity set up

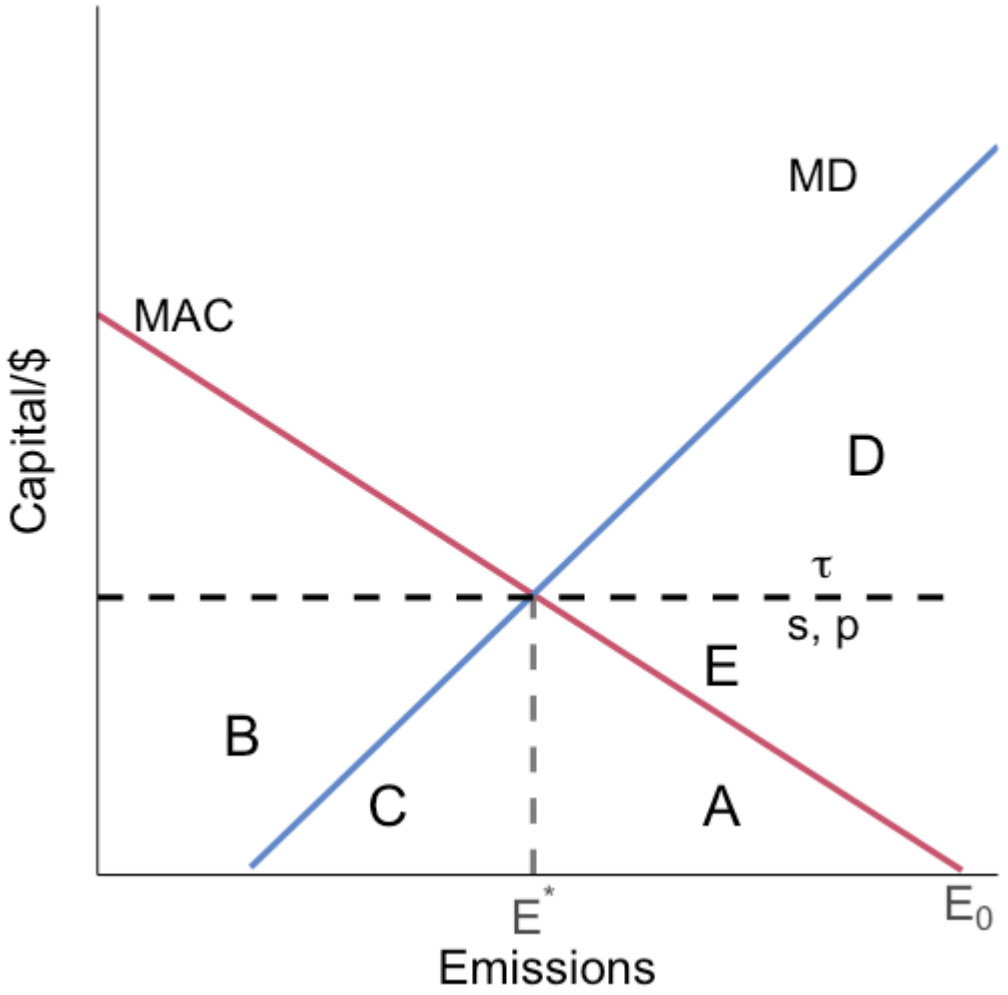


First let's look at **equity**

How do the costs and benefits of the policies fall on different groups?

From here on we will roll the tax and permit system into 1: they are actually identical in terms of their impacts if all permits are auctioned

The distributional outcomes



	Tax/Permits	Standard	Subsidy	Ranking
Firm	-(A+B+C)	-A	E	Sub > Std > Tax
Households	A+D+E	A+D+E	A+D+E	Indifferent
Government	B+C	0	-(E+A)	Tax > Std > Sub
Total	D+E	D+E	D+E	

The total welfare gain is the same for all policies

The difference is in the **distribution**

The standard strikes a middle ground out of the three

Output effects

So far we have assumed that actual firm output is not affected by abatement/emission decisions

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This won't always be true in the real world

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Key identity:

$$E = \left(\frac{E}{q}\right) \times q$$

Firms can reduce total emissions by lowering emissions intensity E/q, output q, or both

Output effects: the firm's choices

The firm chooses output $q > 0$ and emissions intensity $e = E/q$, so $E = eq$

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Let $R(q)$ be revenue and $C_0(q)$ production cost before abatement

Let $a(e)$ be abatement cost per unit of output, so total cost is

$$C_0(q) + qa(e)$$

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Let $a(e)$ be abatement cost per unit of output, so total cost is

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Unregulated intensity is e_0, with $a(e_0) = 0$

For $e < e_0$, assume $a'(e) < 0$: cleaner production costs more

Deriving the MAC

Holding q fixed, MAC is the cost of avoiding one more unit of emissions

$$MAC(E,q) = -\frac{\partial[qa(E/q)]}{\partial E}$$

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$$MAC(E,q) = -\frac{\partial[qa(E/q)]}{\partial E}$$

Applying the chain rule gives

$$MAC(E,q) = -qa'(E/q)\frac{1}{q} = -a'(e)$$

Output effects: choosing intensity

Hold output fixed and use the familiar abatement rules

- A tax saves τ for each unit of emissions avoided
- A subsidy pays s for each unit of emissions avoided

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For this comparison, the standard limits emissions *per unit of output*, such as one ton per MWh

This is an **intensity standard**, $e \leq \bar{e}$, whereas the earlier cap limited total emissions

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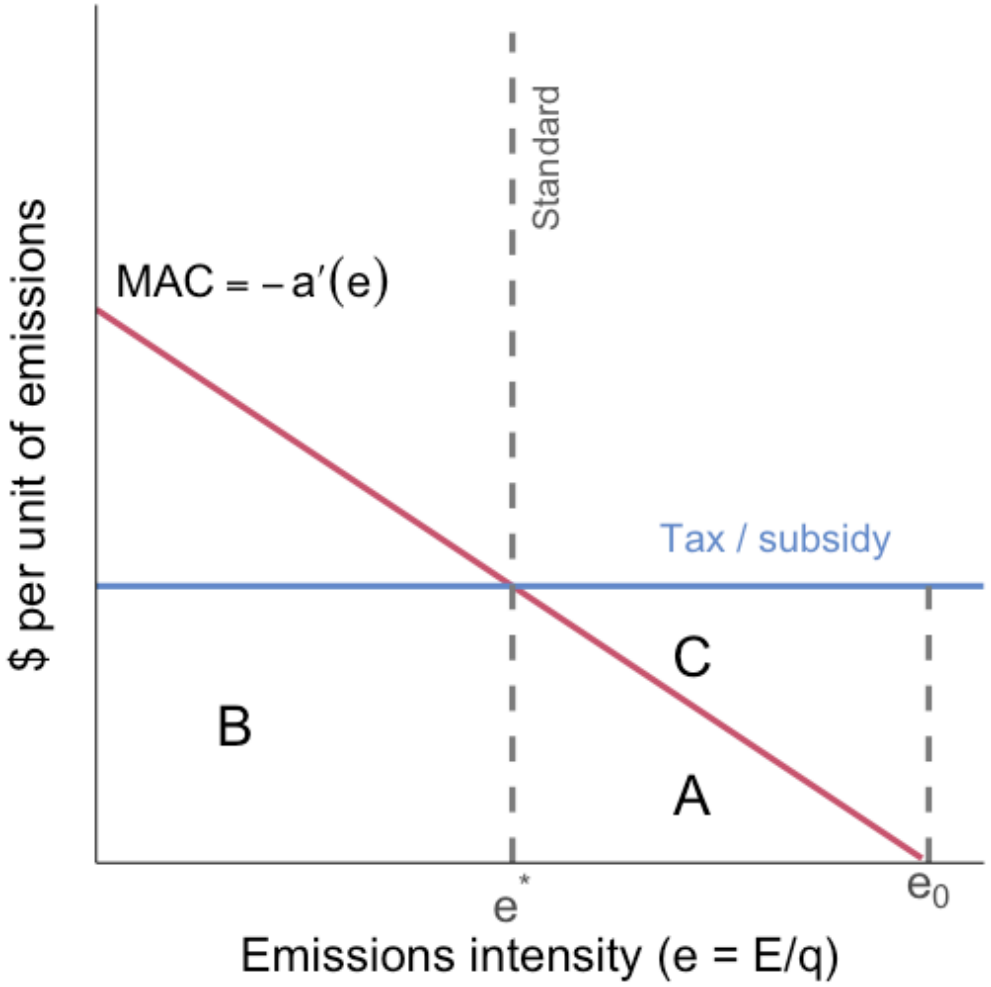
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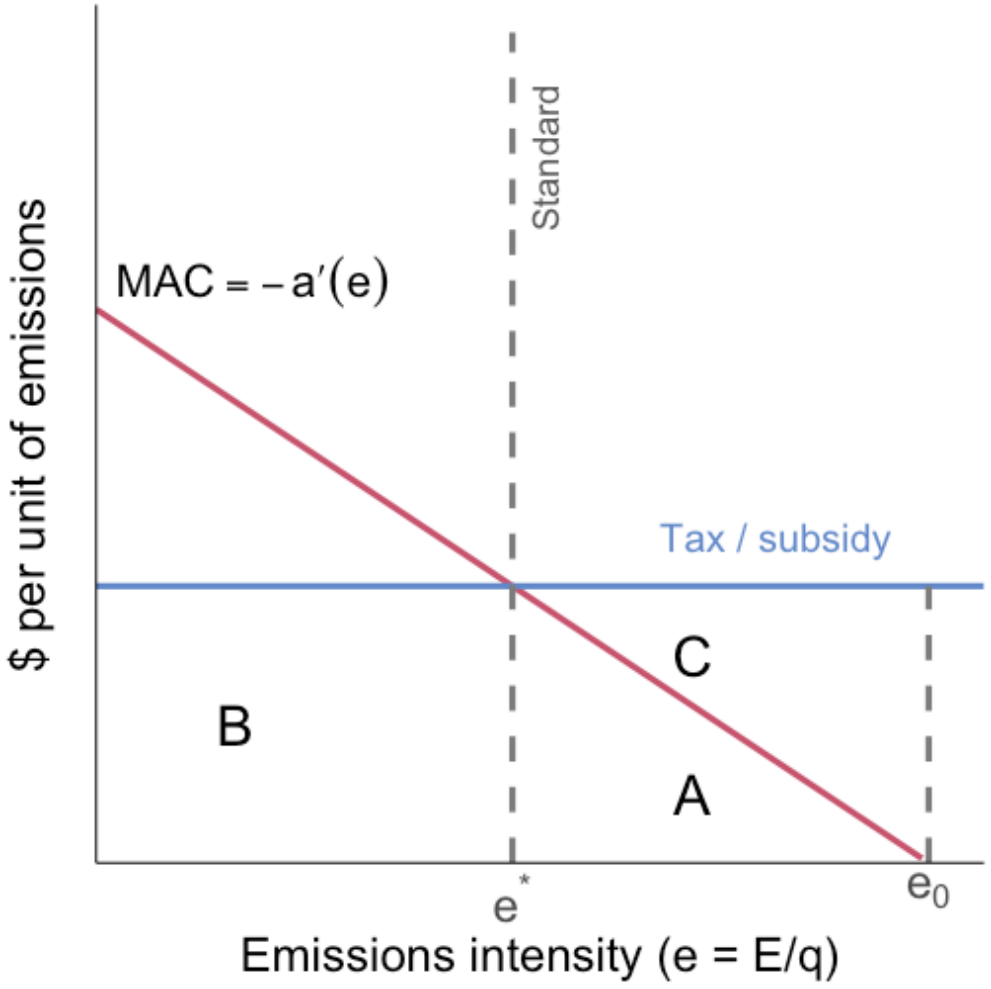
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Output effects: intensity then output



All areas are dollars per unit of output

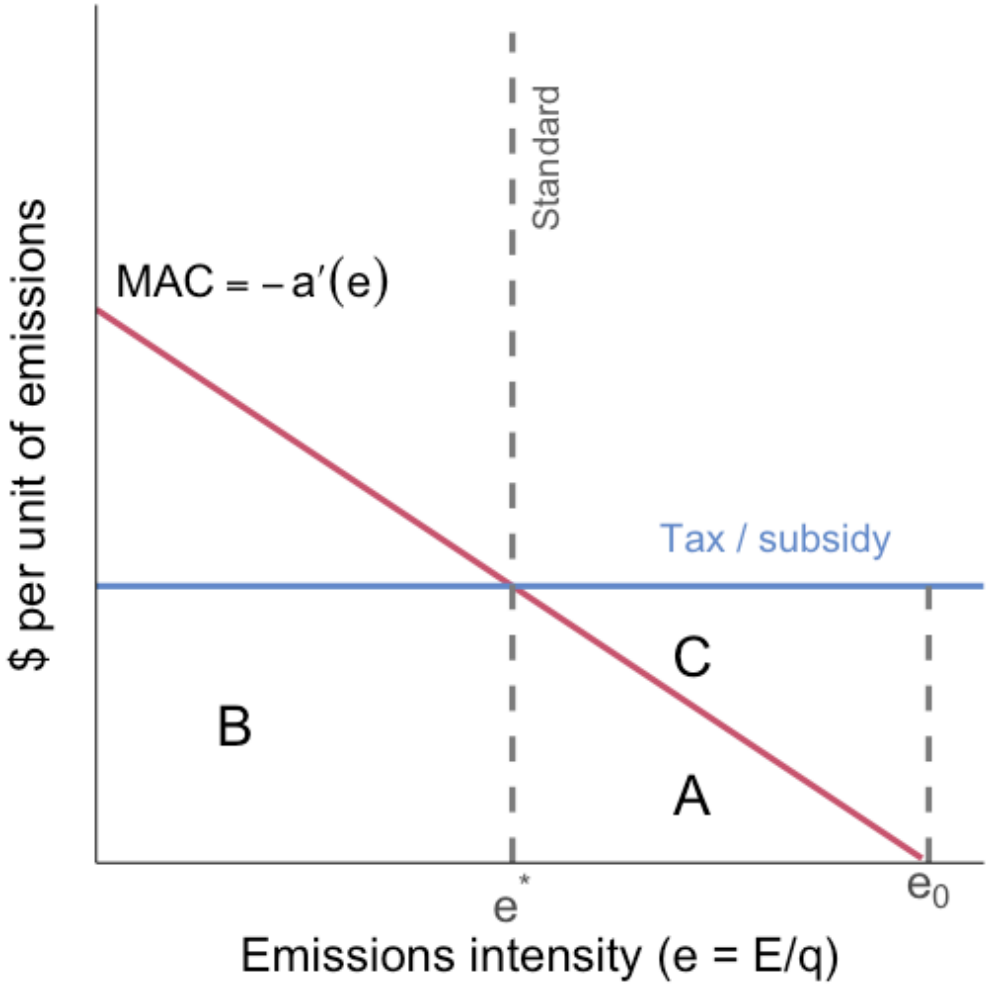
Output effects: intensity then output



All areas are dollars per unit of output

The firm chooses e^* where MAC equals the tax rate

Output effects: intensity then output

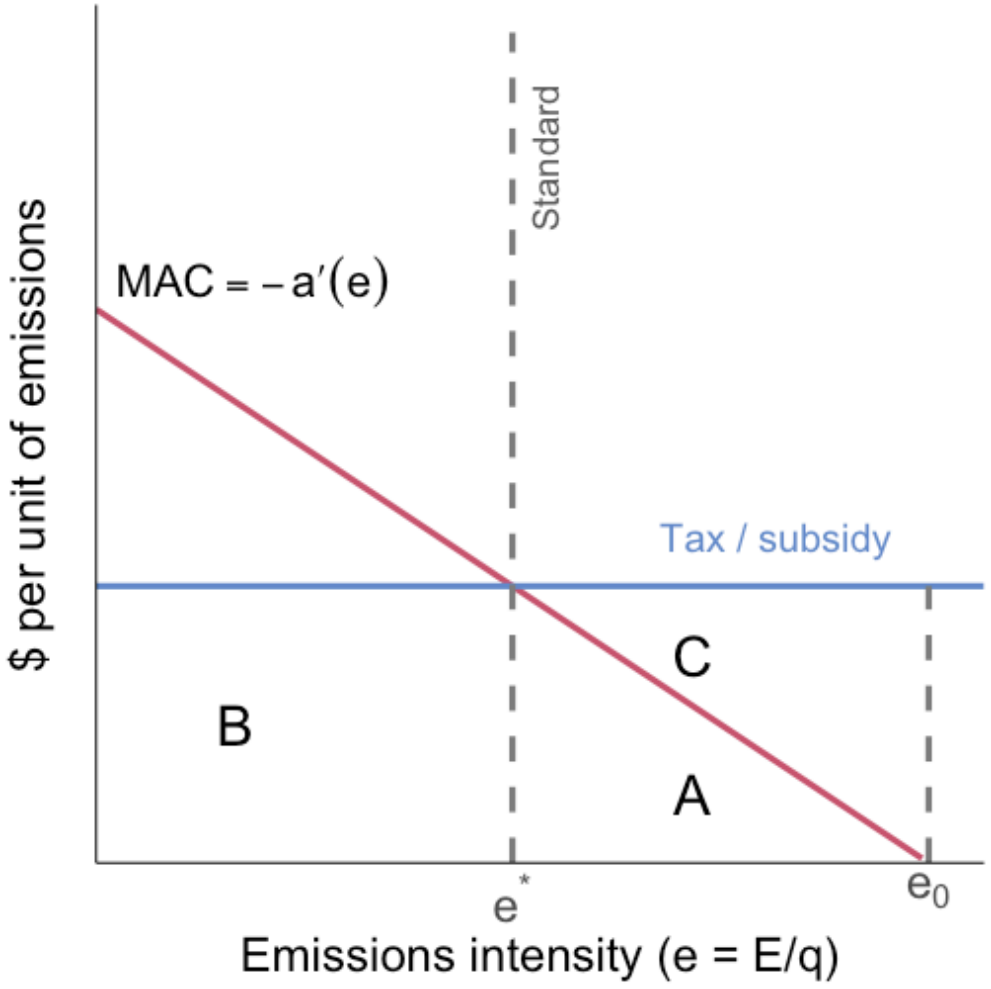


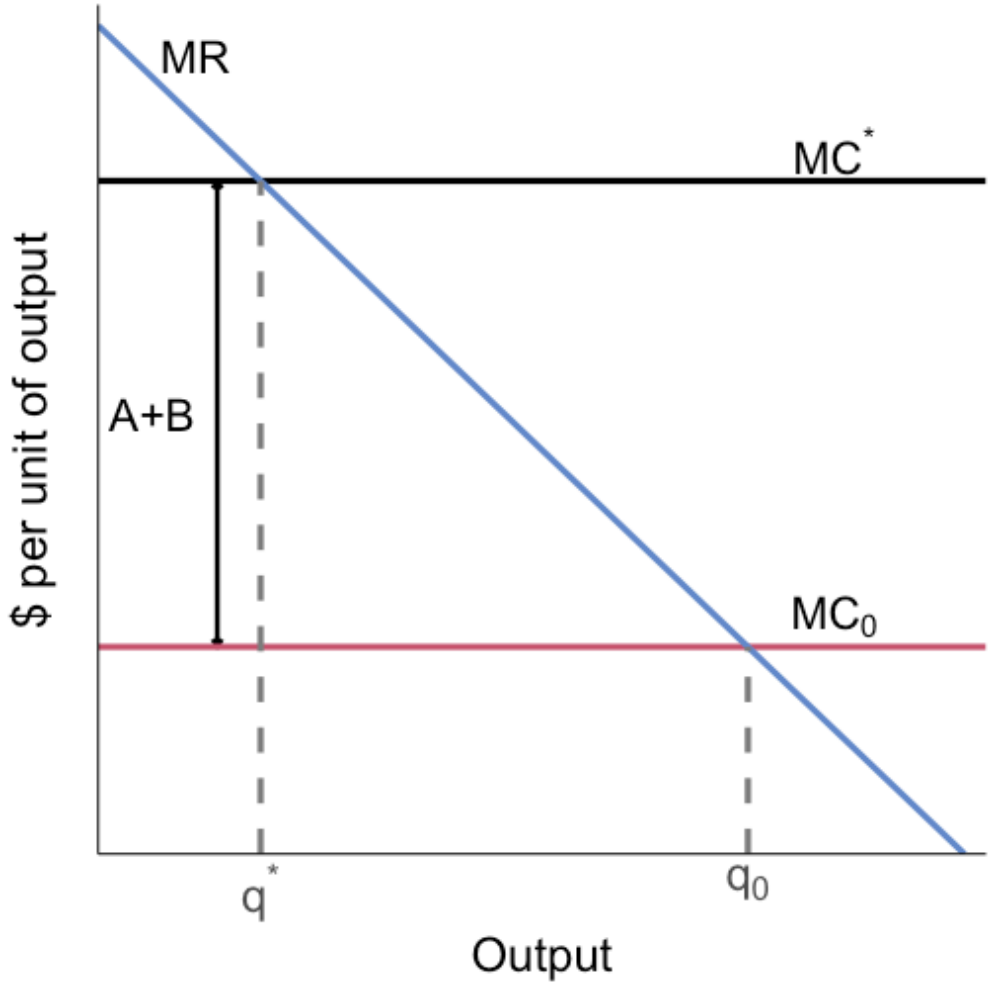
All areas are dollars per unit of output

The firm chooses e^* where MAC equals the tax rate

Cost per unit of output: A in abatement + B in tax

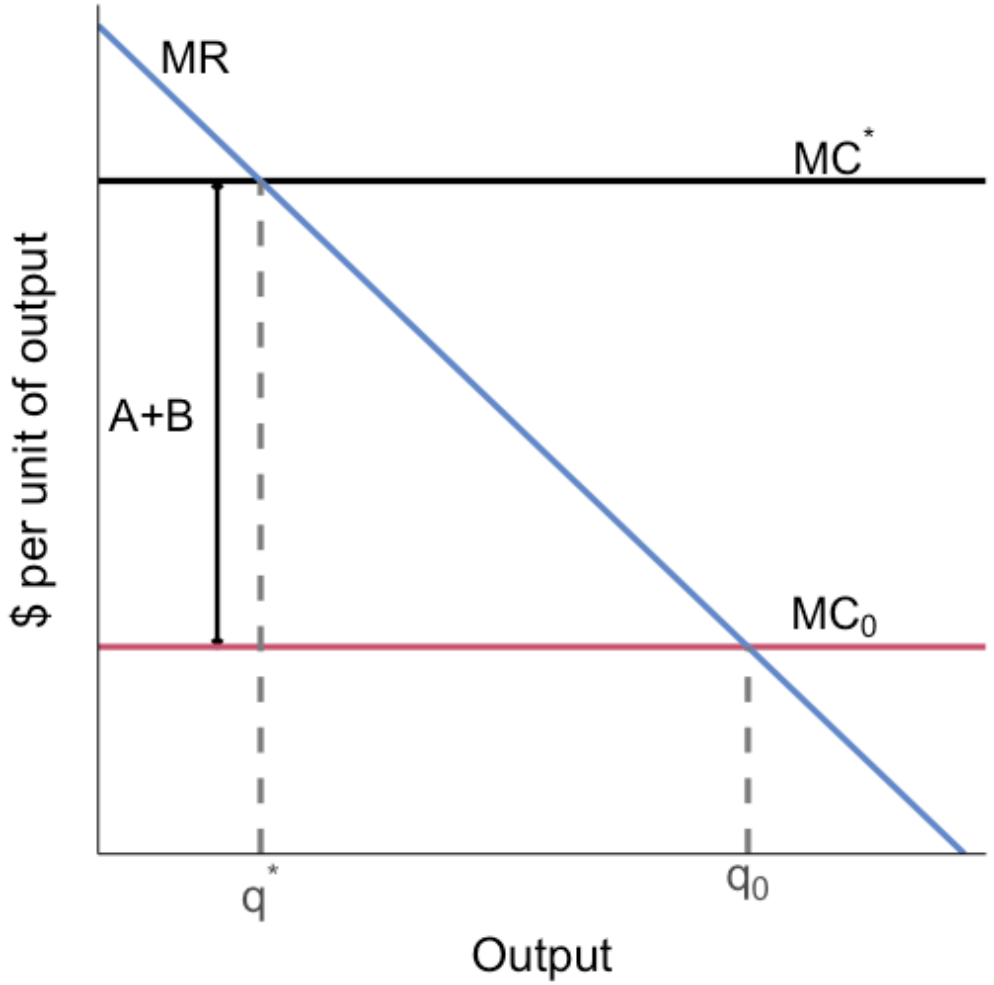
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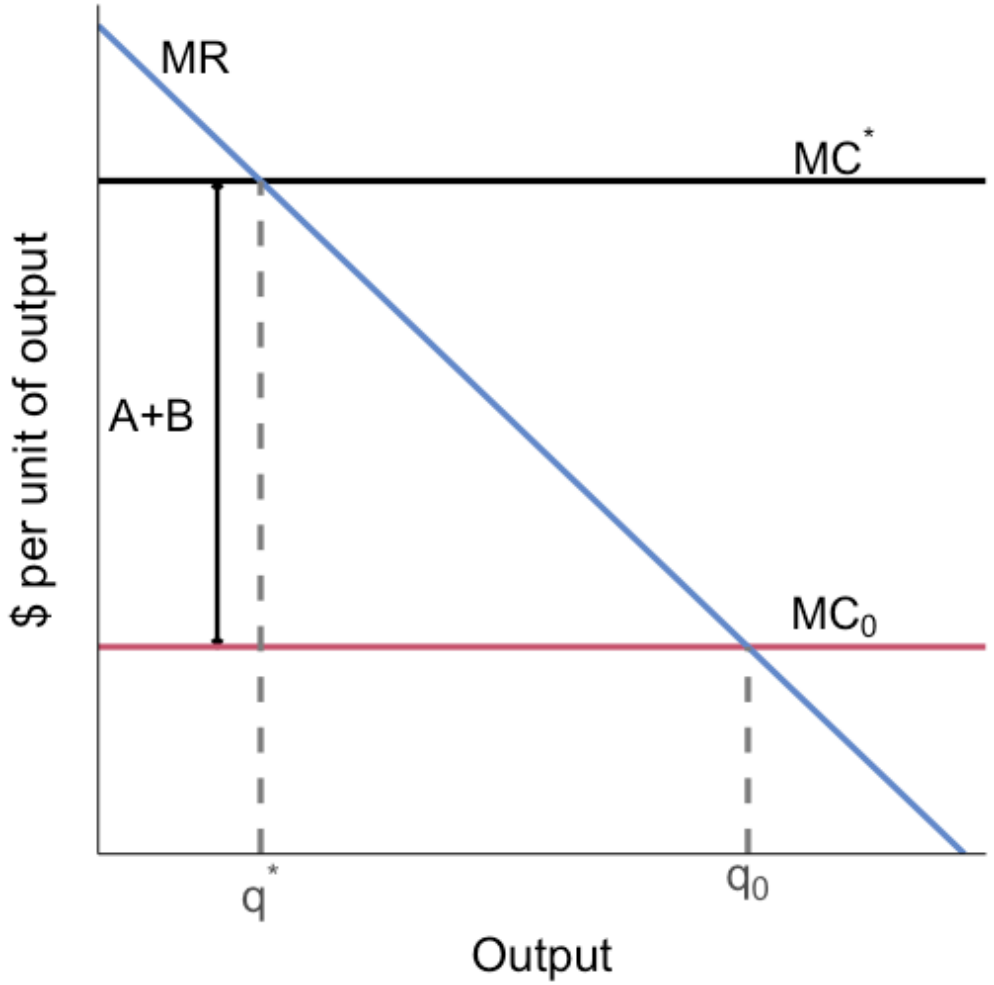
Abatement and tax payments raise the cost per unit of output by A+B



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At MR = MC, profit-maximizing output falls from q_0 to q^*

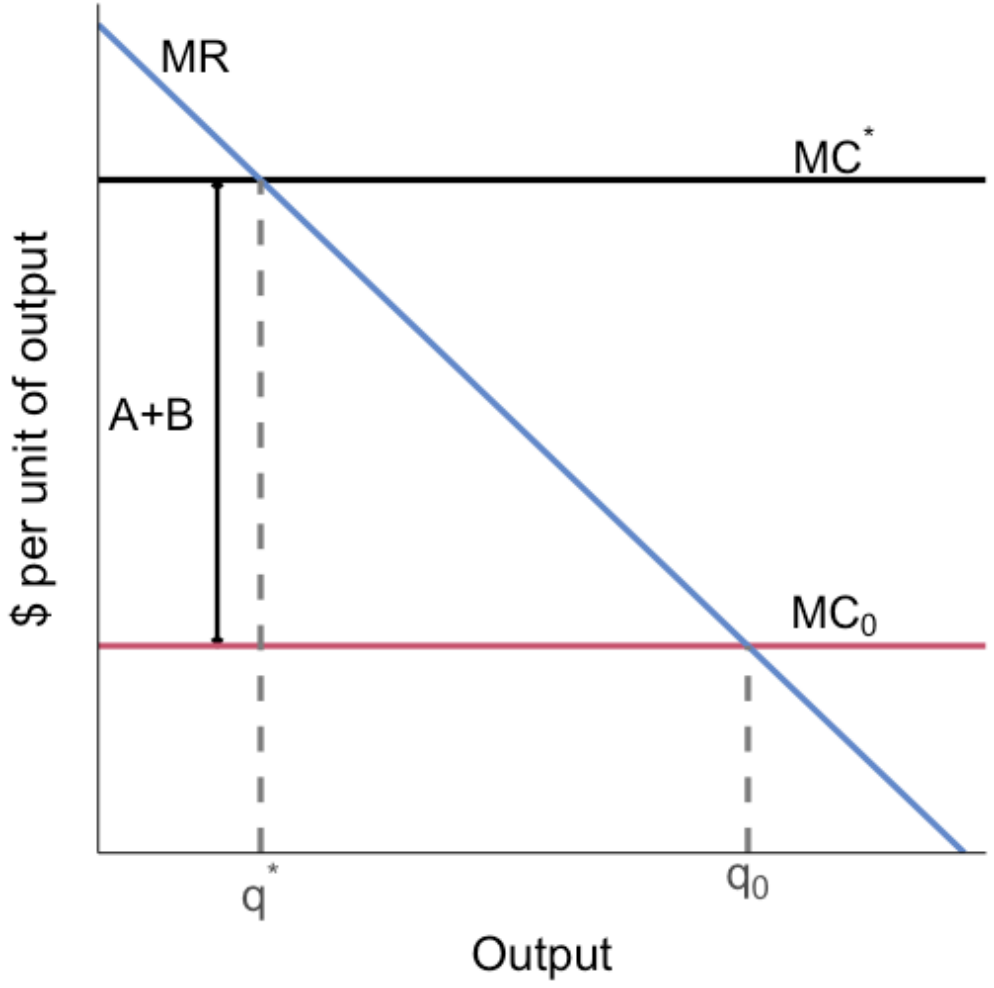


Output effects: intensity then output

Abatement and tax payments raise the cost per unit of output by A+B

At MR = MC, profit-maximizing output falls from q_0 to q^*

Lower intensity and lower output both reduce total emissions



The output results

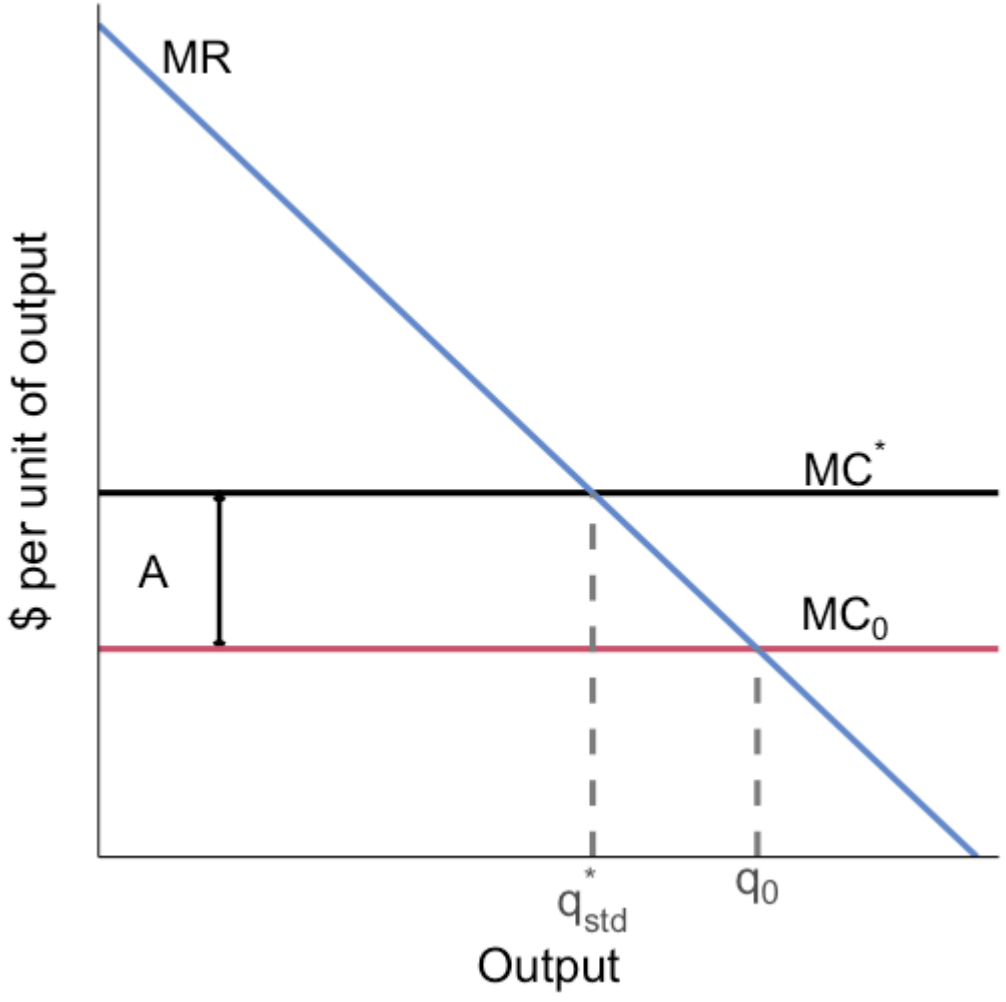
Emission tax:

- The firm chooses intensity e^* where MAC equals the tax rate
- Abatement cost A plus tax payment B raises output MC by A+B
- Profit-maximizing output satisfies $MR = MC$, so higher MC lowers output
- Both lower intensity and lower output reduce total emissions

The output results

Emissions-intensity standard:

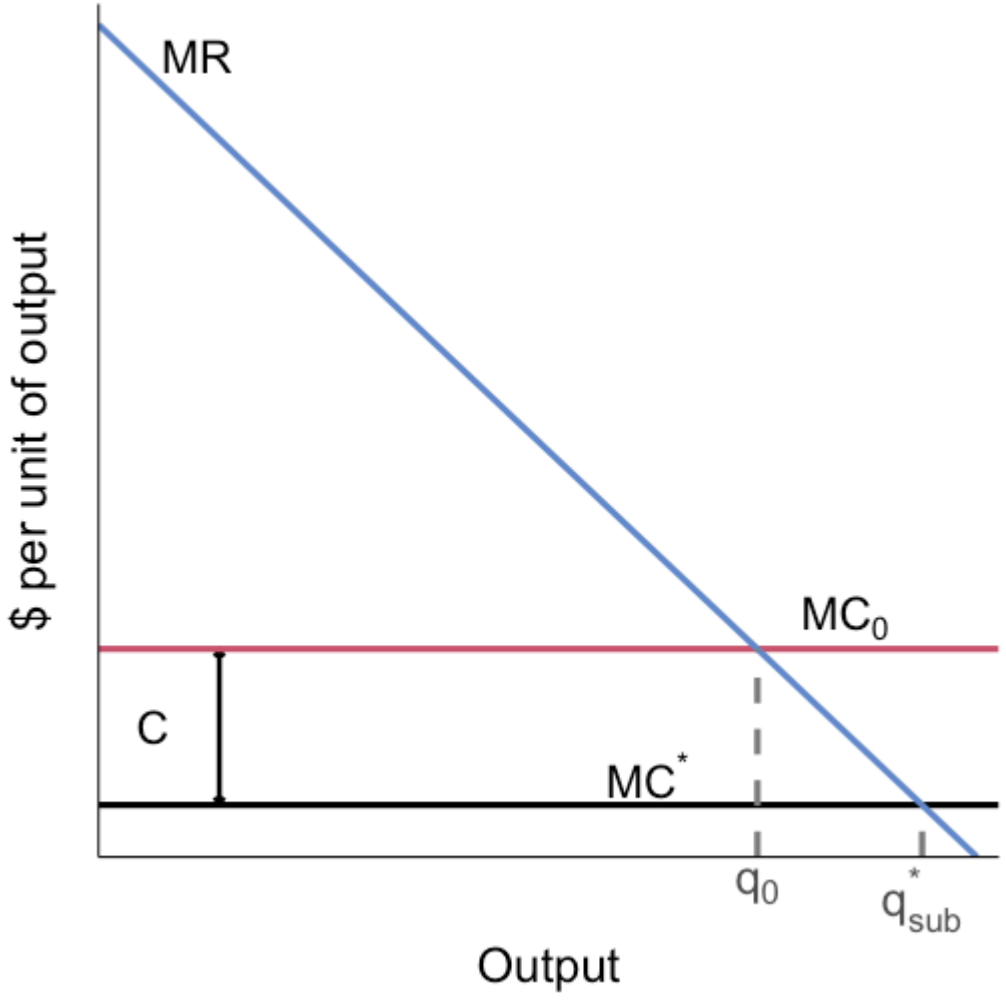
- Require $e \leq e^*$, where $e = E/q$
- The firm pays A in abatement cost per unit of output, with no tax payment
- This raises the marginal cost of output by A
- Output and total emissions fall, but by less than under the tax



The output results

Abatement subsidy:

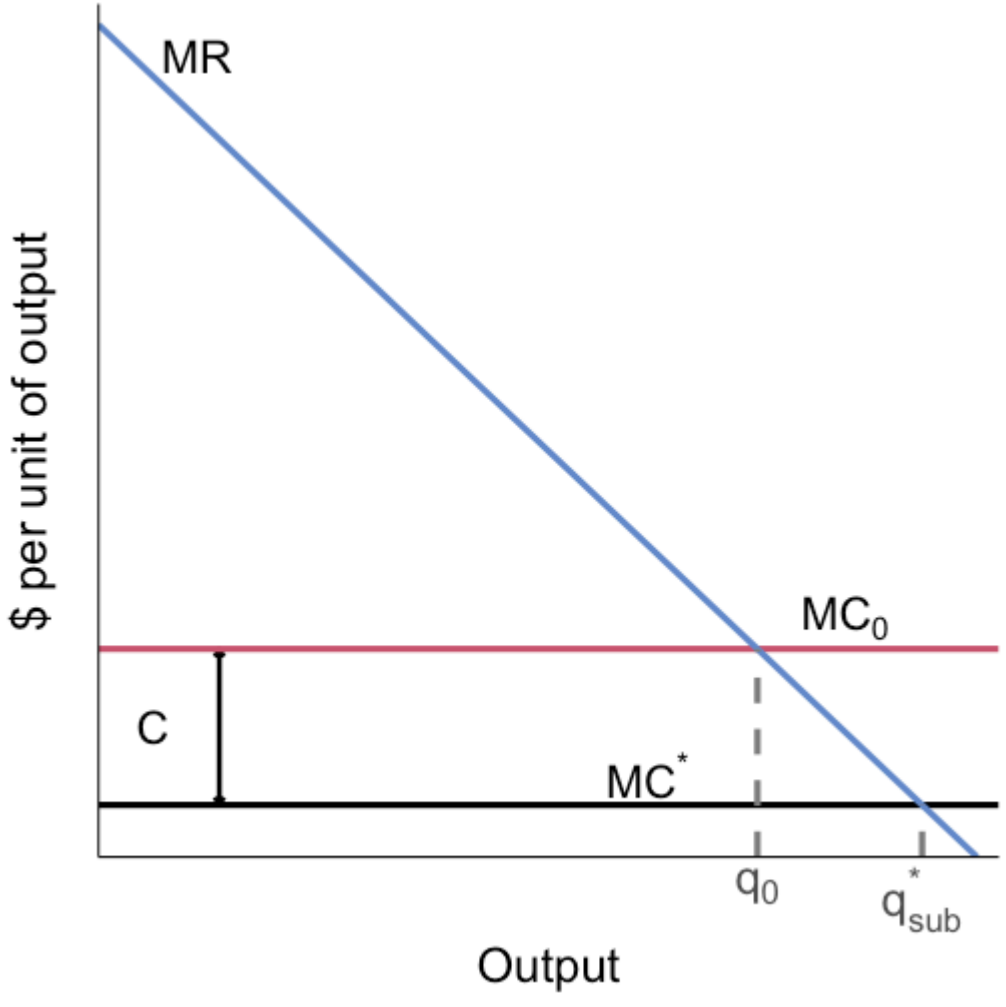
- Pay $s(e_0 - e)$ per unit of output relative to baseline intensity e_0
- Gross subsidy A+C minus abatement cost A leaves net gain C per unit of output
- Output MC falls by C, so output rises
- Total emissions $E = e^* q^*$ can rise or fall



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Firms have incentives to try to cheat!

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Pigouvian policies will not work as well for **non-point sources** like cars or farms

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Point sources like power plants are much easier to handle with Pigouvian policies like taxes

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This might make things more politically difficult to pass

Administration

When does C&C / technology standards make sense?

1. If there's a dominant technology where there's benefits to coordination or scale economies from production of the technology
2. High costs of monitoring/enforcement
3. High admin costs and little heterogeneity across firms

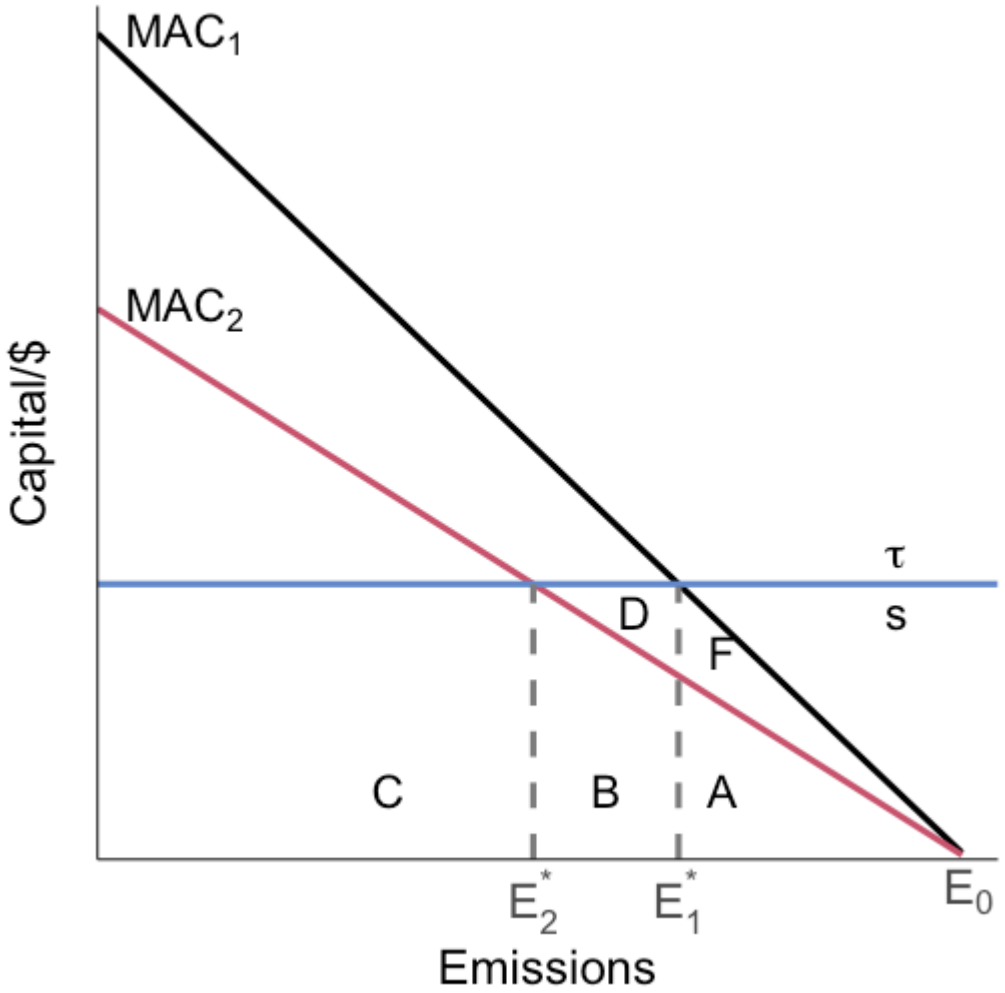
Dynamic incentives

Firms may innovate or adopt new technologies that lower their marginal abatement costs, shifting MAC down

What incentives do different policy instruments give firms to make these improvements?

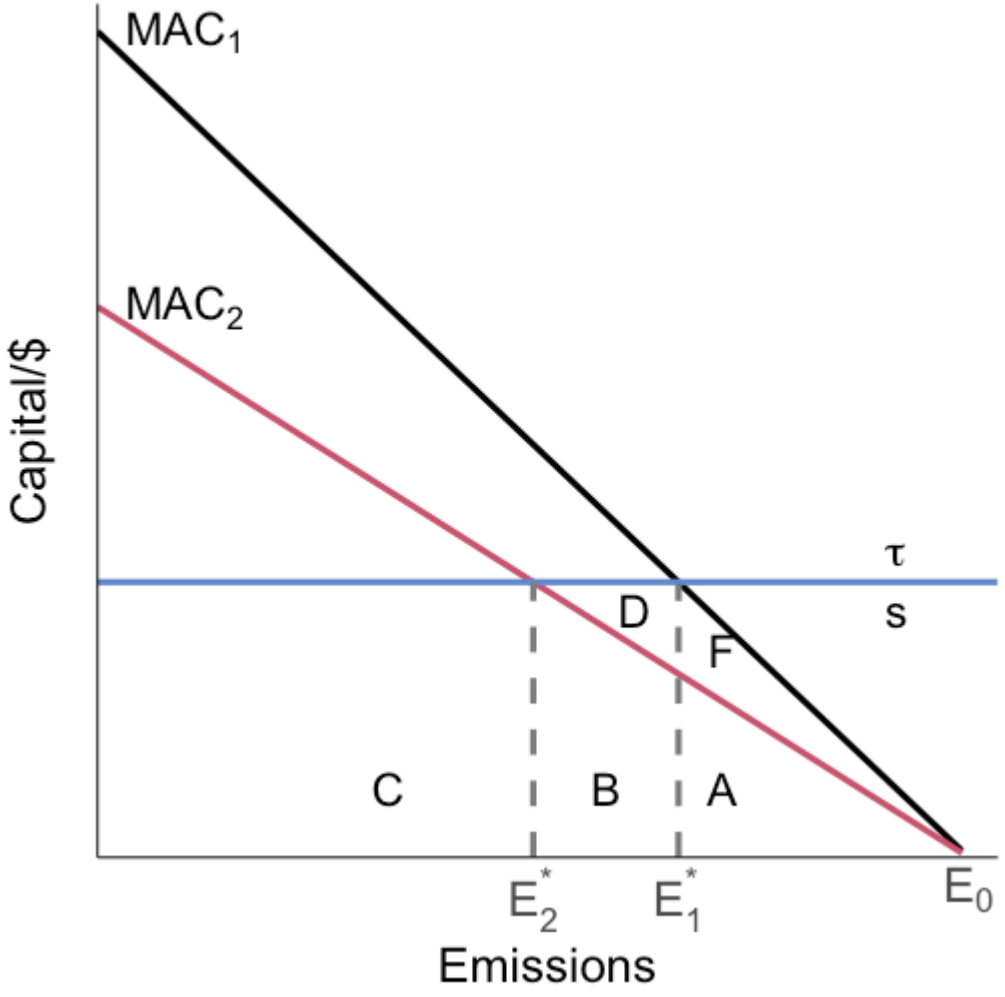
To determine this, we need to work out how a lower MAC changes the firm's abatement costs and tax or subsidy payments

Dynamic incentives



What are the gains to the firm from moving from MAC_1 to MAC_2?

Dynamic incentives



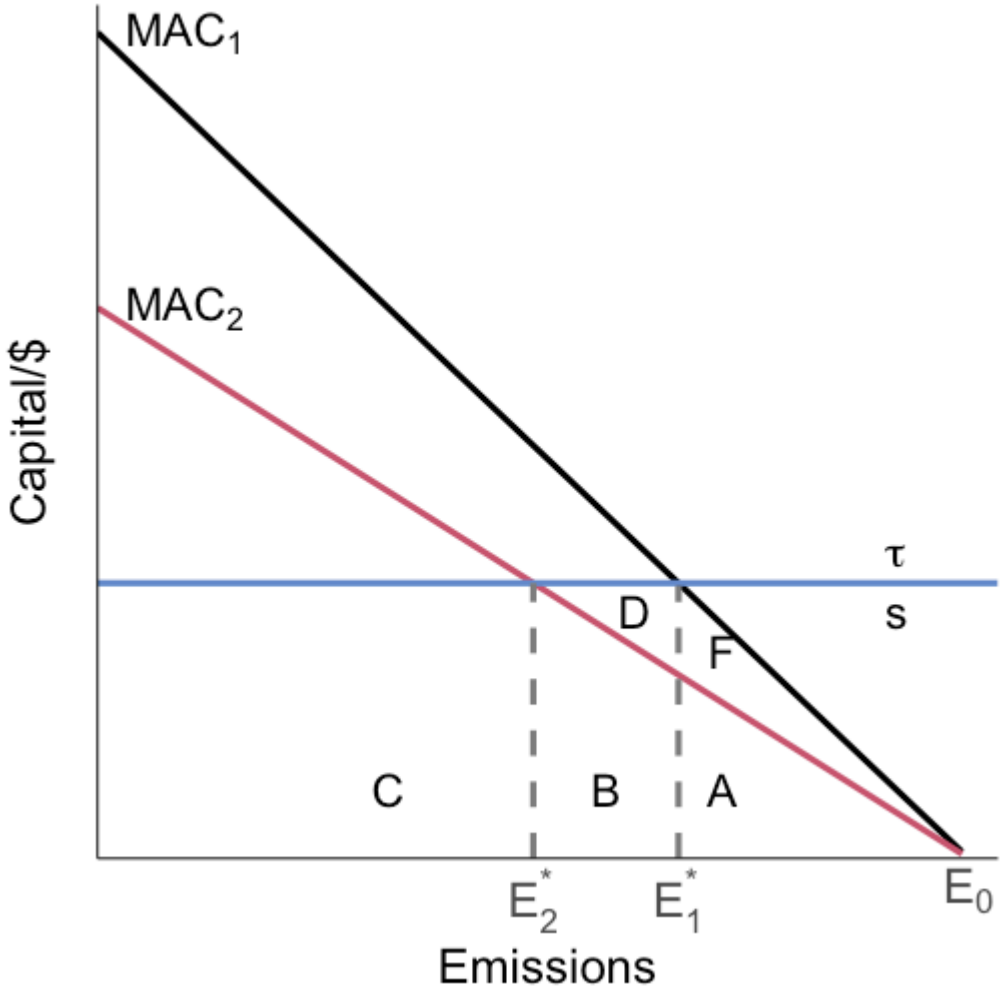
What are the gains to the firm from moving from MAC_1 to MAC_2?

Standard: F (abatement cost reduction)

Emission Tax: F + D (abatement cost and tax payment reduction)

Abatement Subsidy: F + D (abatement cost reduction and abatement subsidy increase)

Dynamic incentives



What are the gains to the firm from moving to MAC_2?

Taxes and subsidies give greater incentives to innovate!

Once a firm meets a standard, there's no additional incentive beyond reducing abatement costs

Taxes and subsidies give the firm extra benefits for further reductions

How do carbon markets work?

